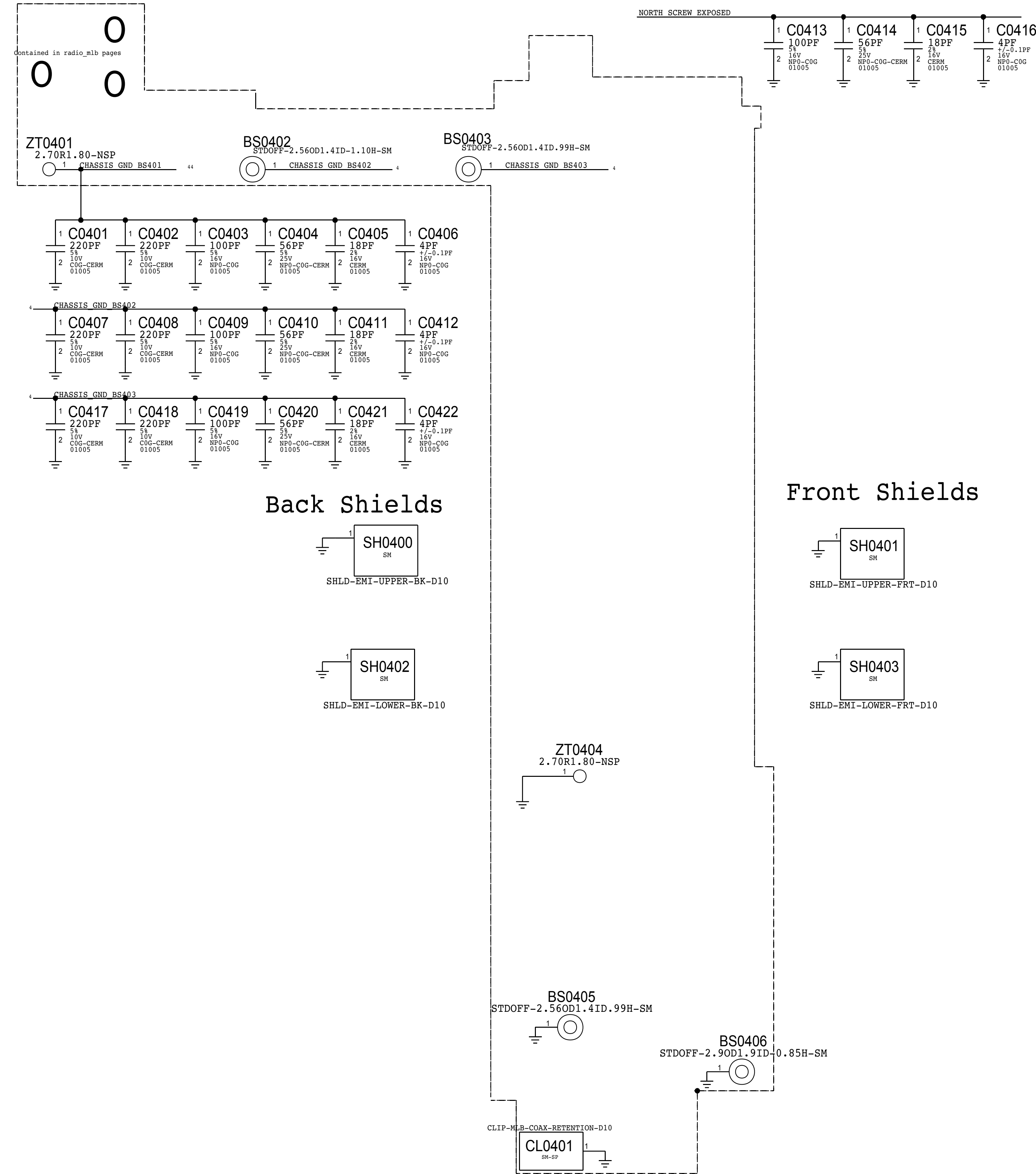
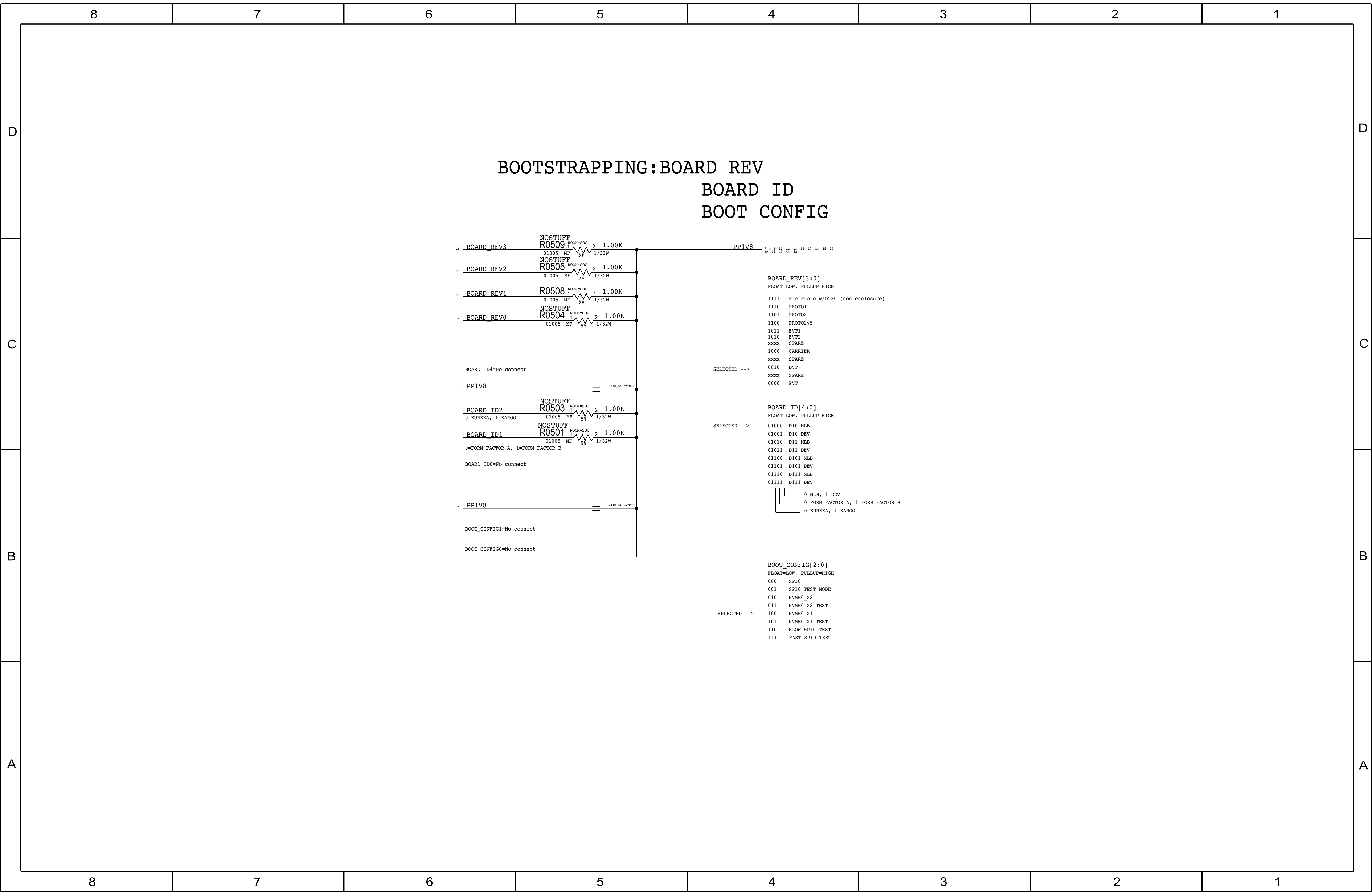


Current as of D10 MCO 056-01342-78



TOP SIDE



12

BOARD_REV0

NOSTUFF

R0504

ROOM=SO

01005

MF

5

2

1.00K

1/32W

BOARD_ID4=No connect

11

PPIV8

MAKE_BASE=TRUE

11

BOARD_ID2

NOSTUFF

R0503

ROOM=SO

01005

MF

5

2

1.00K

1/32W

0=EUREKA, 1=KAROO

11

BOARD_ID1

NOSTUFF

R0501

ROOM=SO

01005

MF

5

2

1.00K

1/32W

0=FORM FACTOR A, 1=FORM FACTOR B

BOARD_ID0=No connect

12

PPIV8

MAKE_BASE=TRUE

BOOT_CONFIG1=No connect

BOOT_CONFIG0=No connect

PPIV8

15

8

9

11

12

13

16

17

18

25

29

BOARD_REV[3:0]

FLOAT=LOW, PULLUP=HIGH

1111 Pre-Proto w/D520 (non enclosure)

1110 PROTO1

1101 PROTO2

1100 PROTO2v5

1011 EVT1

1010 EVT2

xxxx SPARE

1000 CARRIER

xxxx SPARE

0010 DVT

xxxx SPARE

0000 PVT

BOARD_ID[4:0]

FLOAT=LOW, PULLUP=HIGH

01000 D10 MLB

01001 D10 DEV

01010 D11 MLB

01011 D11 DEV

01100 D101 MLB

01101 D101 DEV

01110 D111 MLB

01111 D111 DEV

0=MLB, 1=DEV

0=FORM FACTOR A, 1=FORM FACTOR B

0=EUREKA, 1=KAROO

BOOT_CONFIG[2:0]

FLOAT=LOW, PULLUP=HIGH

000 SPI0

001 SPI0 TEST MODE

010 NVME0_X2

011 NVME0 X2 TEST

100 NVME0 X1

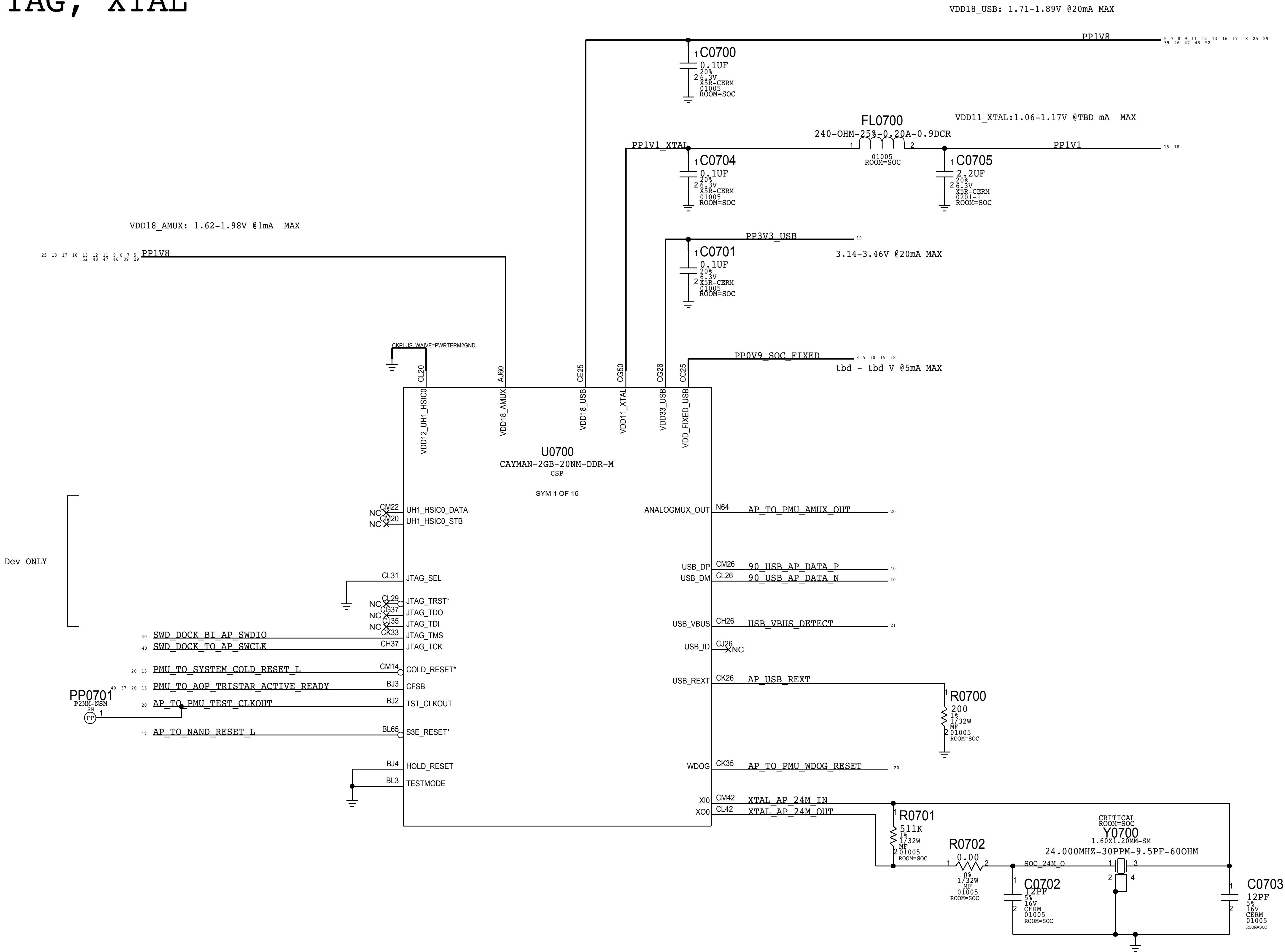
101 NVME0 X1 TEST

110 SLOW SPI0 TEST

111 FAST SPI0 TEST

SELECTED -->

SOC - USB, JTAG, XTAL



SOC - PCIE INTERFACES

D

C

B

A

D

C

B

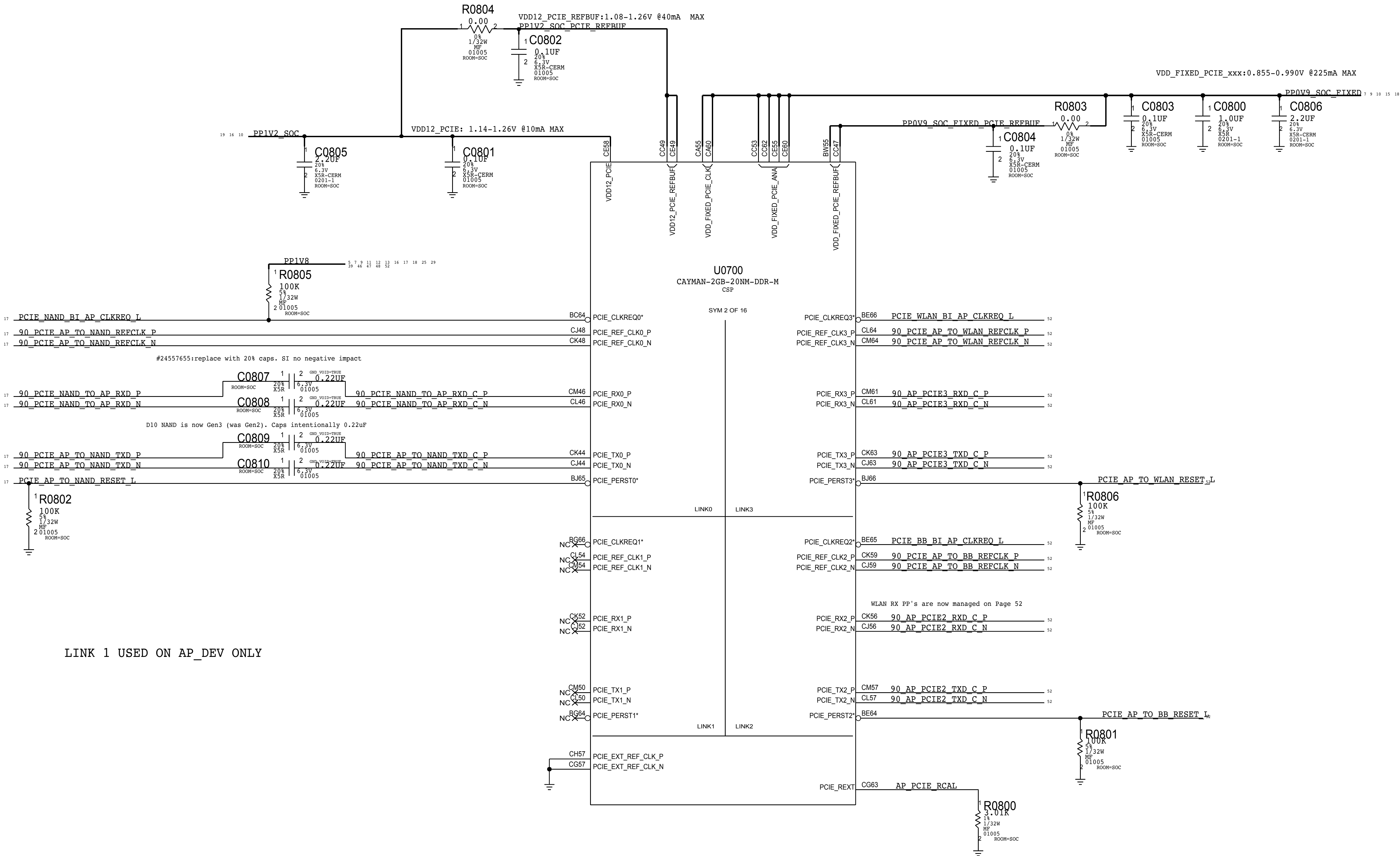
A

PCIE LINK 0

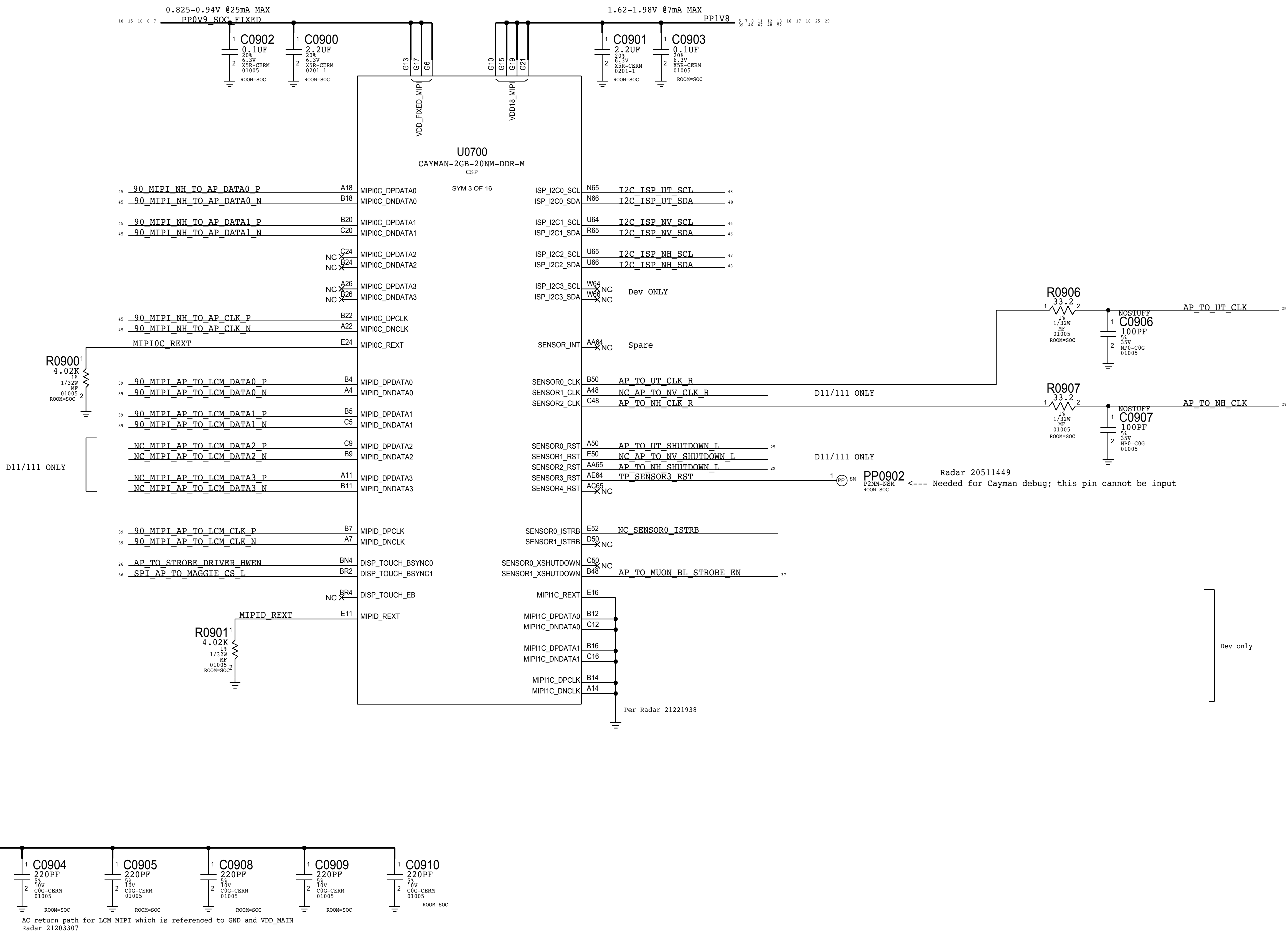
PCIE LINK 1

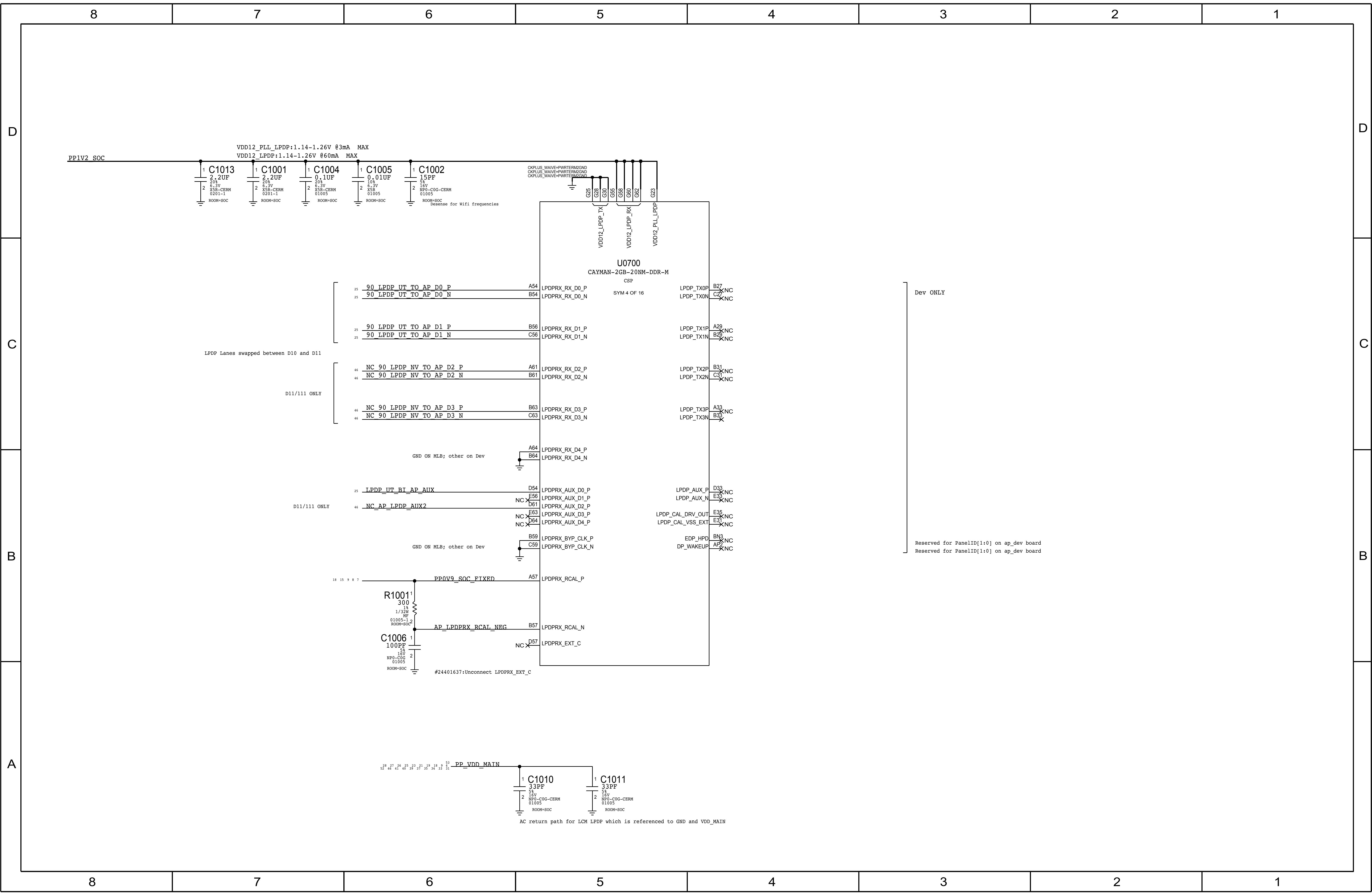
PCIE LINK 3

PCIE LINK 2

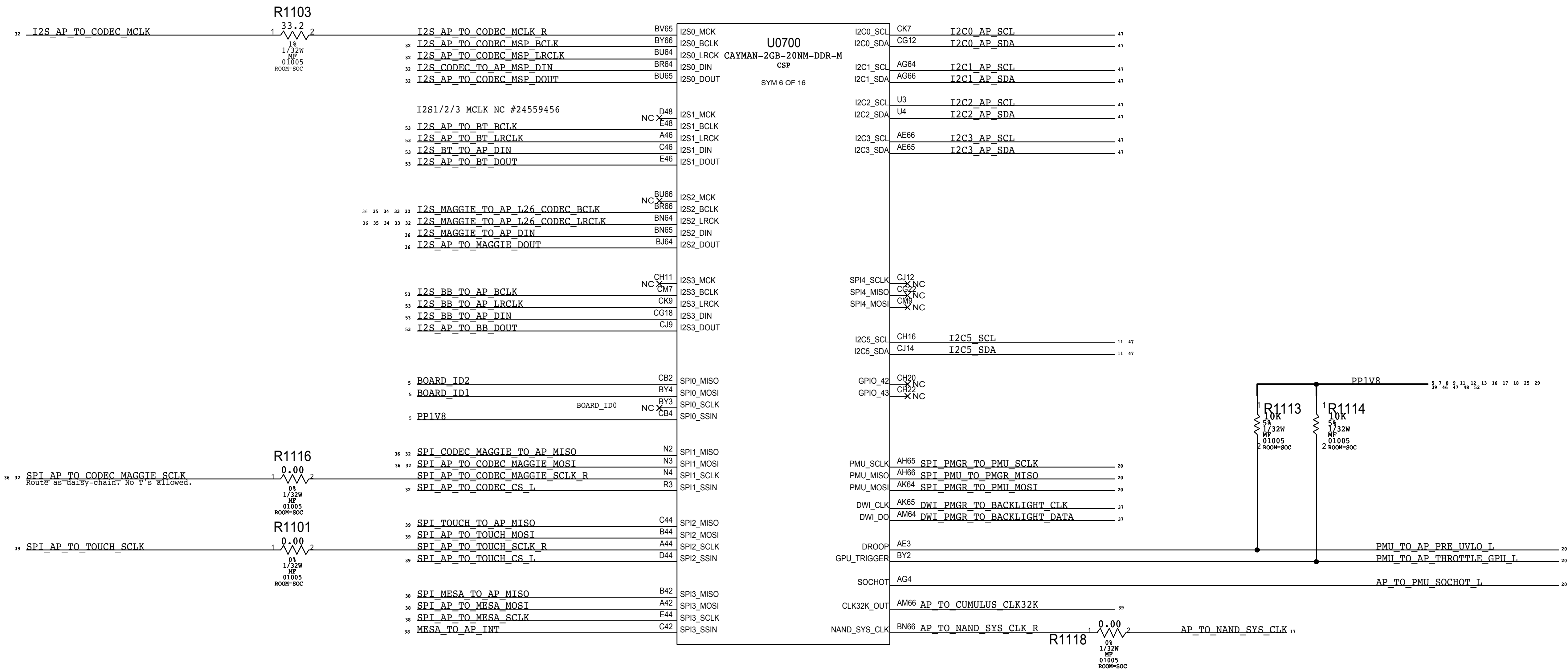


SOC - MIPI & ISP INTERFACES

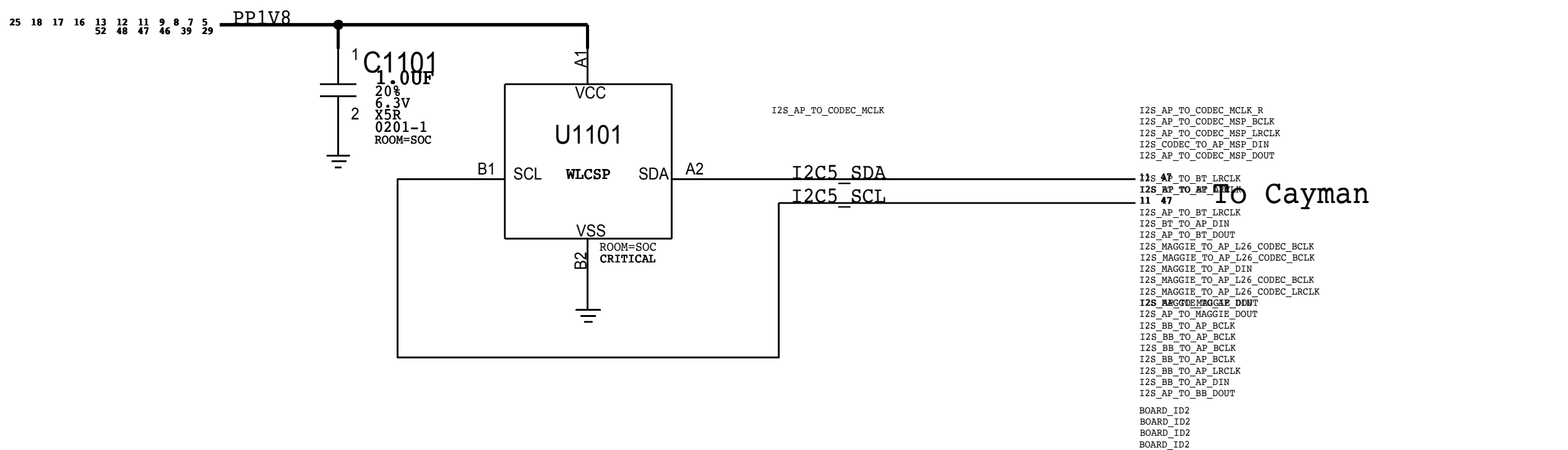




SOC - SERIAL INTERFACES



I2C5
See Radar#25316444 for Details



D



C

C

B

B

A

A

SOC - AOP

8

7

6

5

4

3

2

1

D

D

C

C

B

B

A

A

8

7

6

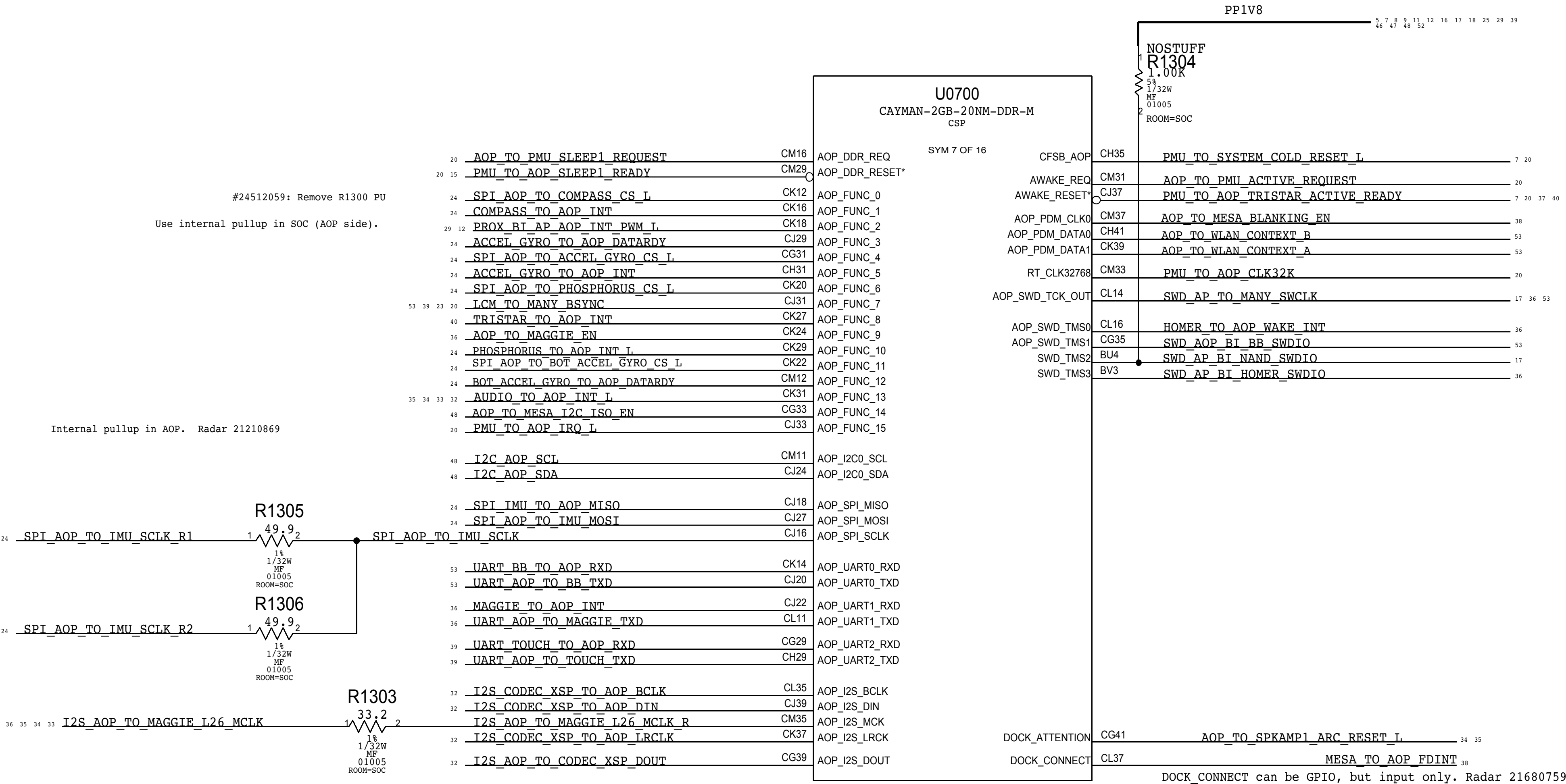
5

4

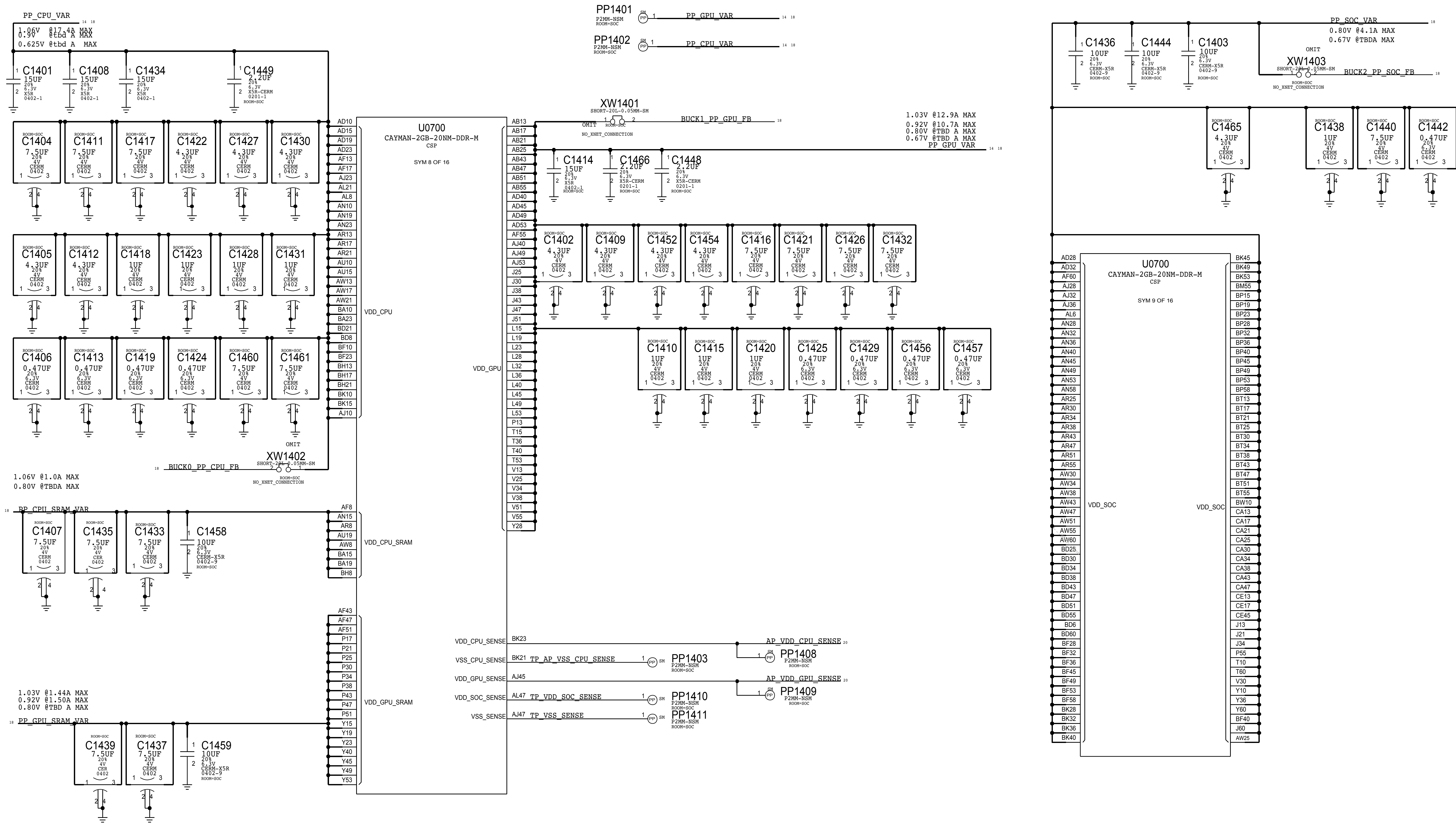
3

2

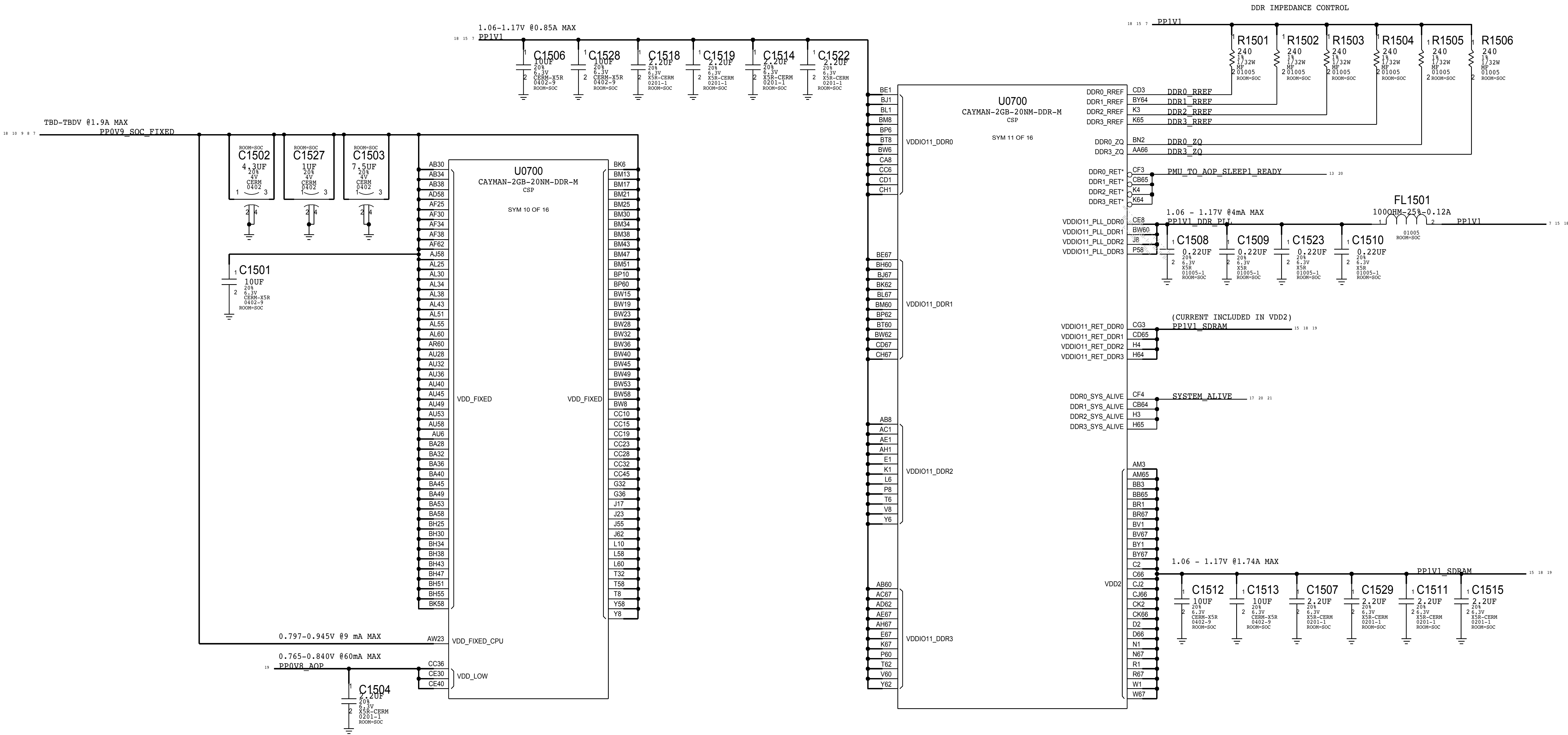
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SOC - CPU, GPU & SOC RAILS



SOC - POWER SUPPLIES



SOC - POWER SUPPLIES

D

C

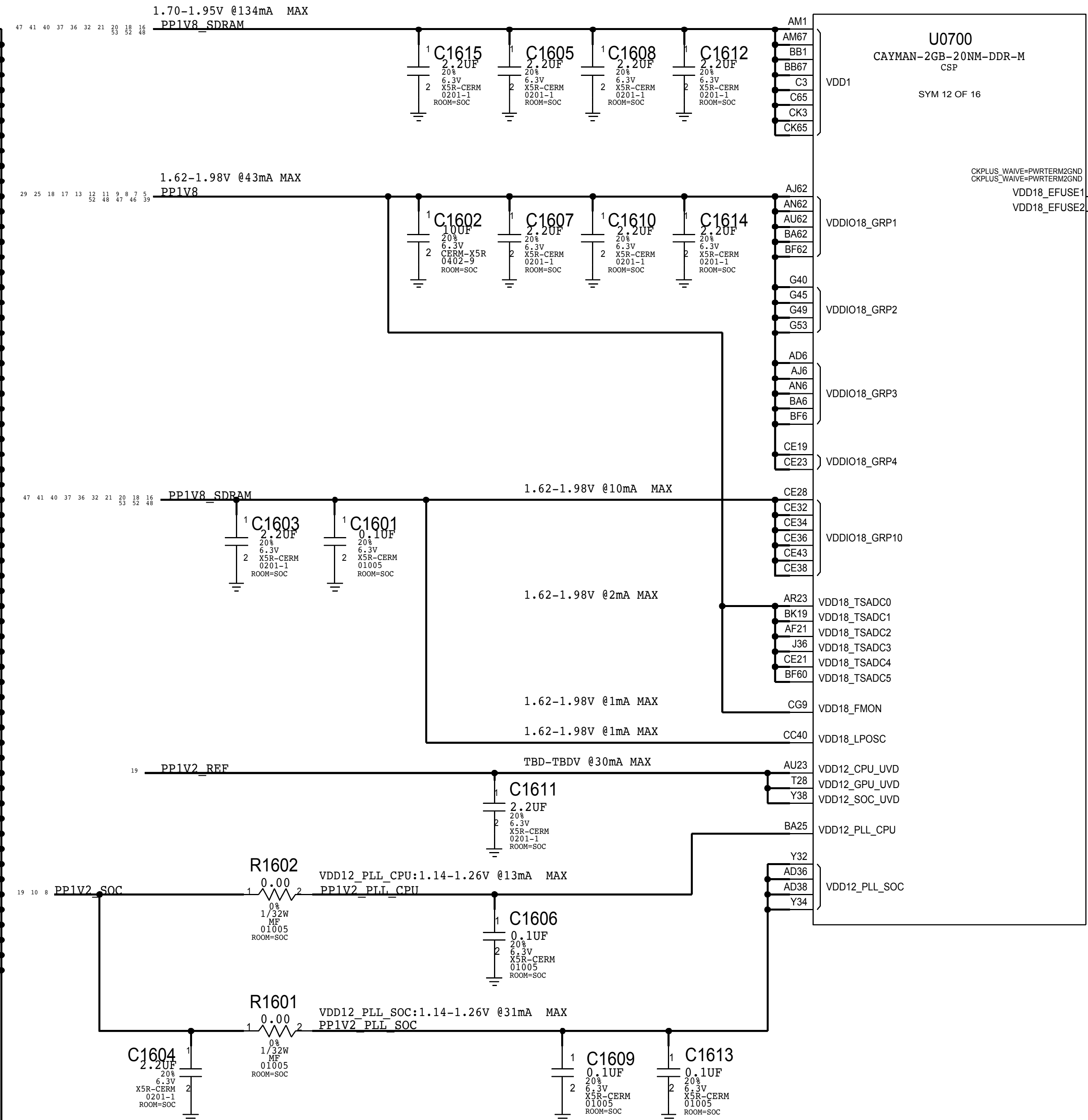
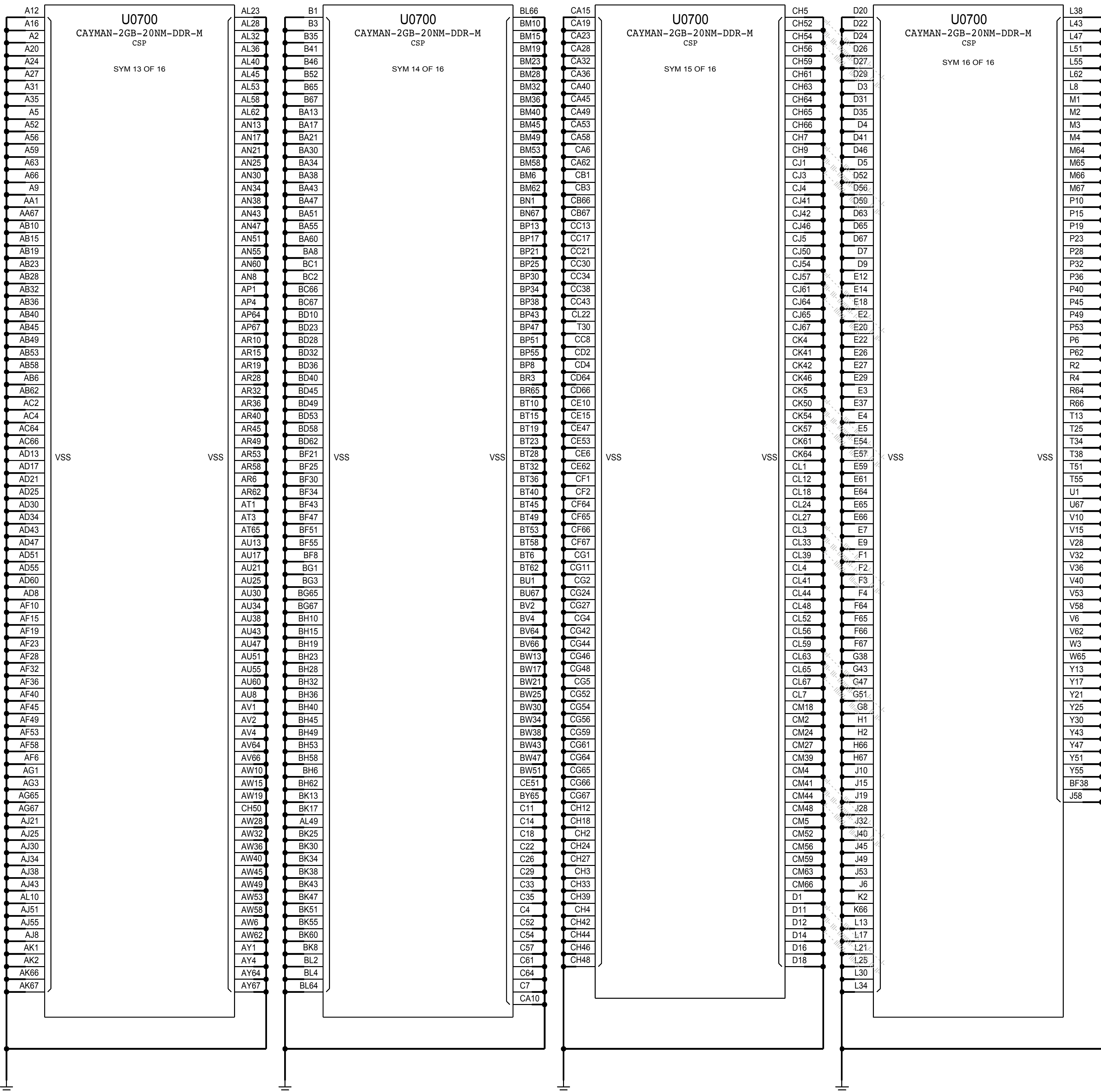
B

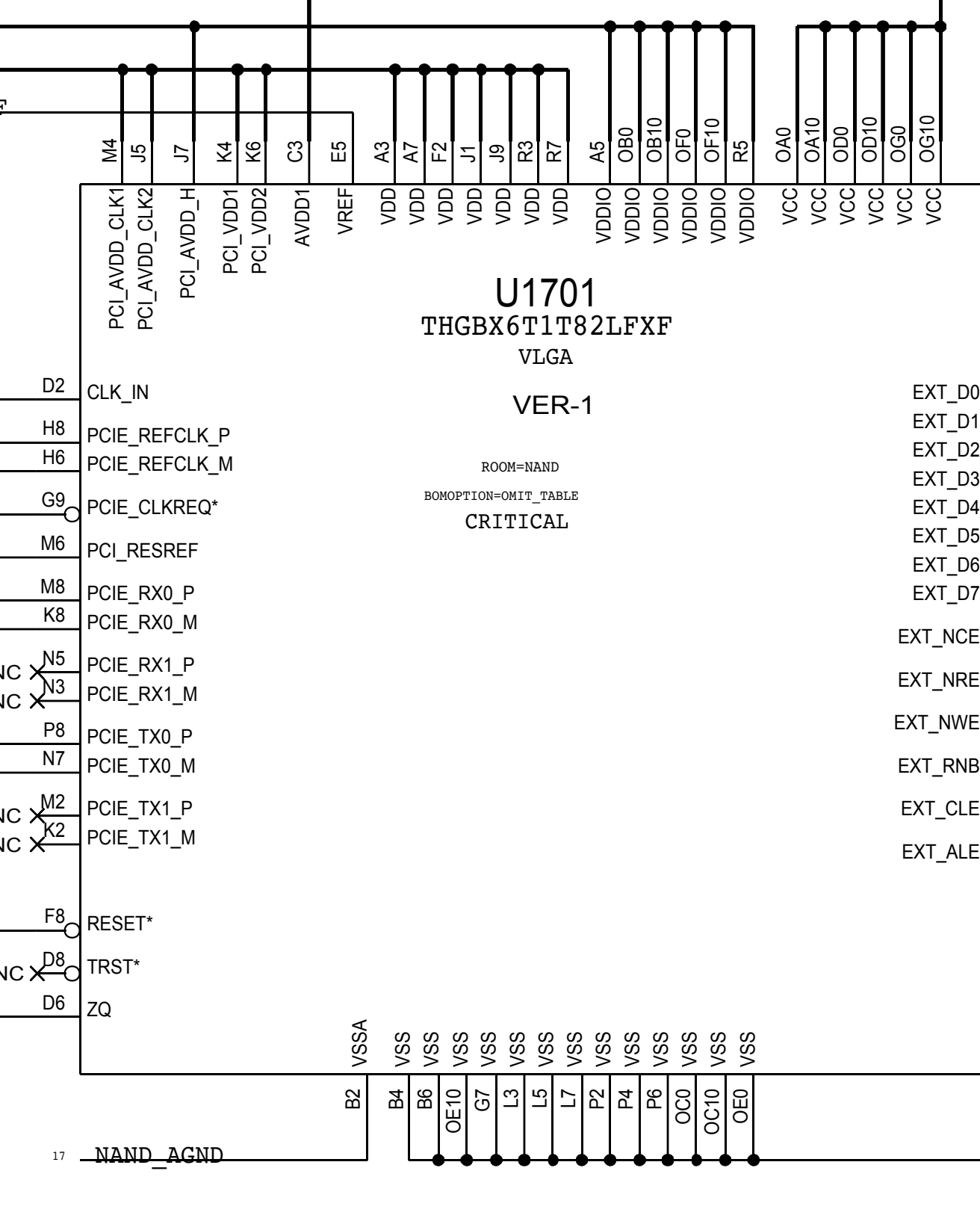
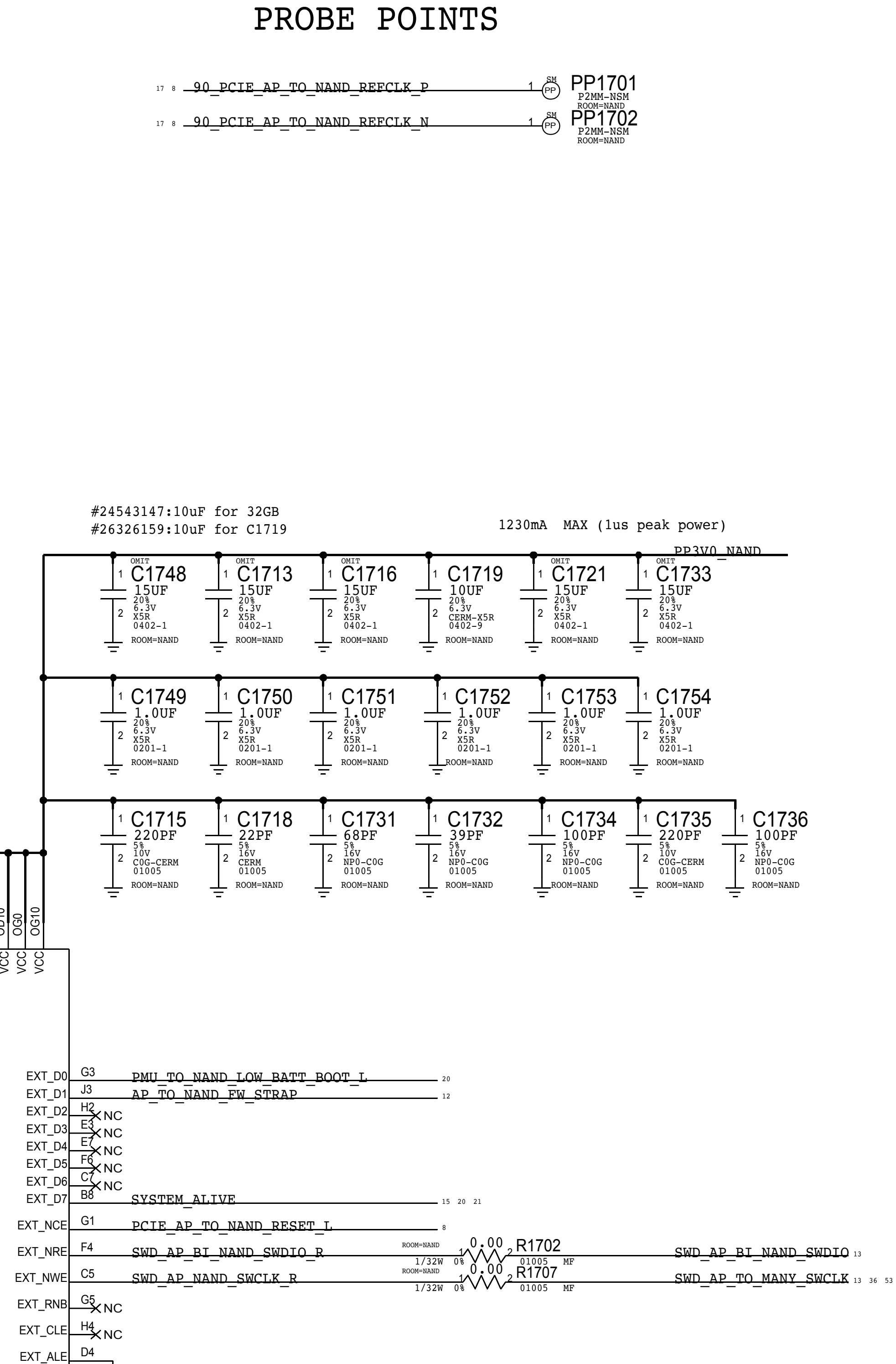
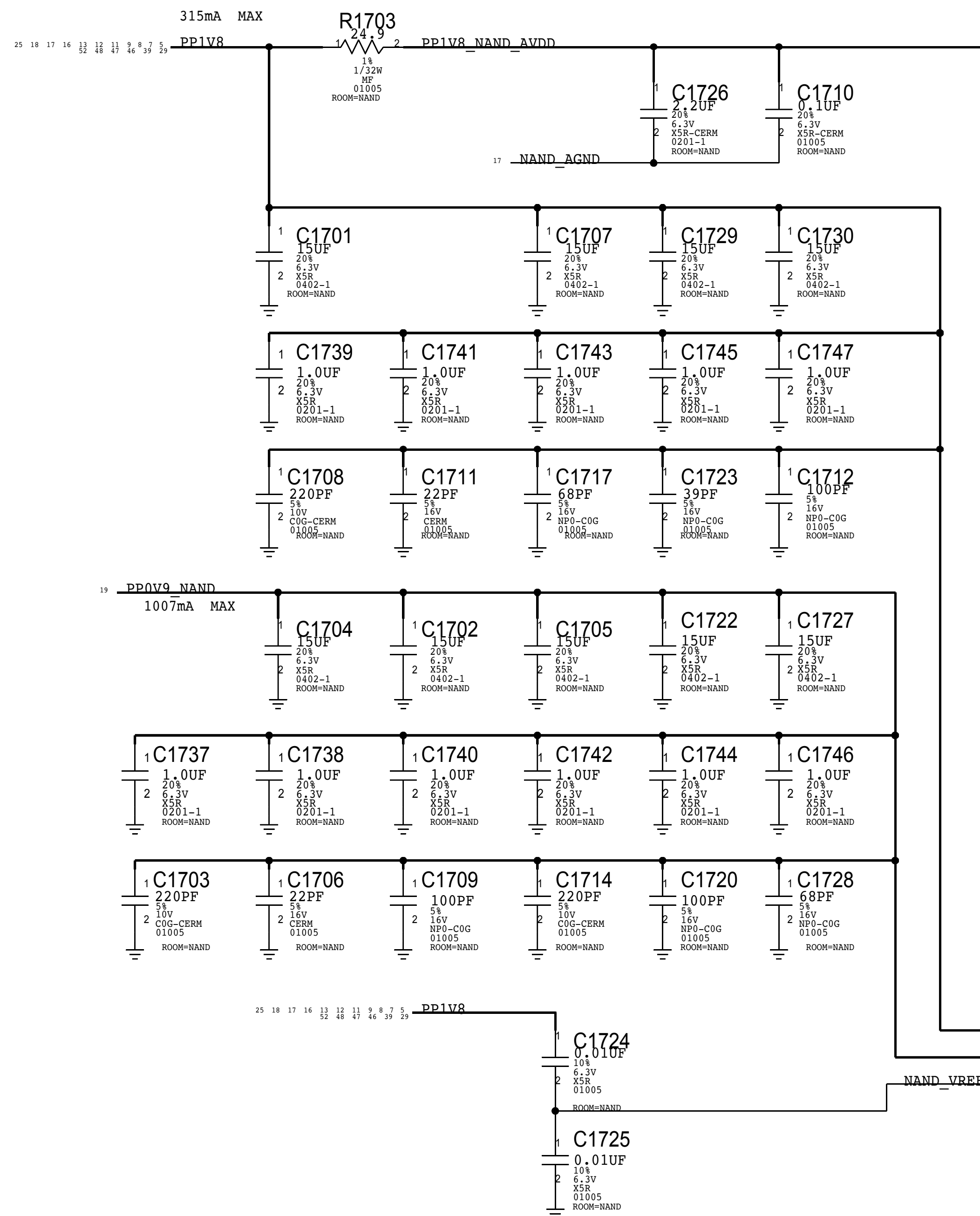
D

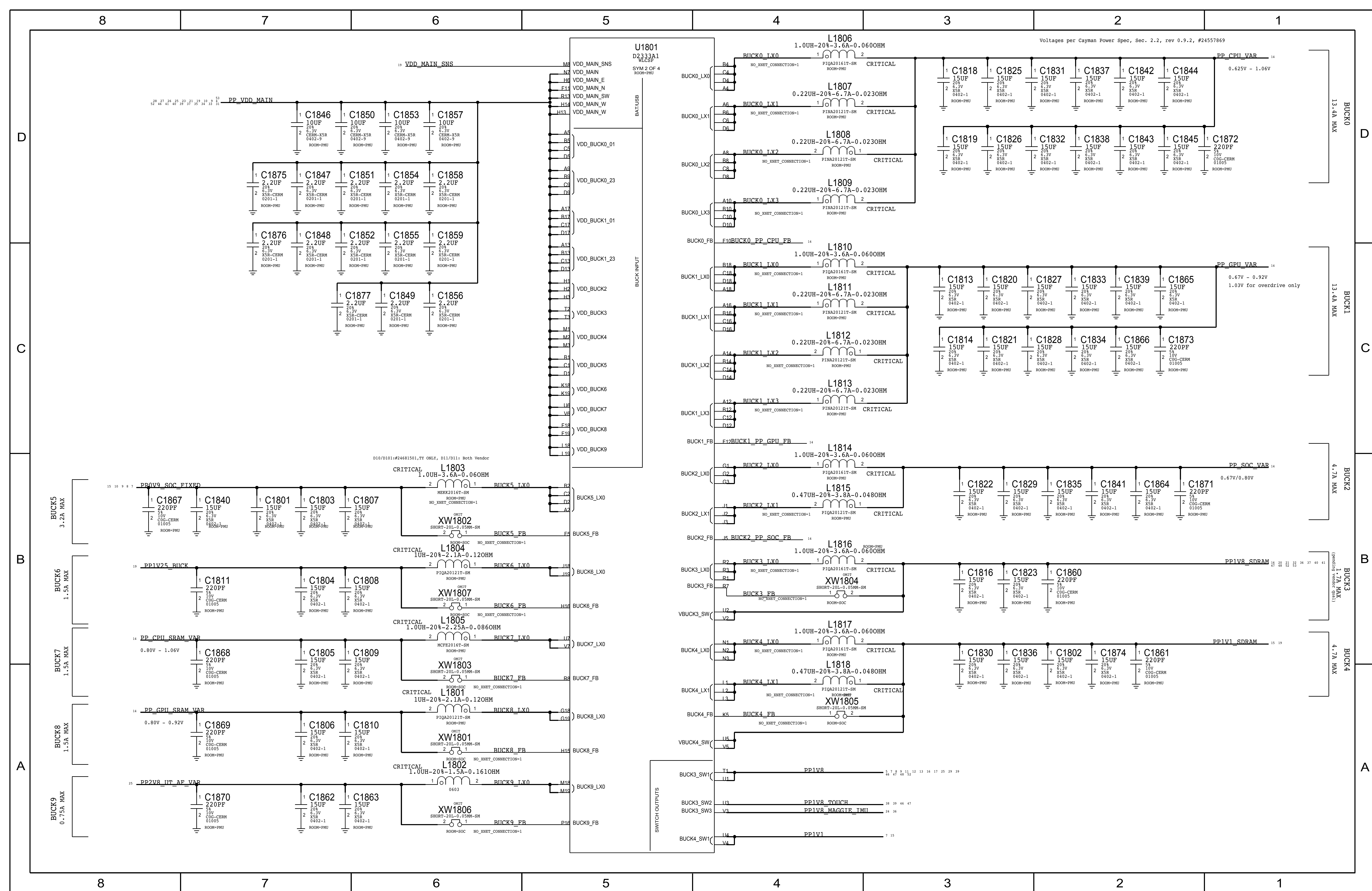
C

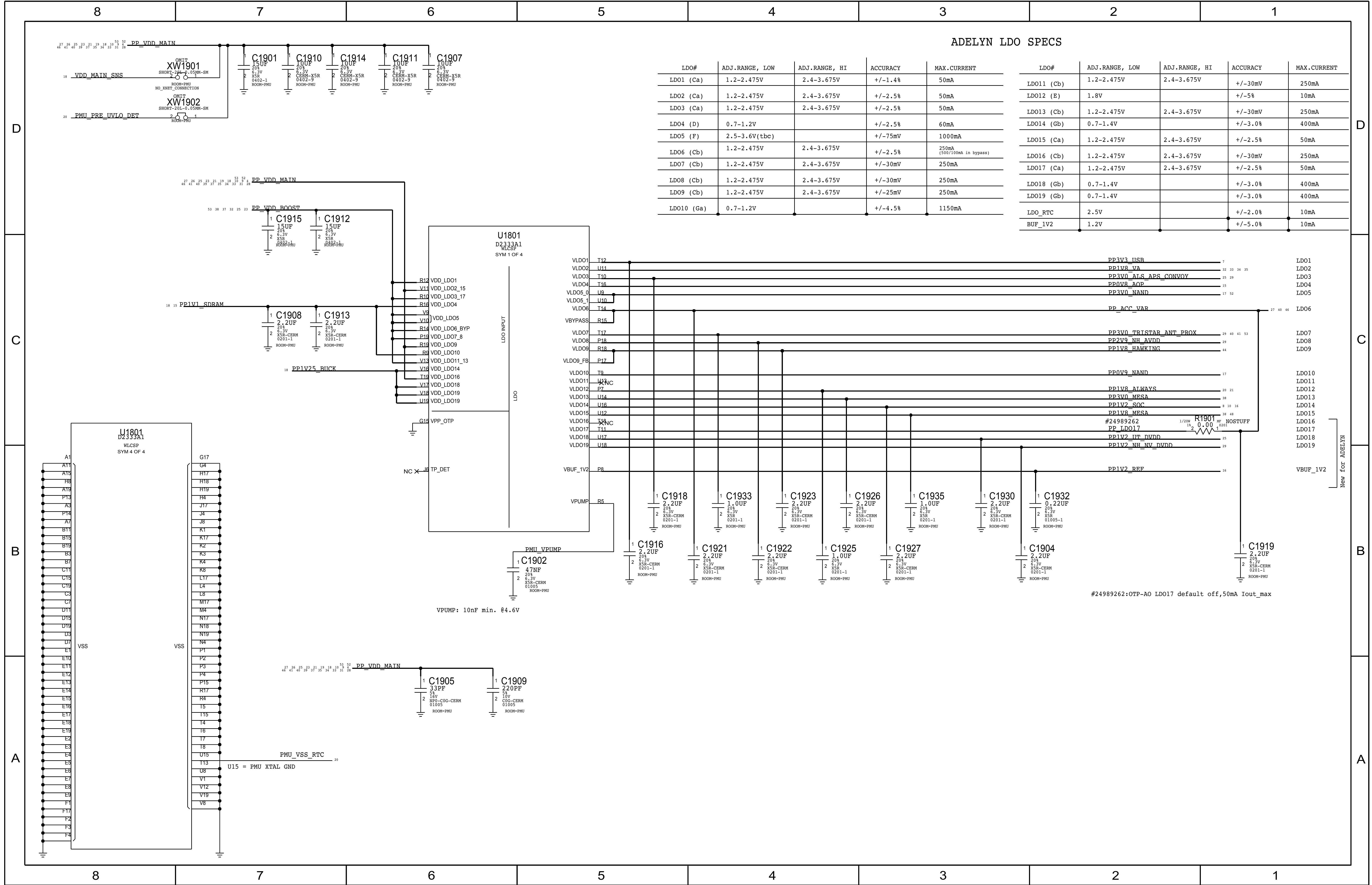
B

A



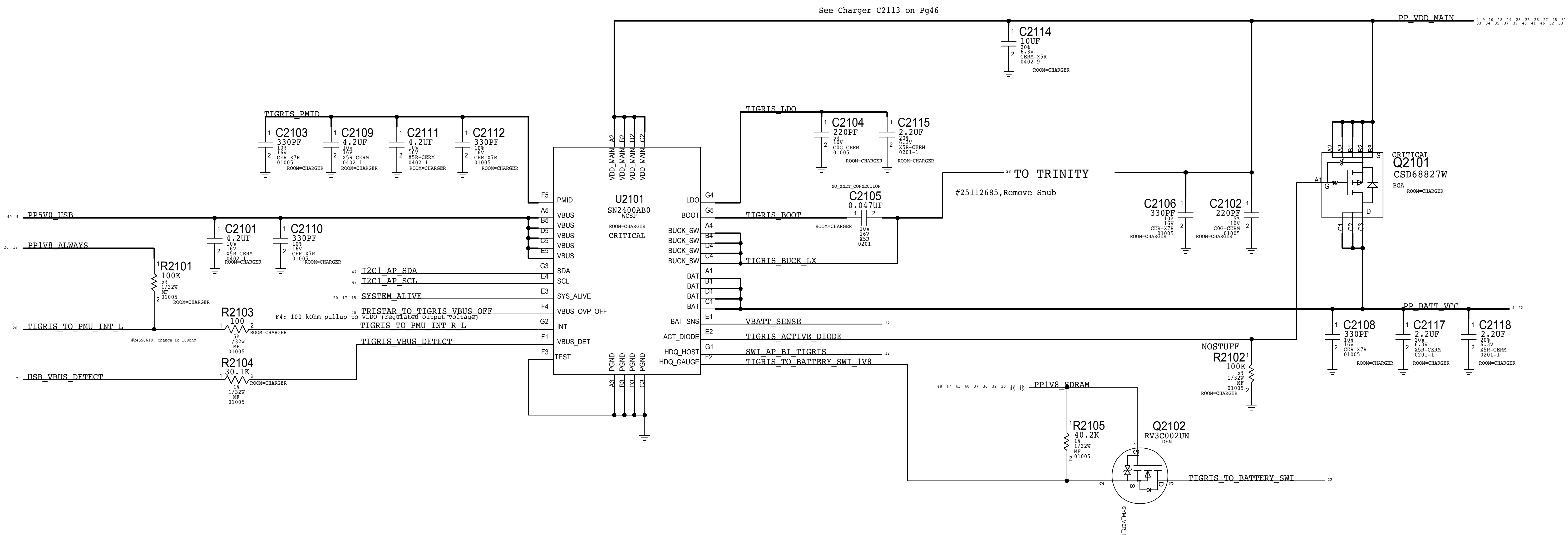


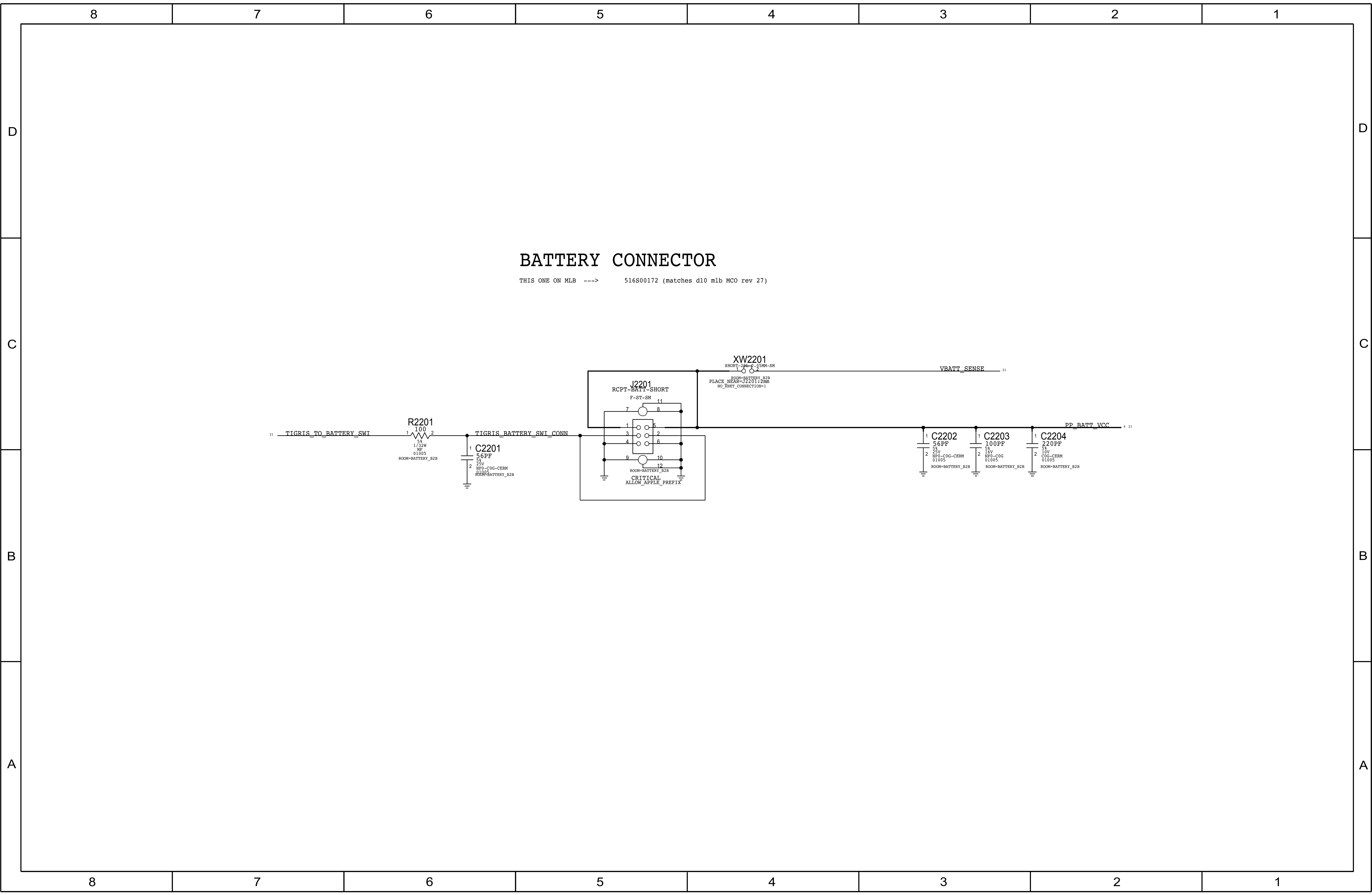


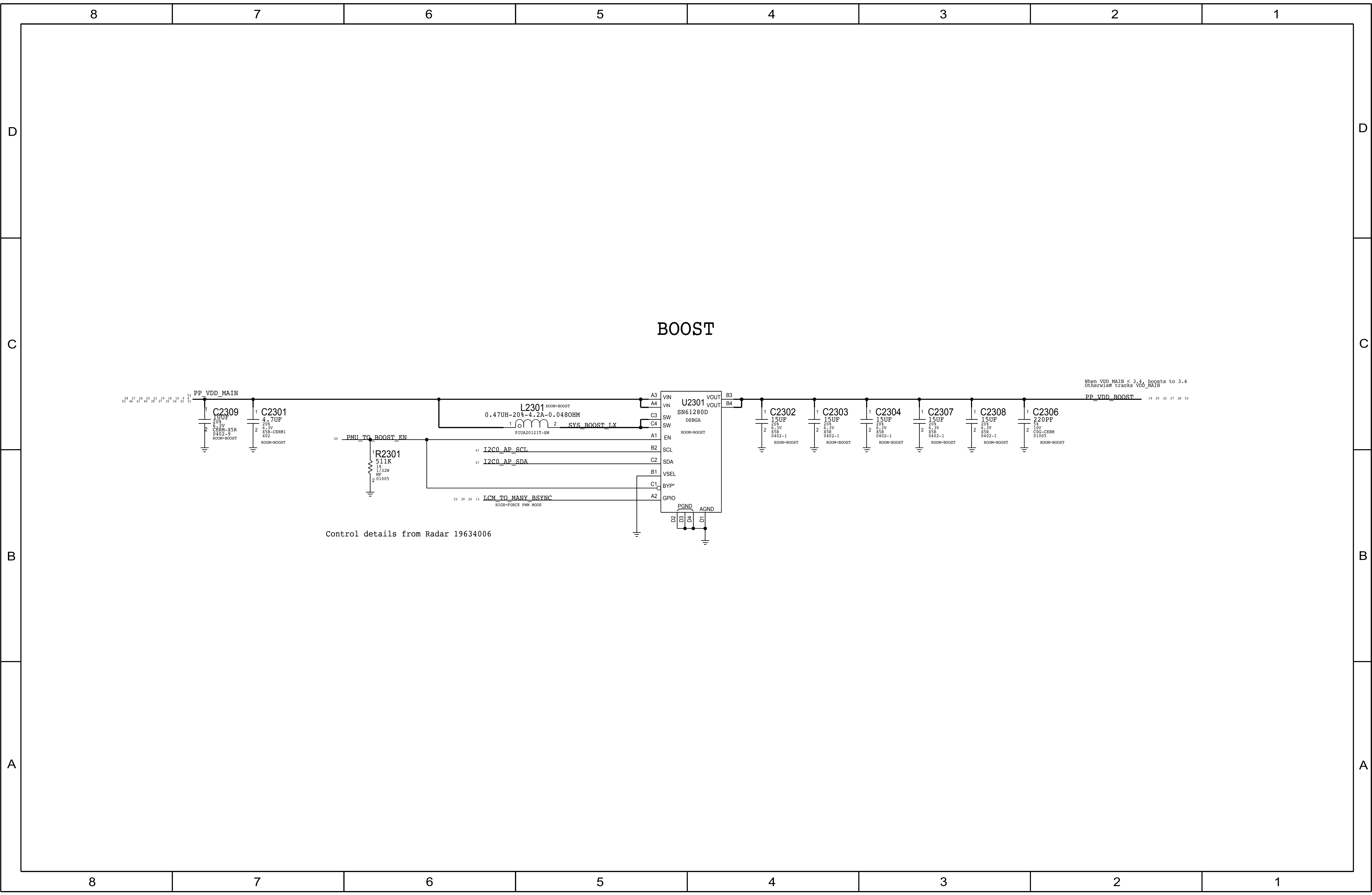




TIGRIS CHARGER

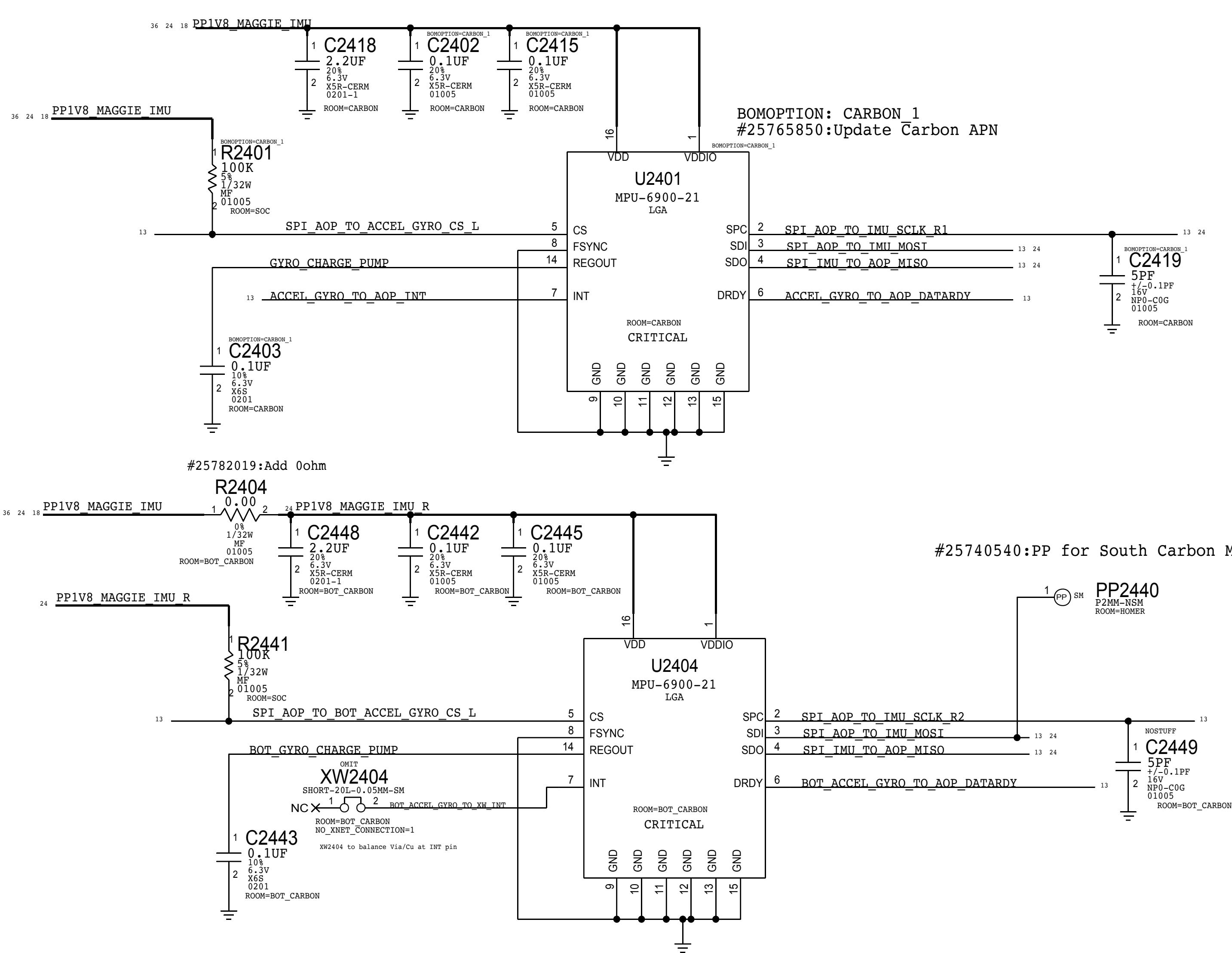




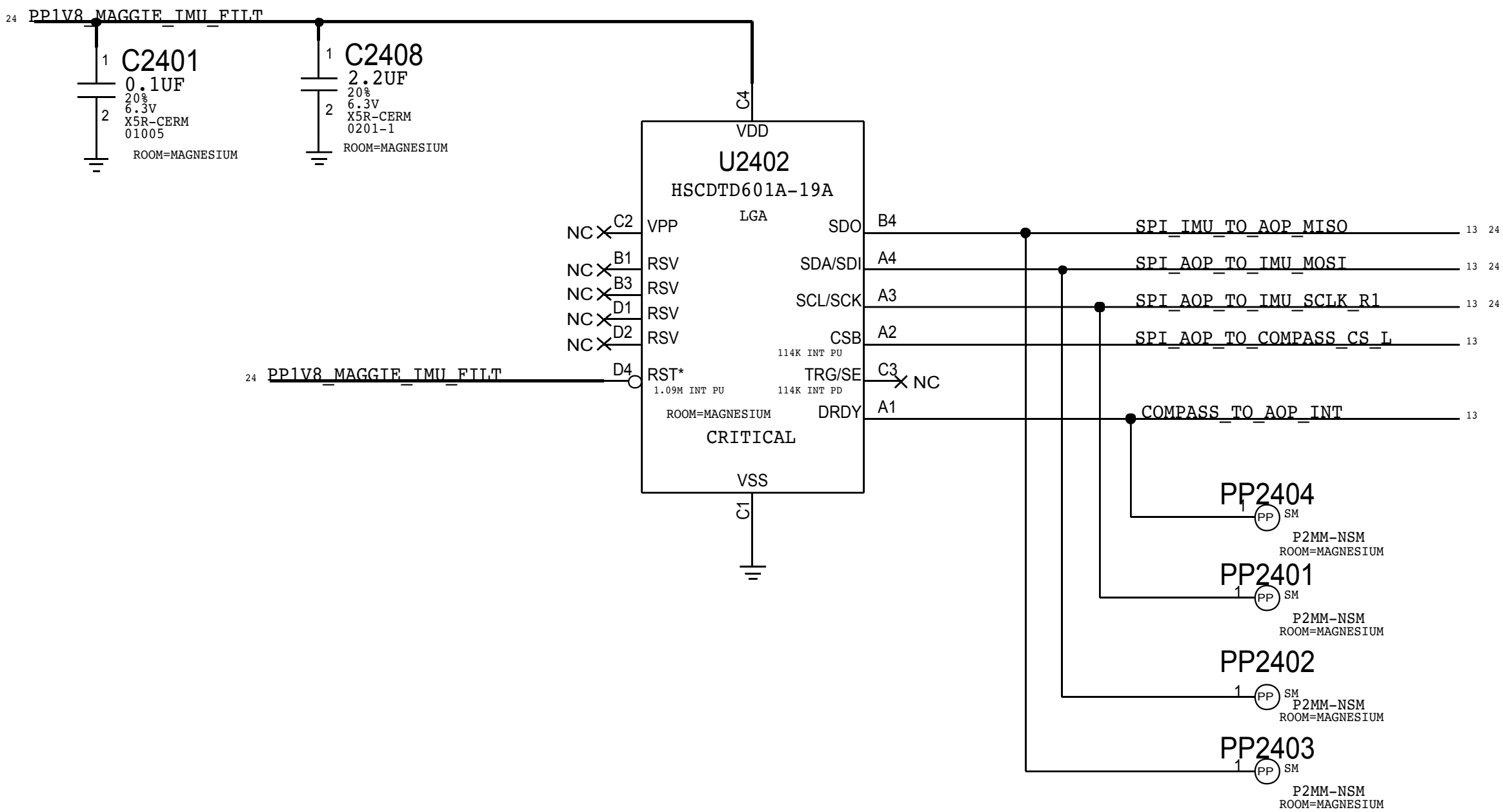


CARBON - ACCEL & GYRO

INVENSENSE, MPU-6800: C2403=0.1UF



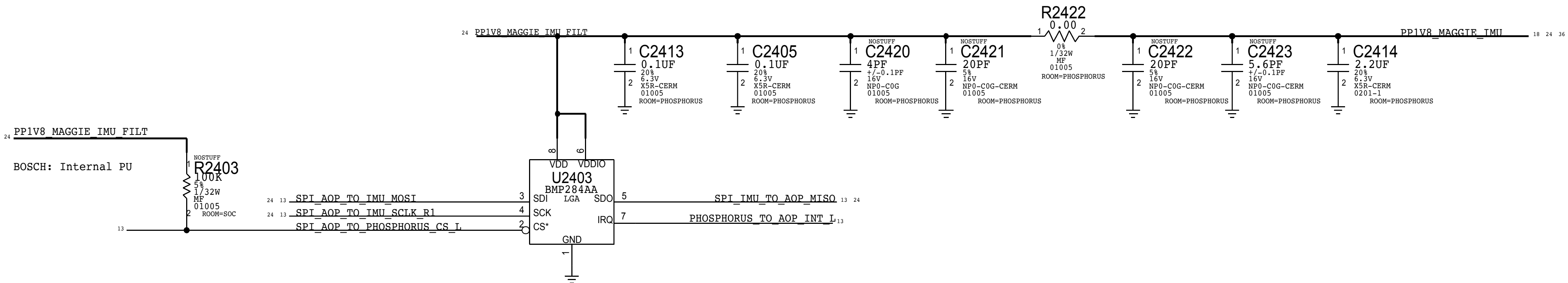
MAGNESIUM - COMPASS



PHOSPHORUS

#24593845, #25691124

BOSCH (APN:338S00188): nostuff C2420/C2421/C2422/C2423 and R2403 PU

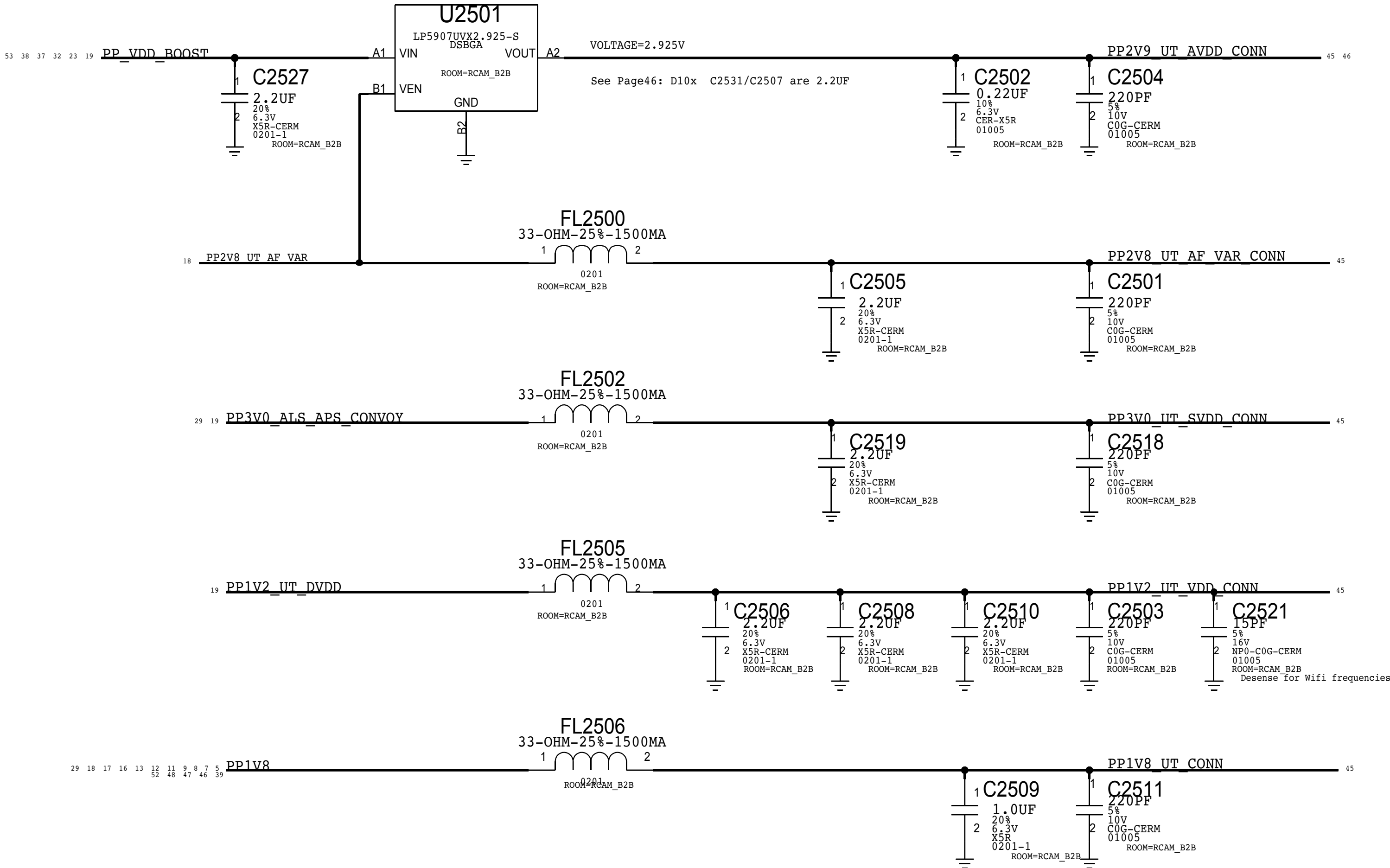


THIS PAGE UNIQUE TO SMALL FORM FACTOR

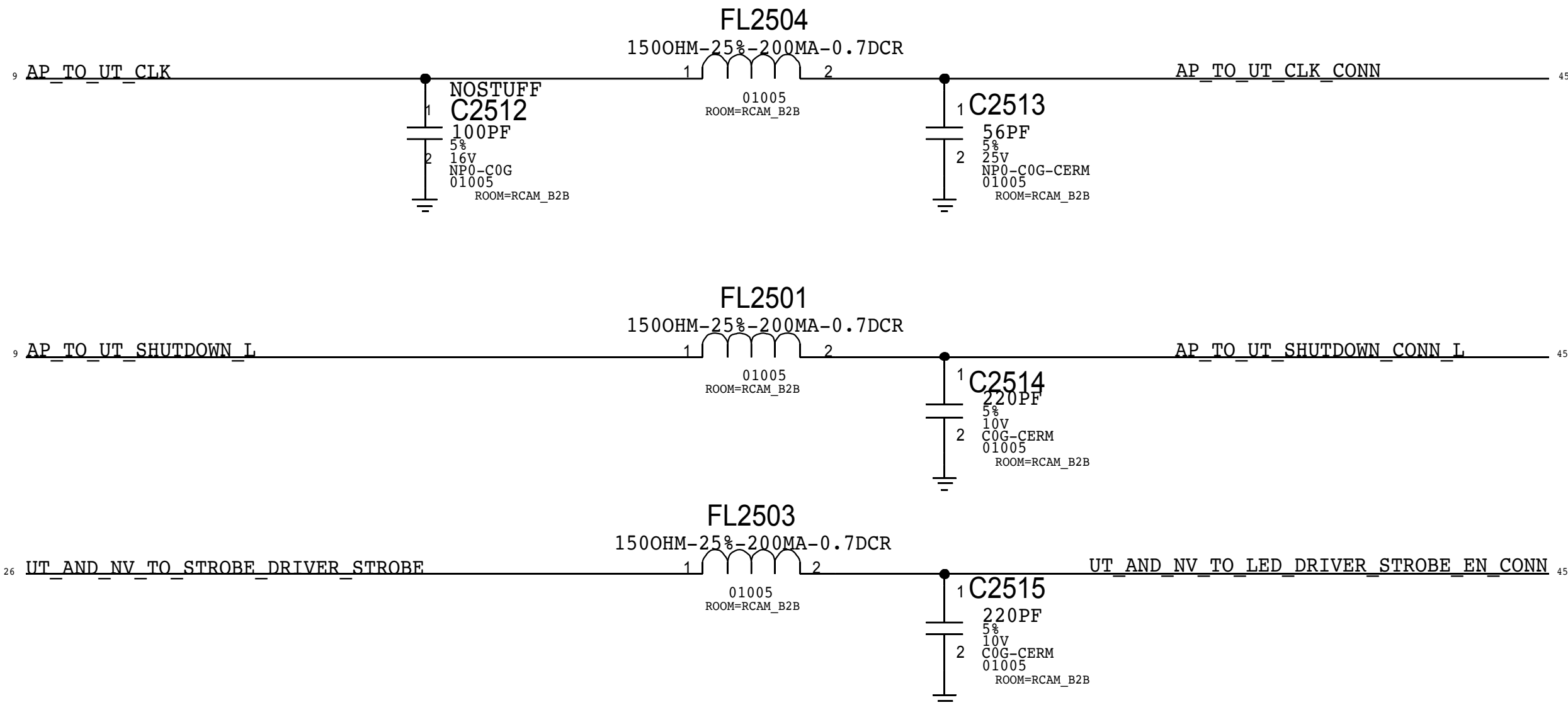
UTAH POWER

NOTE: OUTPUT IMPEDANCE MUST BE >0.005-OHM
IN ORDER TO MEET CAP ESR REQUIREMENT PER LDO SPEC.
VENDOR ALSO RECOMMENDS CIN = COUT FOR STABILITY

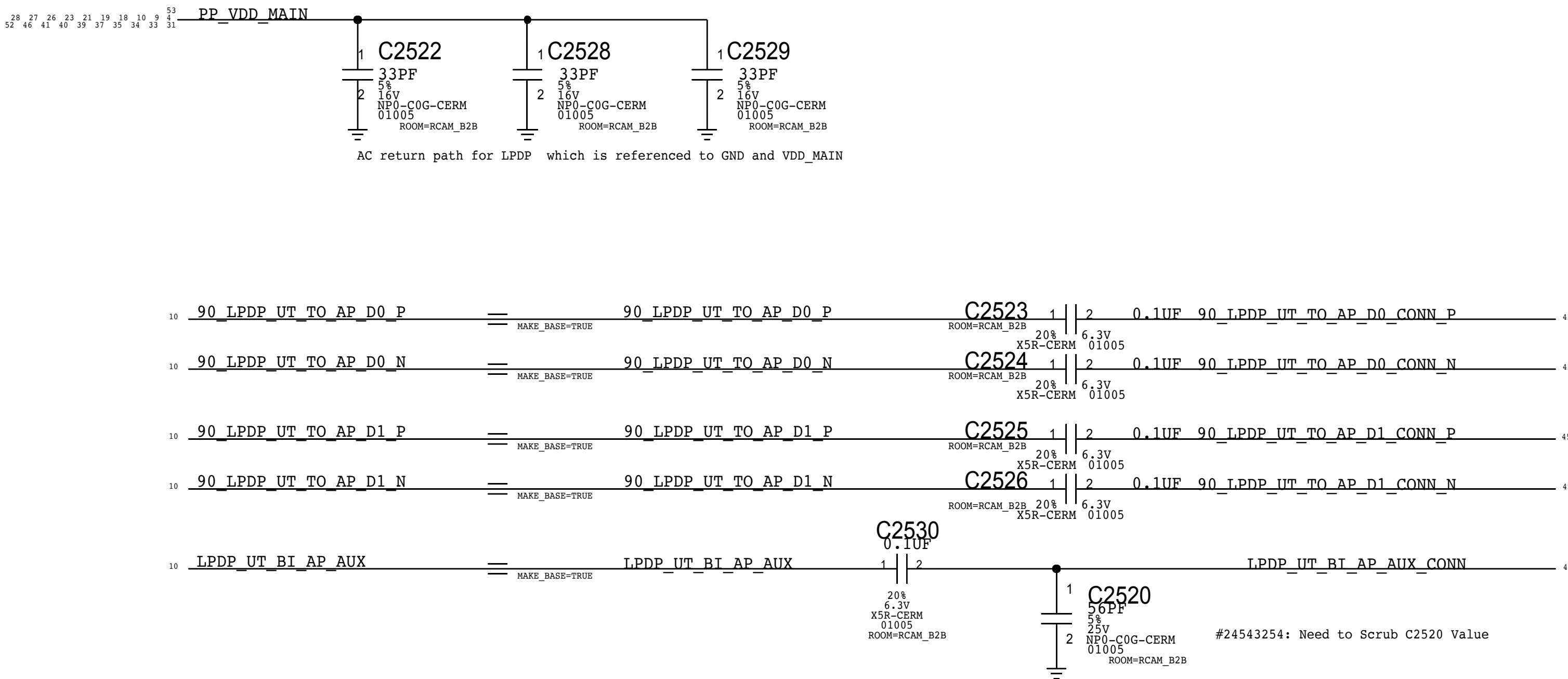
TI:353S00015
ST:353S00889



IO FILTERS

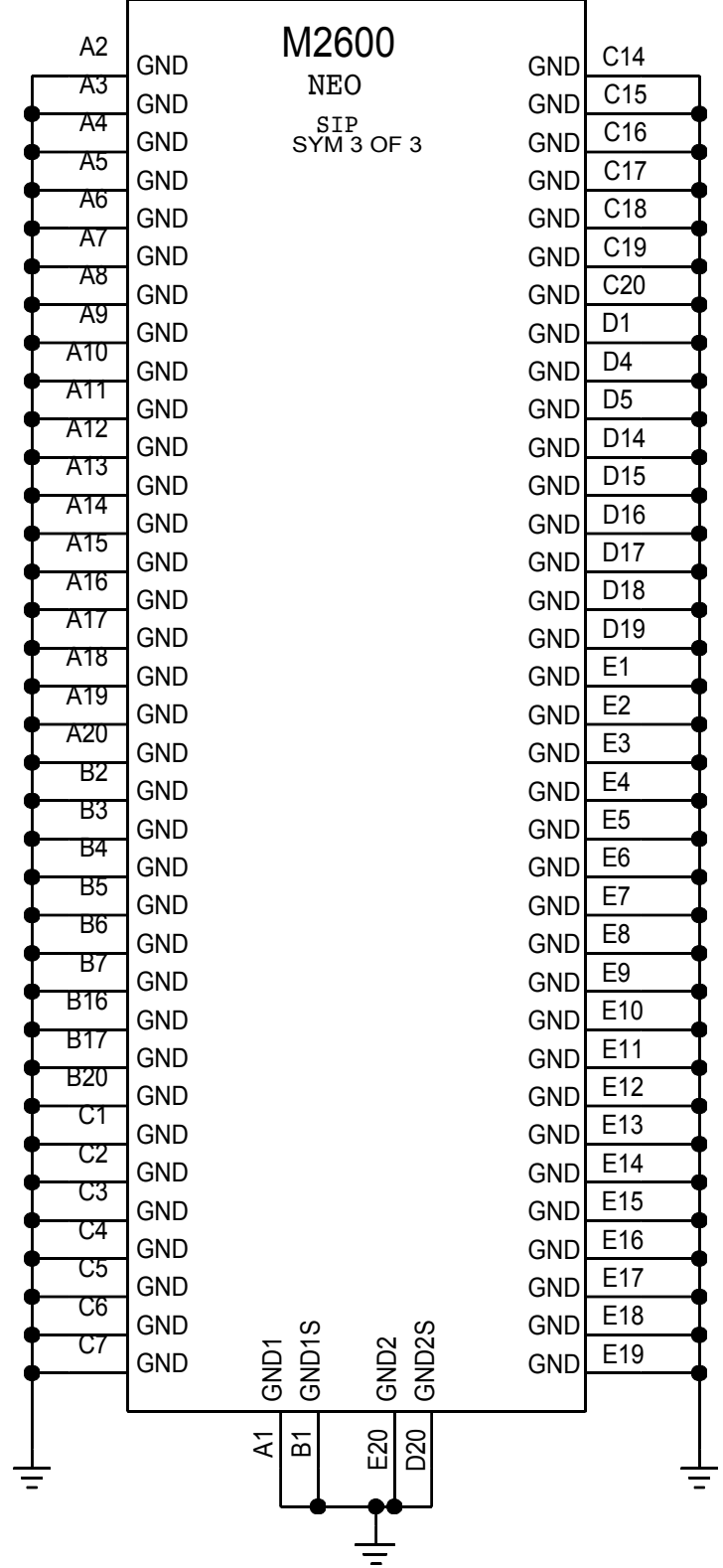
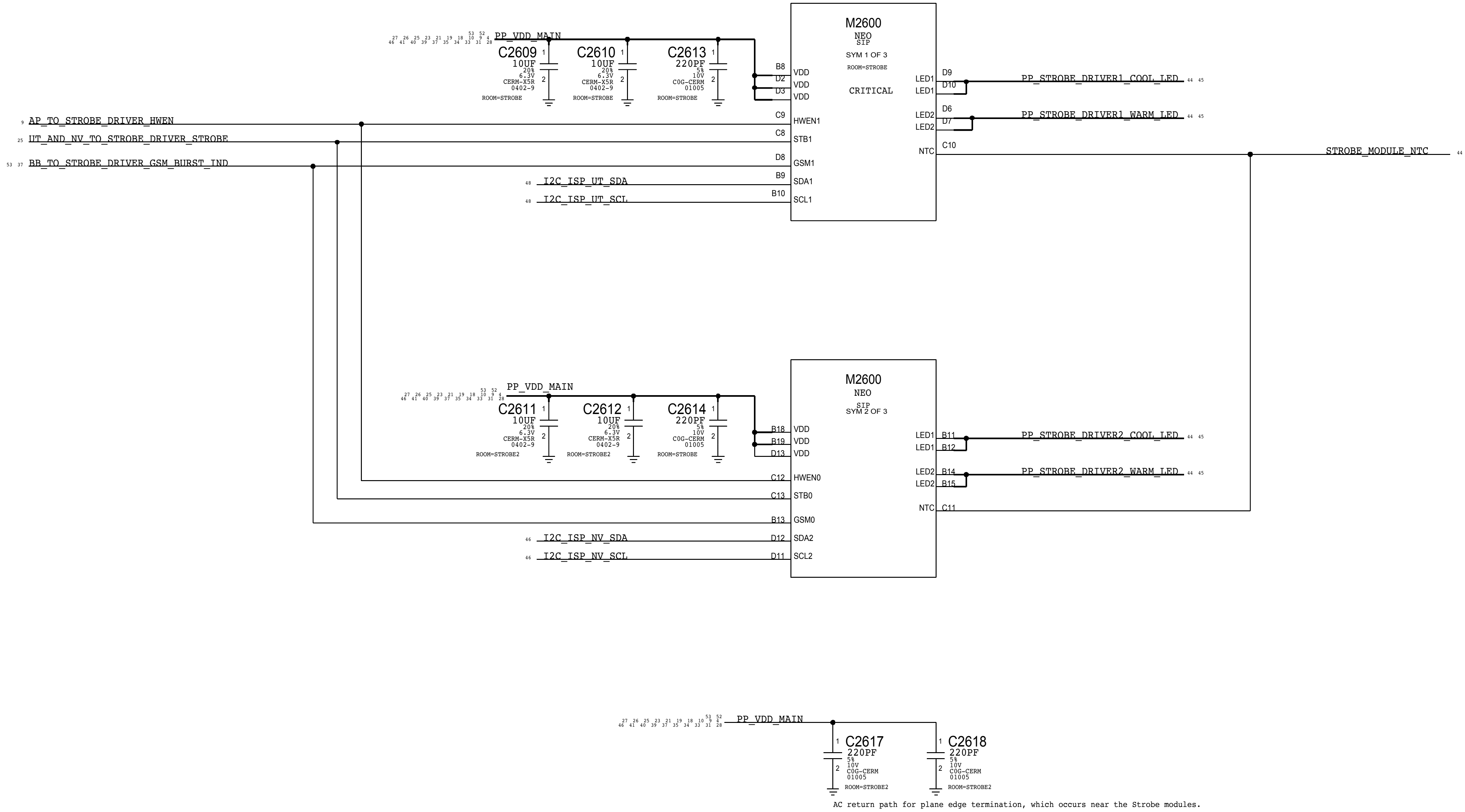


LPDP FILTERS



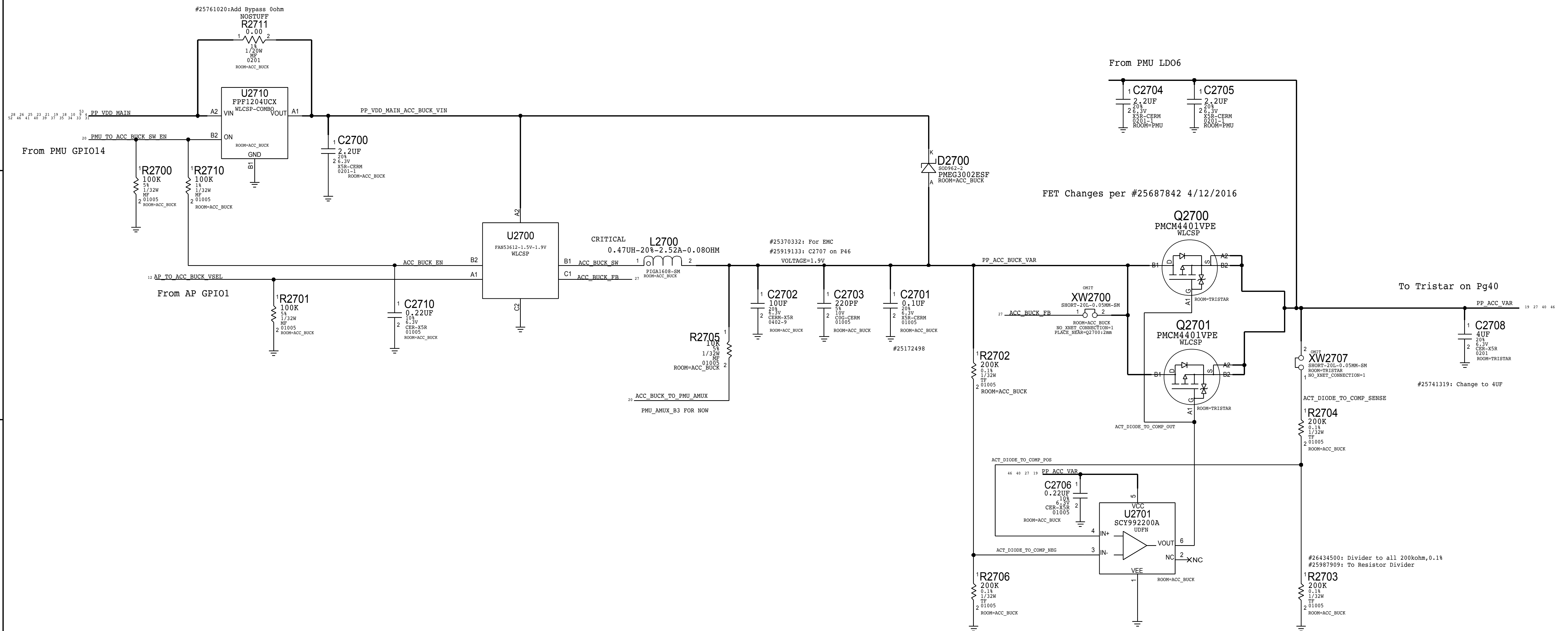
STROBE DRIVERS INSIDE NEO SIP MODULE

D10/sip_neo

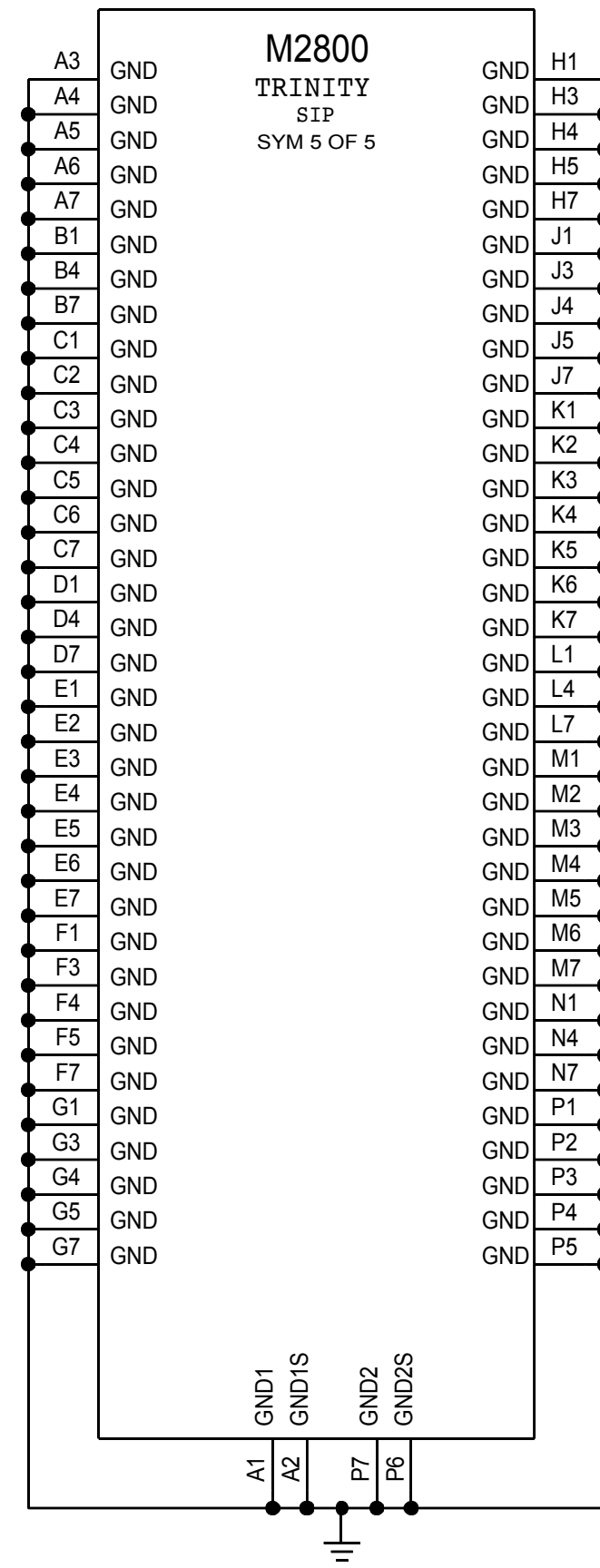
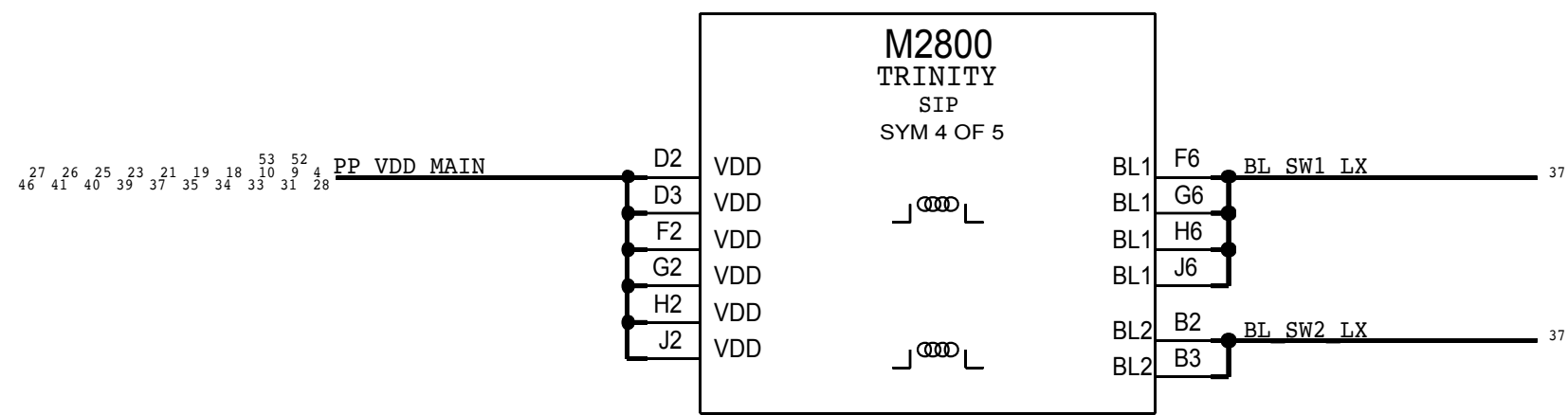
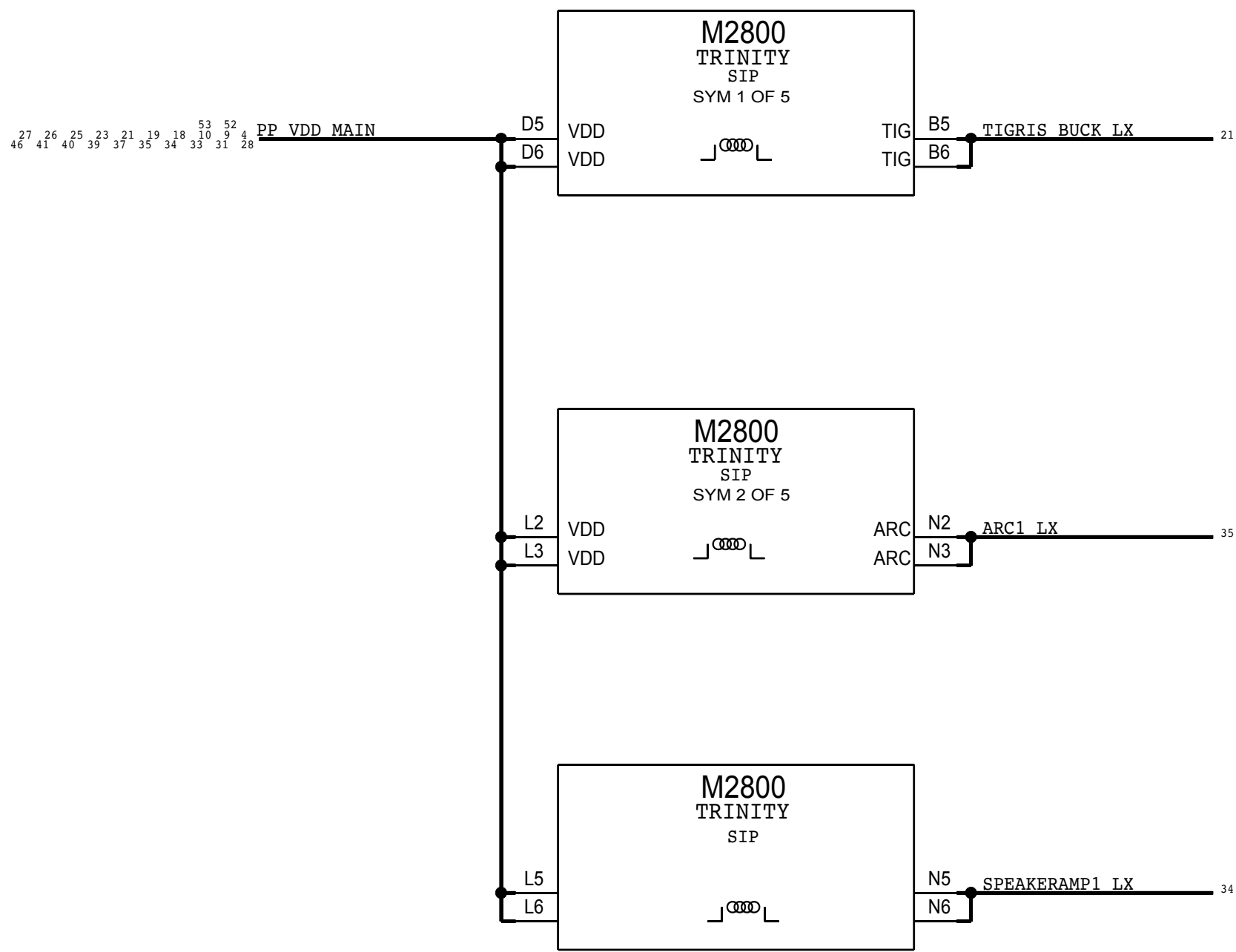


AC return path for plane edge termination, which occurs near the Strobe modules.

ACCESSORY BUCK



THIS PAGE UNIQUE TO SMALL FORM FACTOR



8

7

6

5

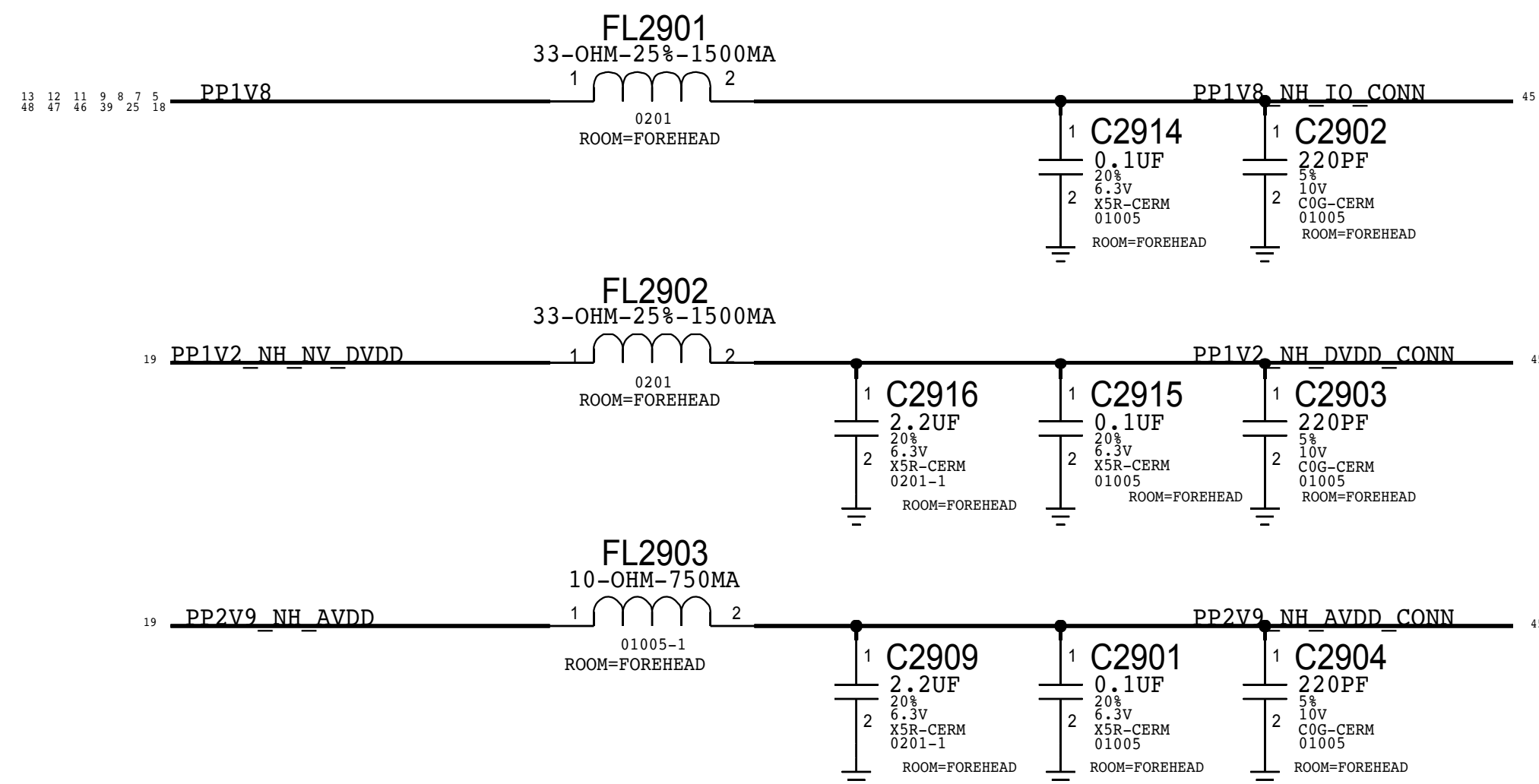
4

3

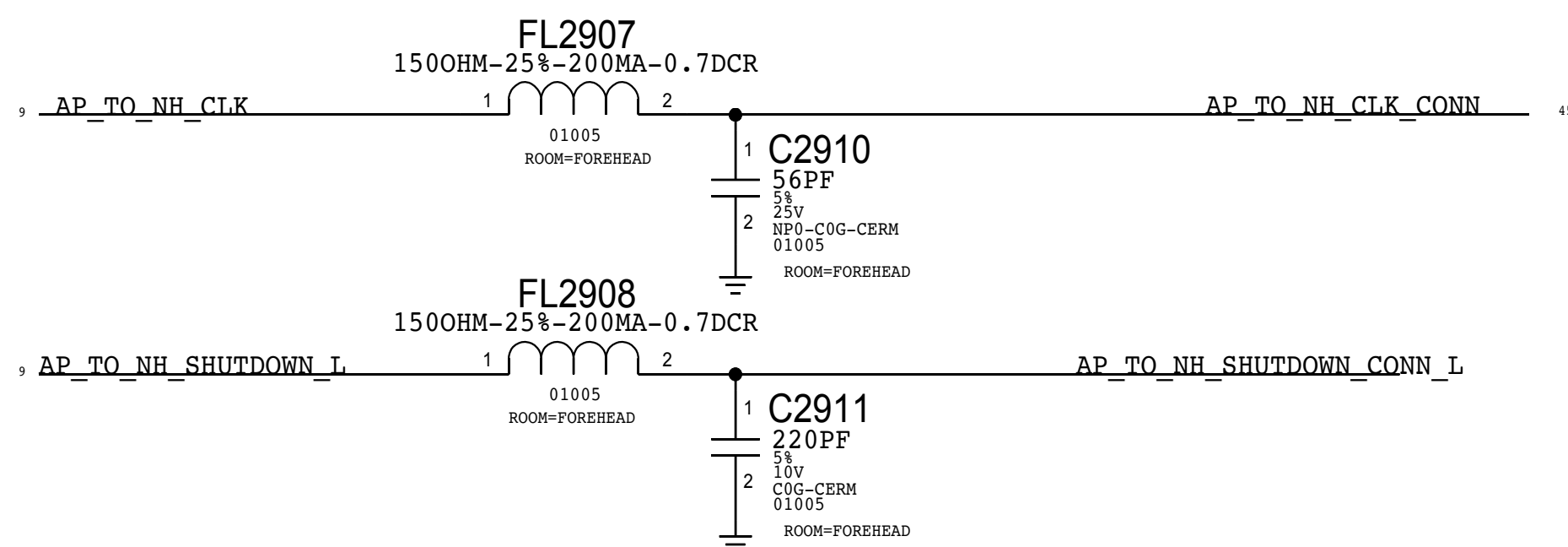
2

1

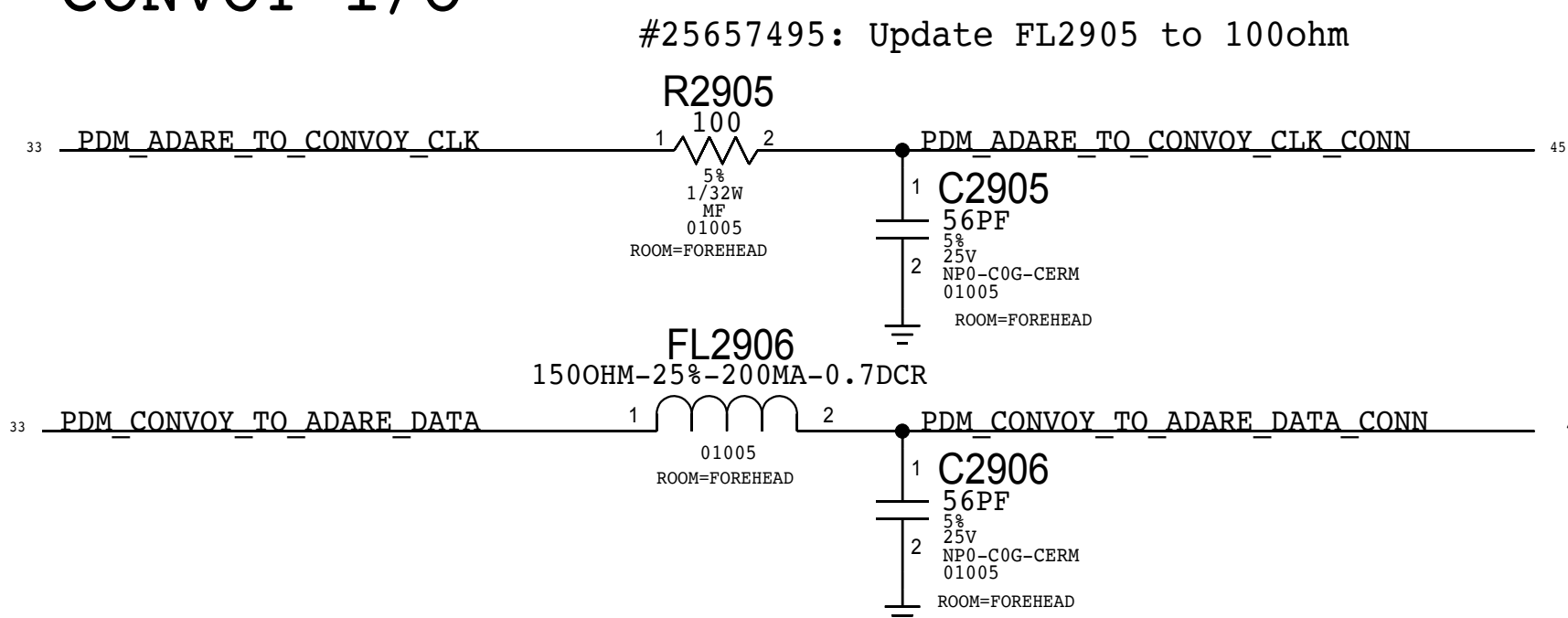
NEW HAMPSHIRE POWER



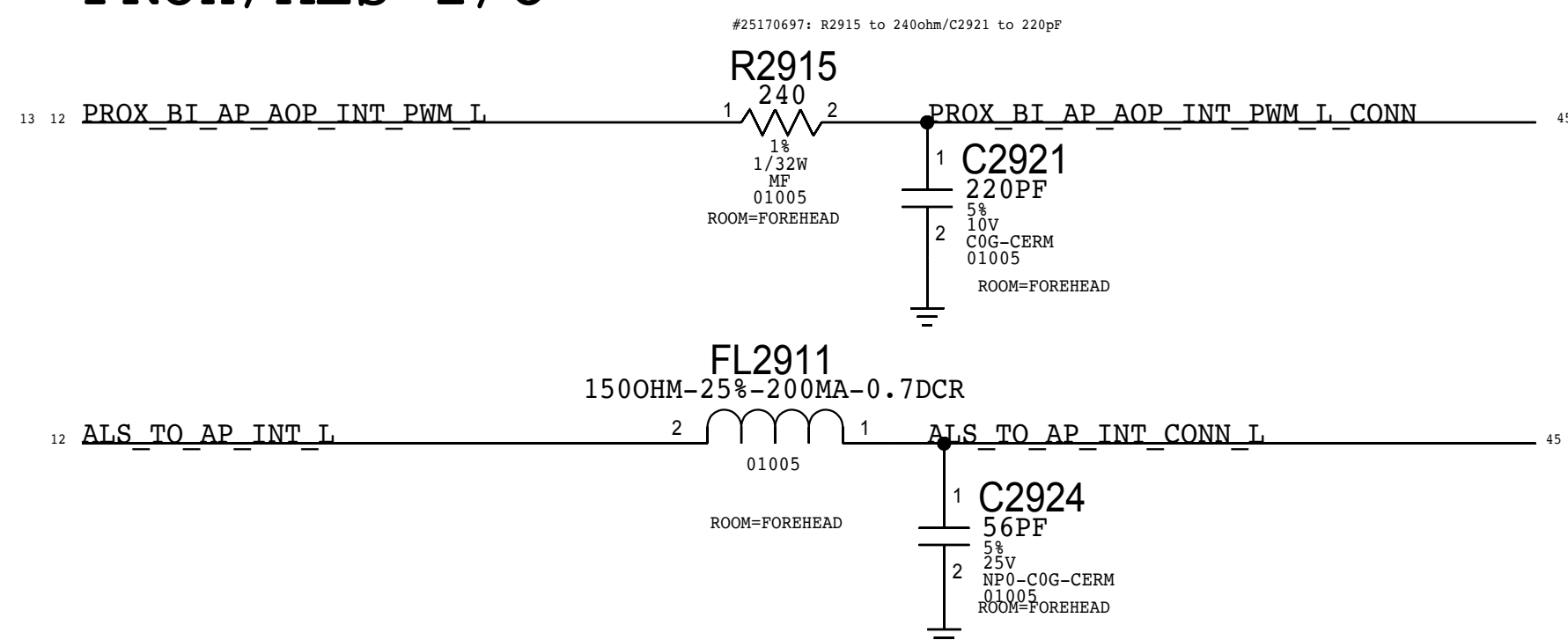
NEW HAMPSHIRE I/O



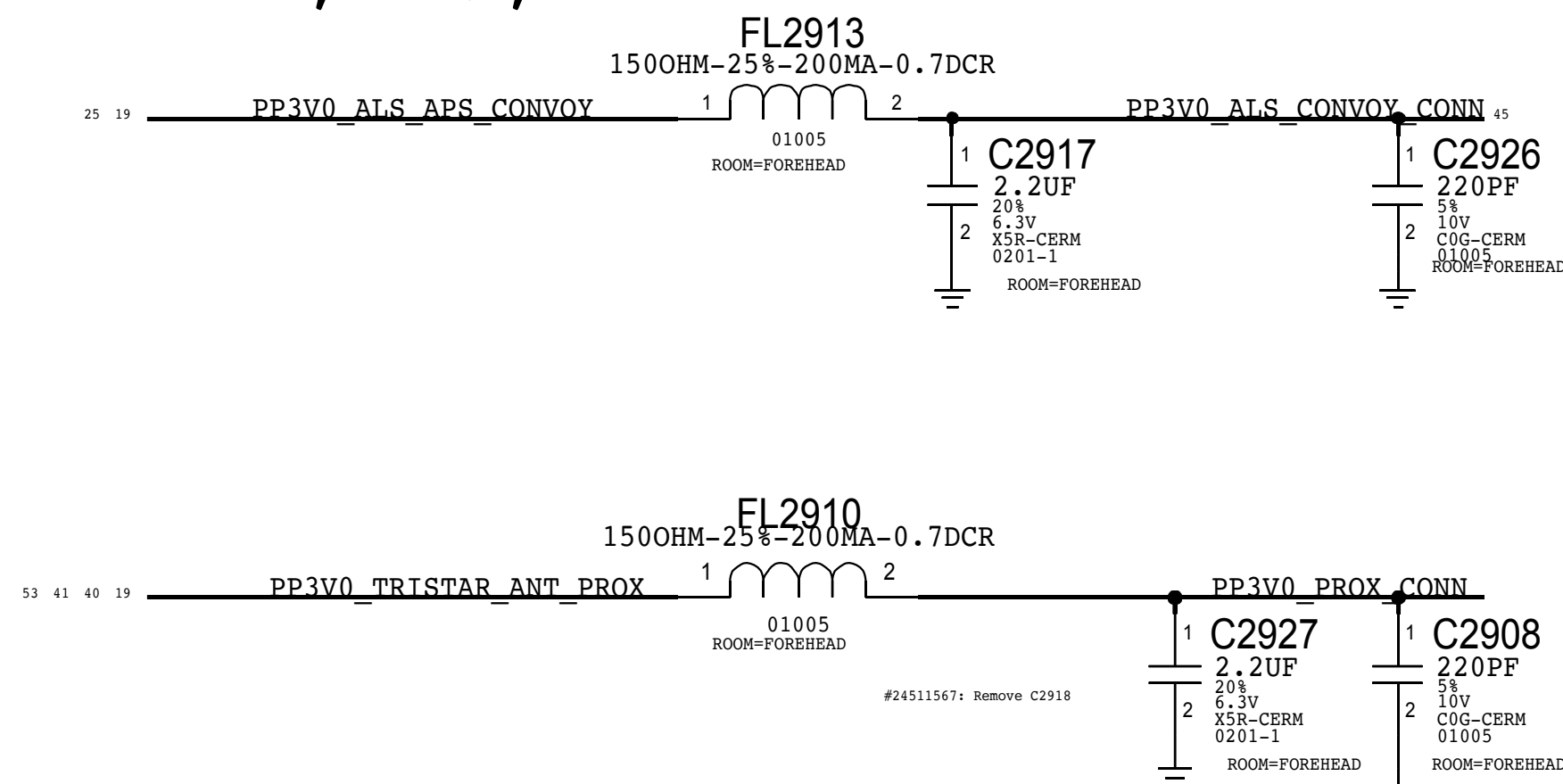
CONVOY I/O



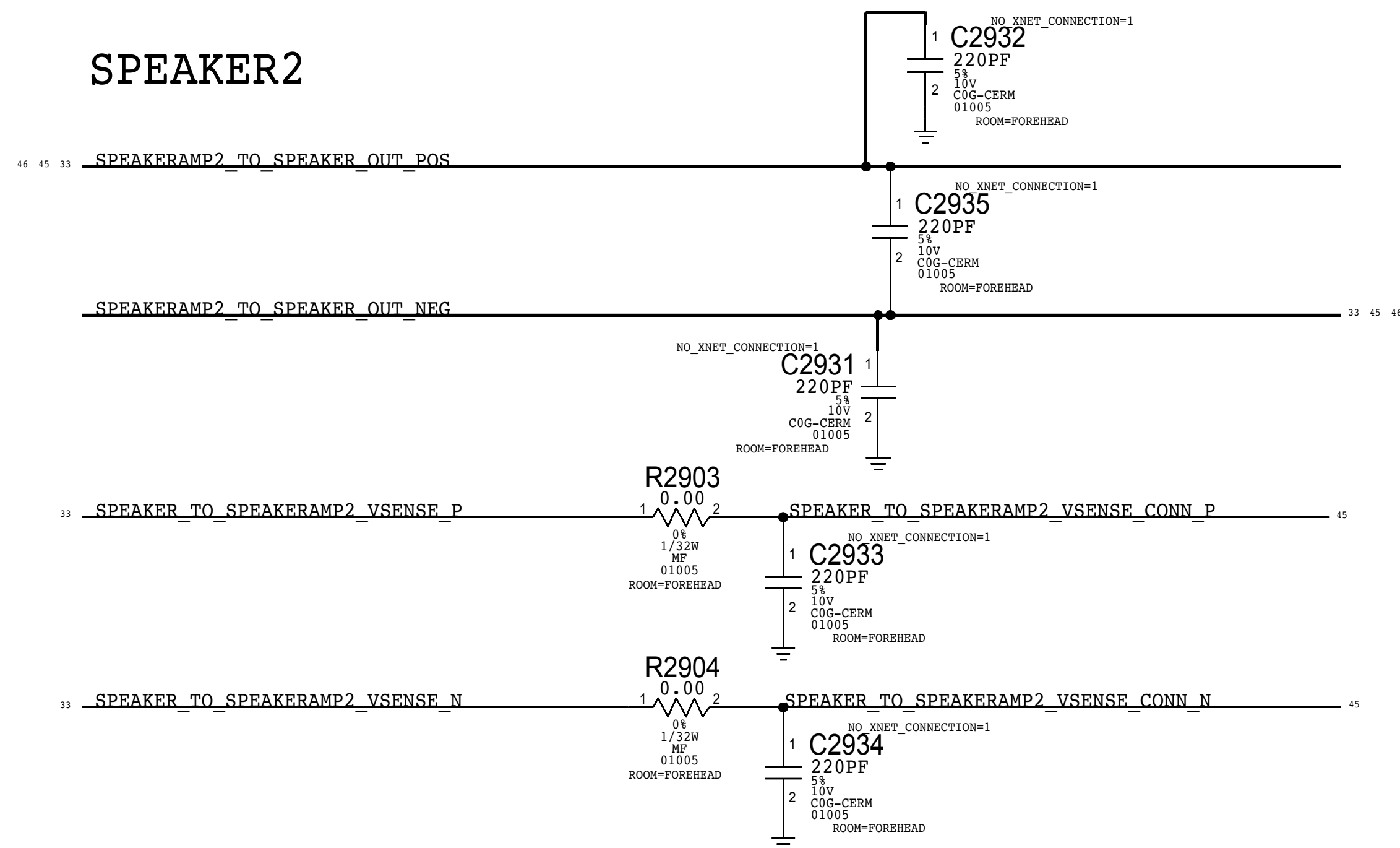
PROX/ALS I/O



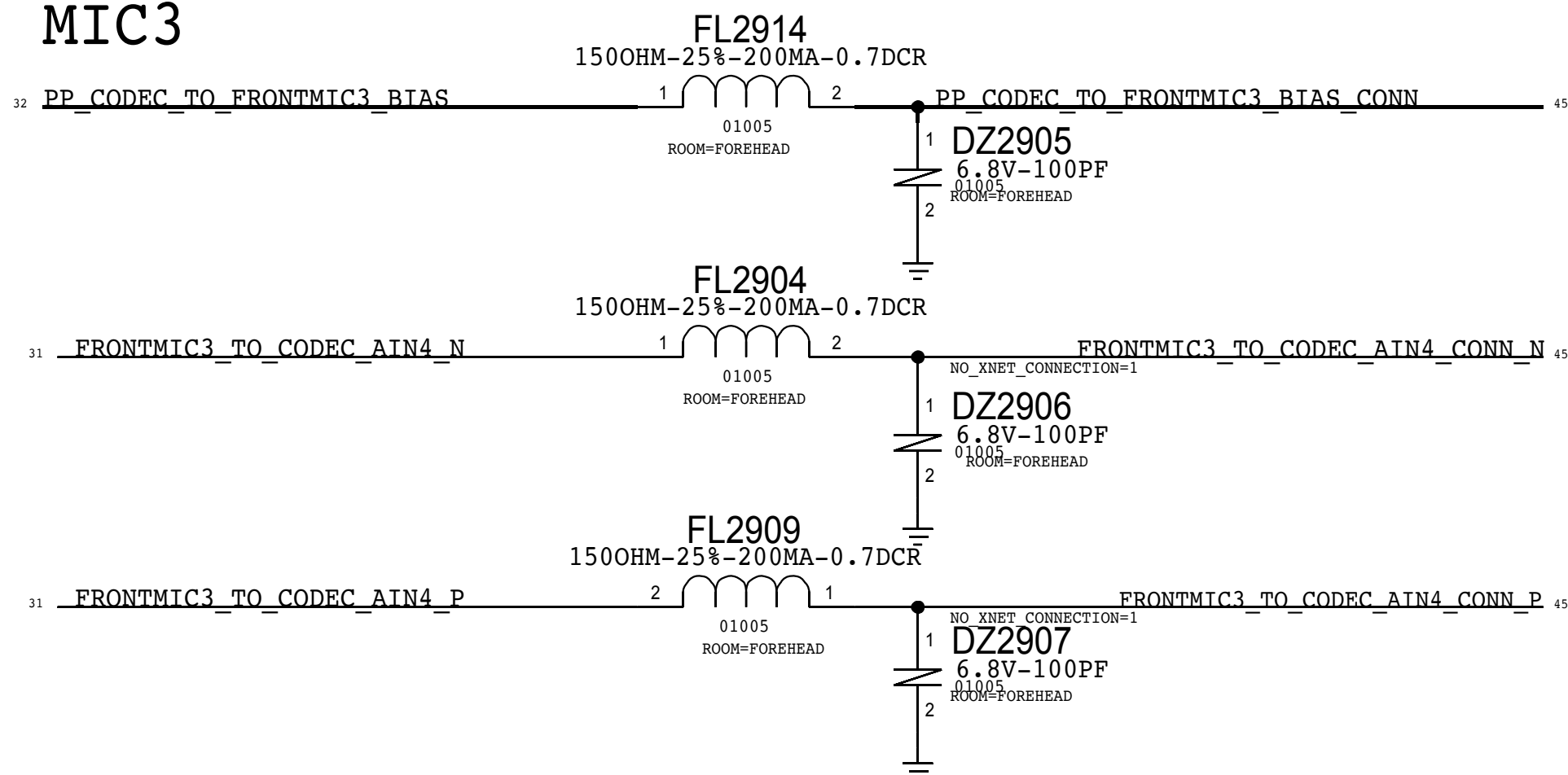
PROX, ALS, & CONVOY POWER



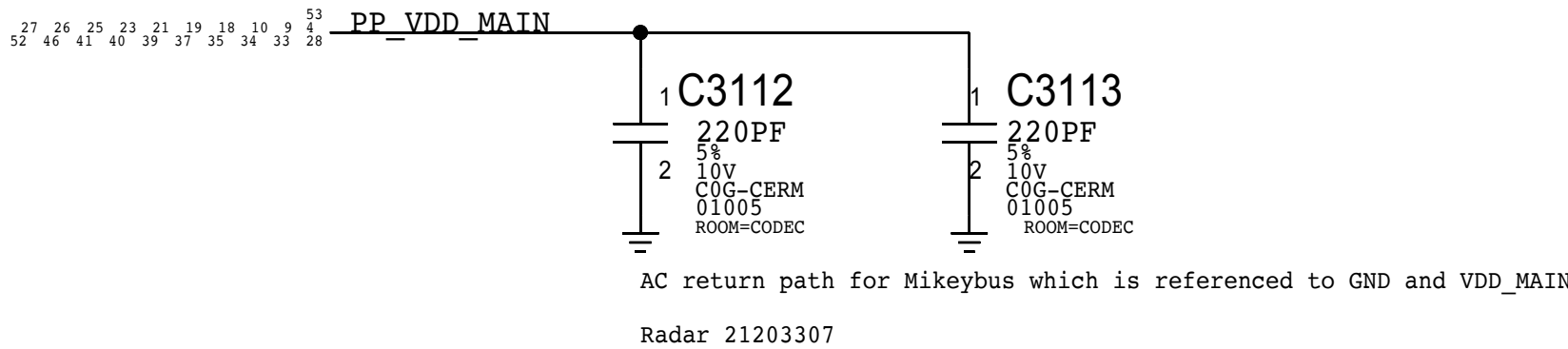
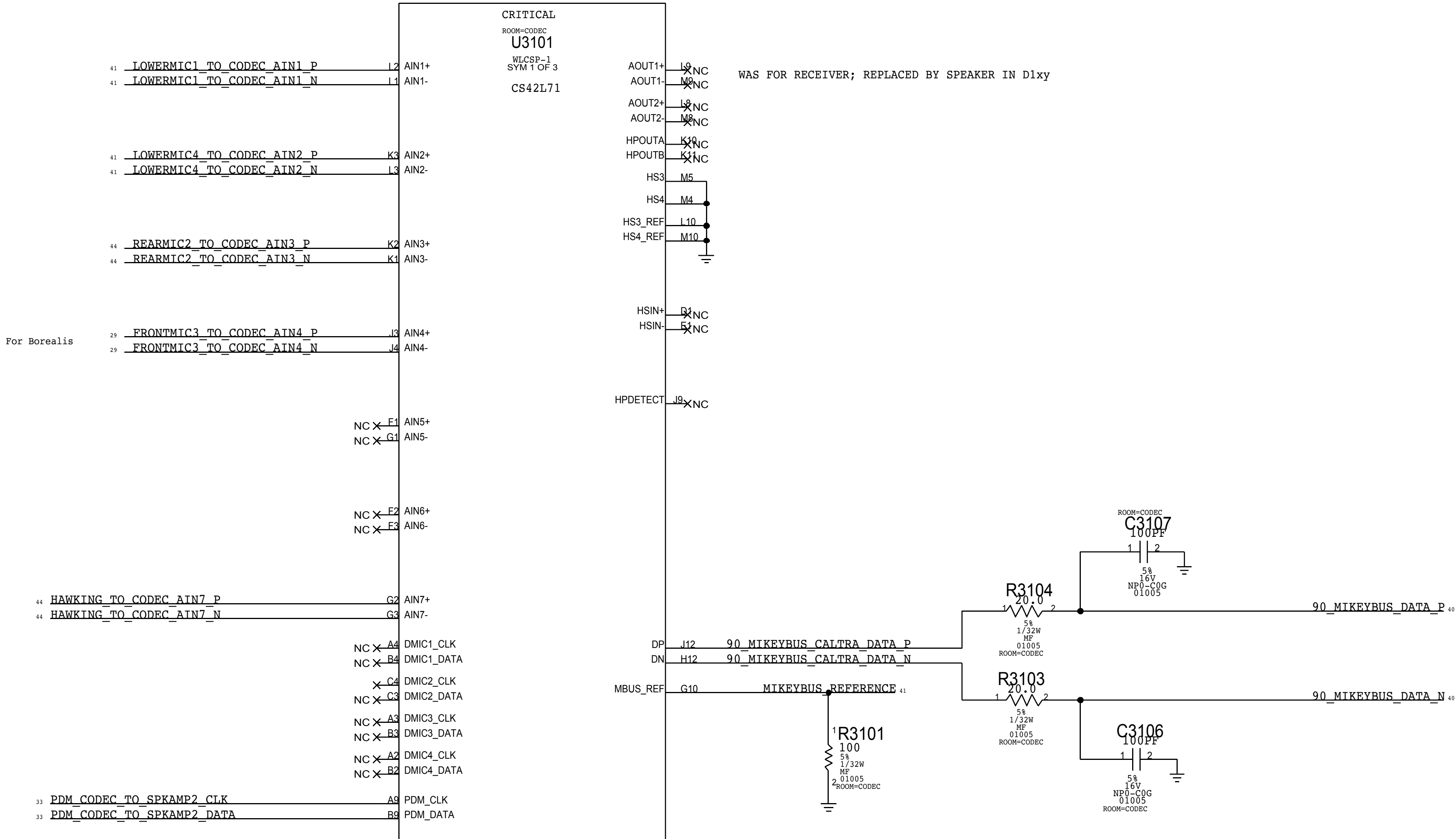
SPEAKER2



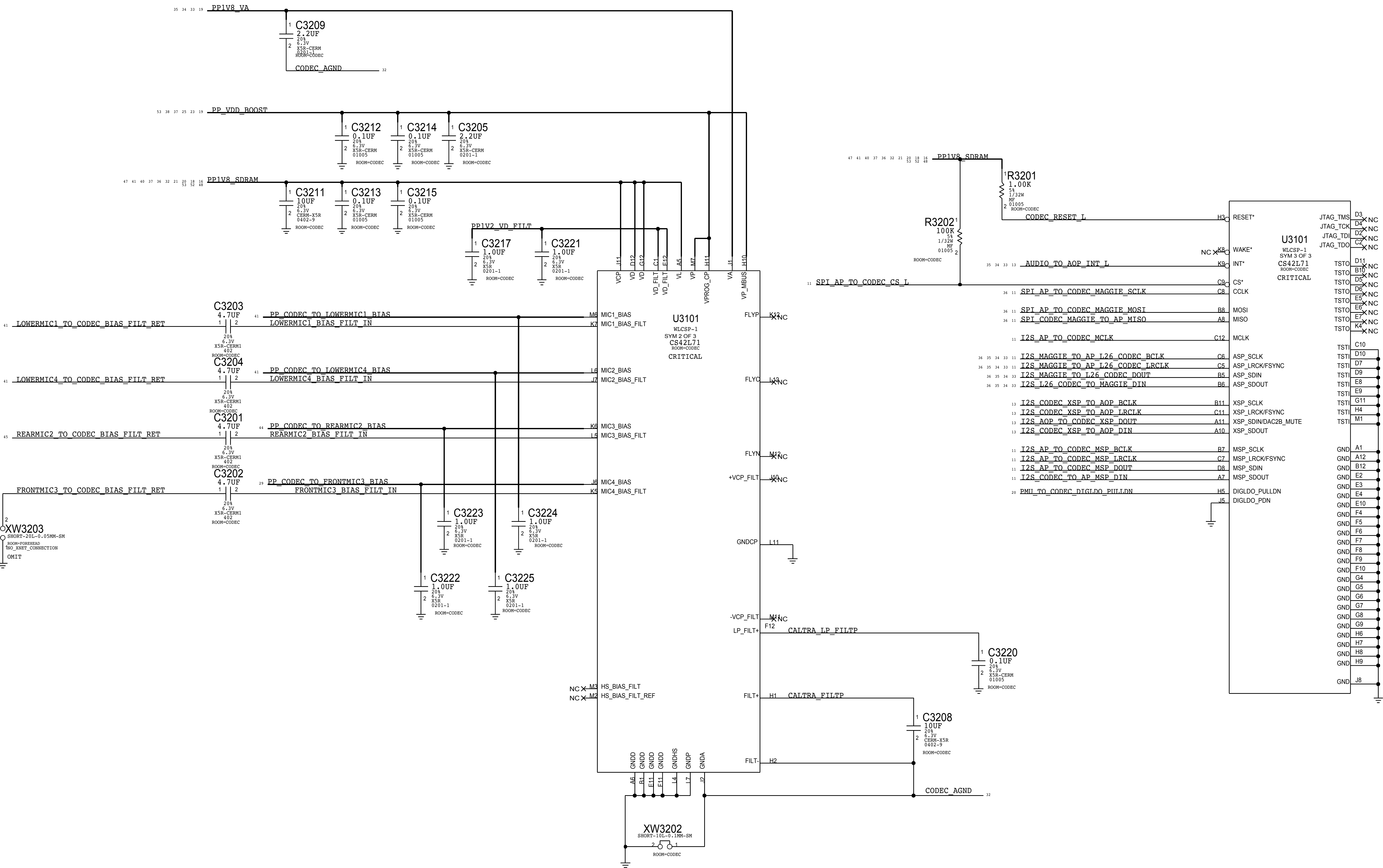
MIC3



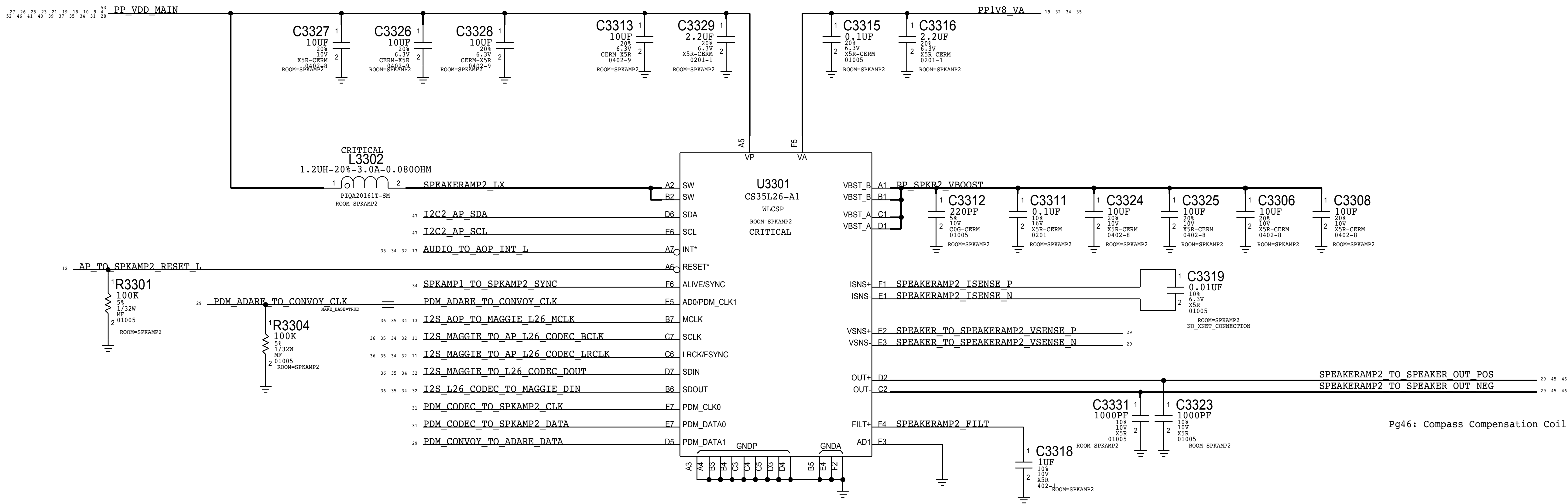
CALTRA AUDIO CODEC (ANALOG INPUTS & OUTPUTS)



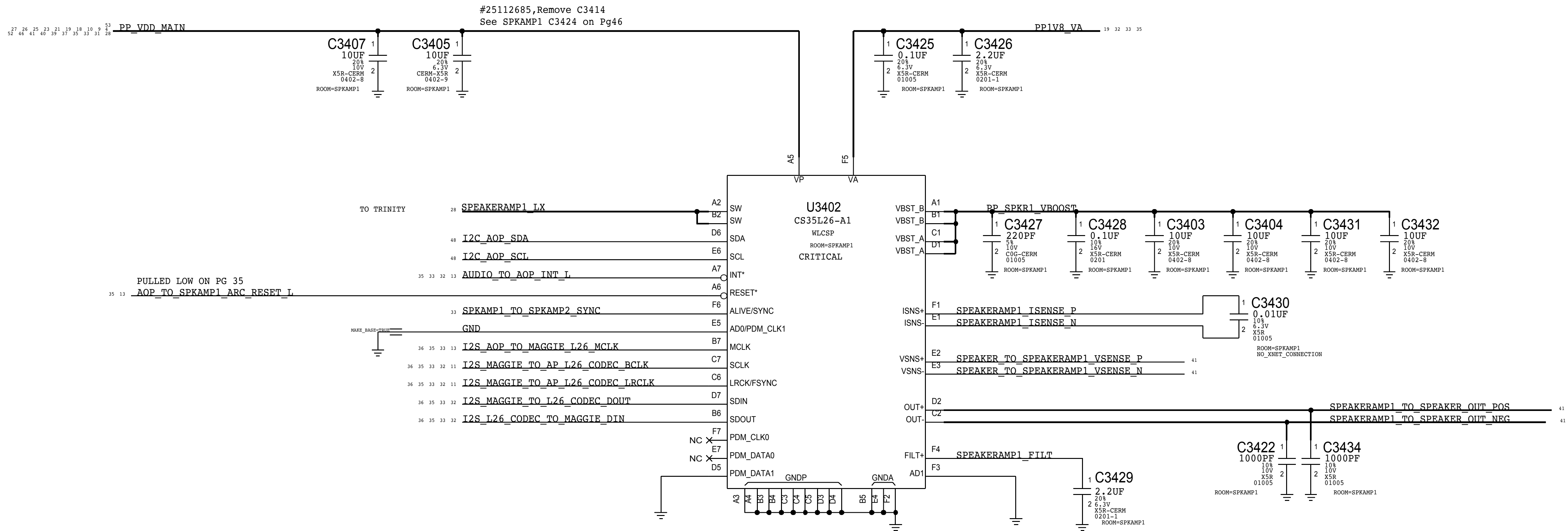
CALTRA AUDIO CODEC (POWER & I/O)



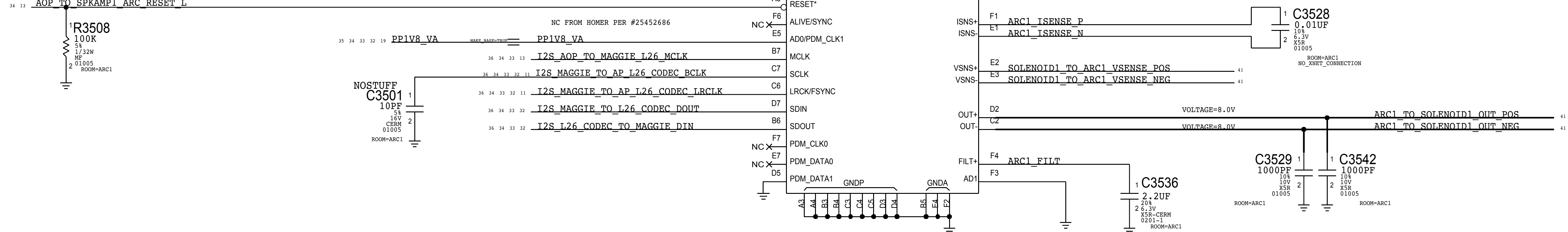
SPEAKER AMPLIFIER 2 (North)



SPEAKER AMPLIFIER 1
(South)



ARC D



D

C

B

A

D

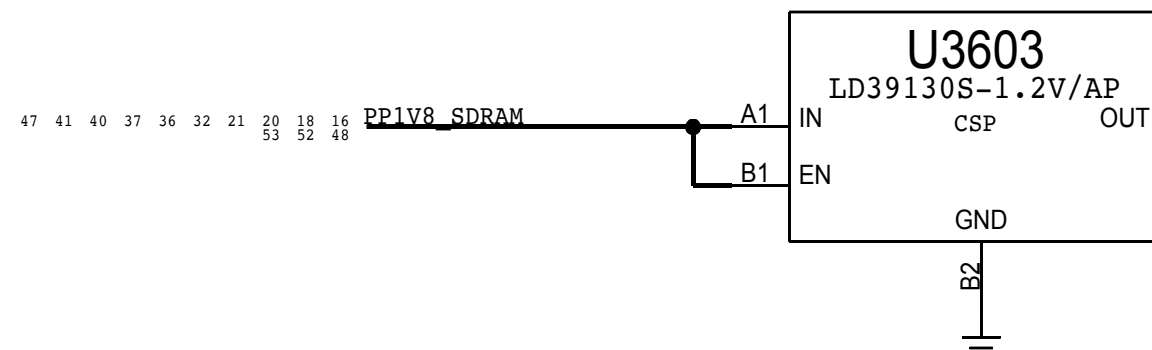
C

B

A

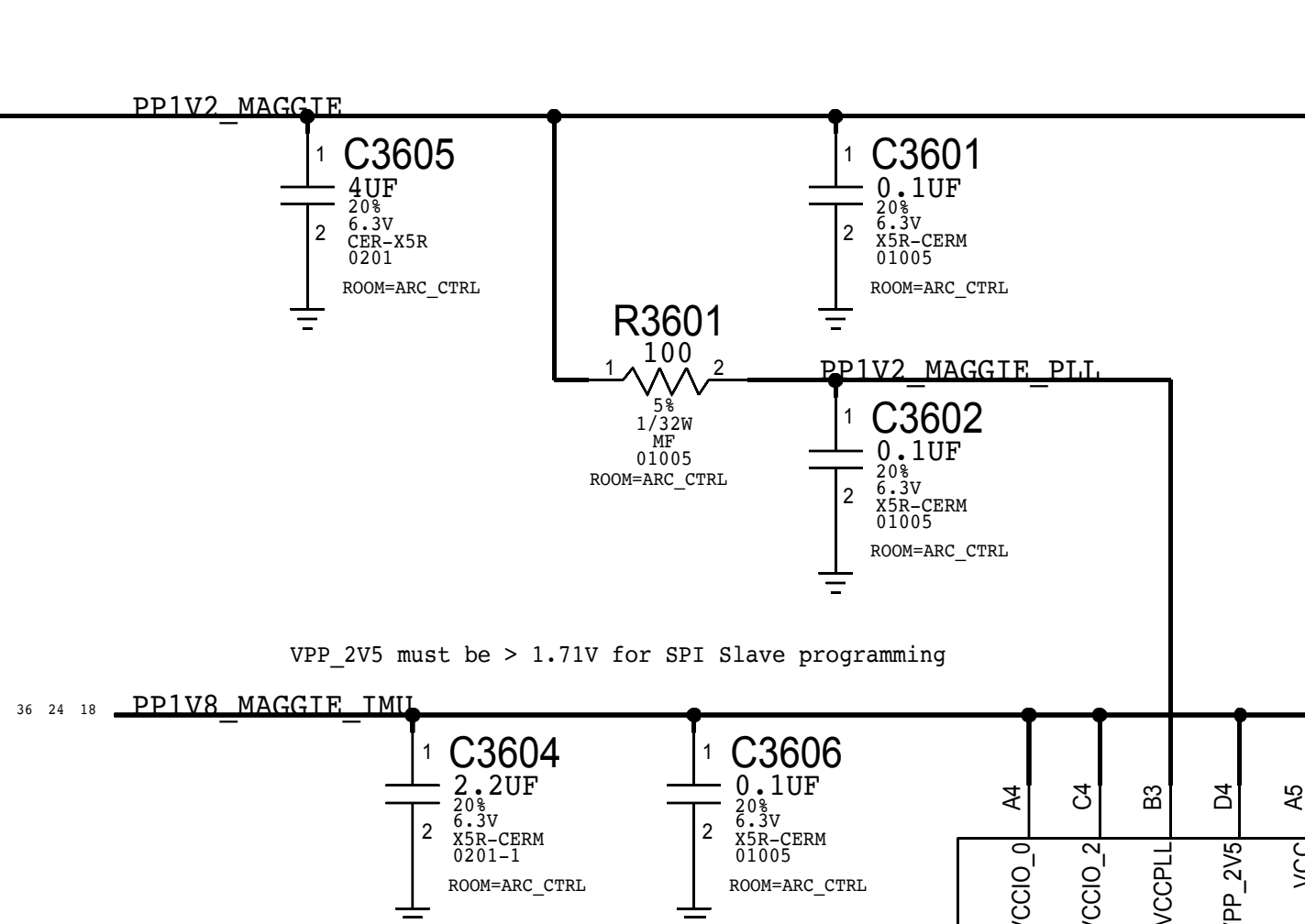
MAGGIE LDO

APN: 353S00842



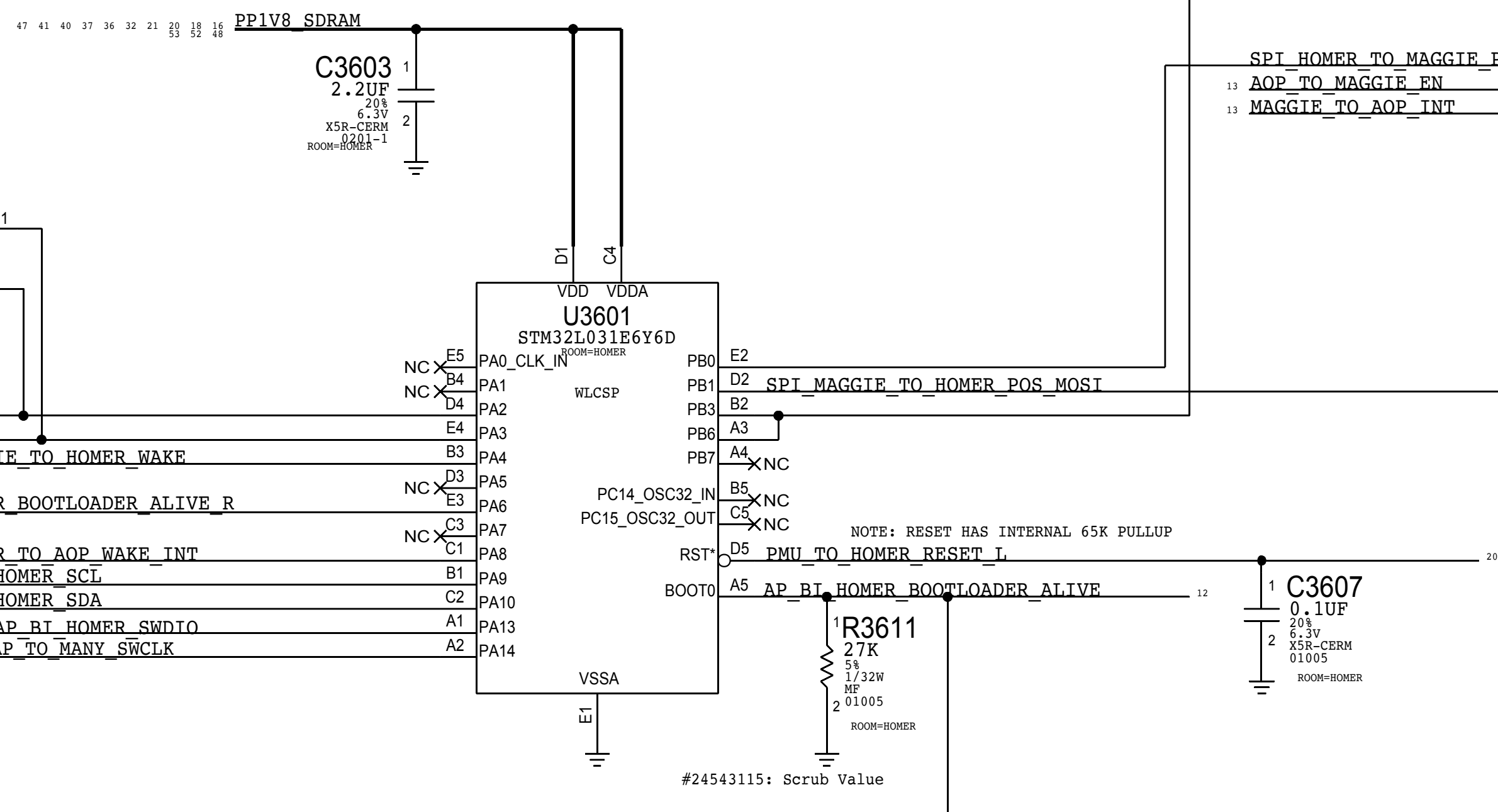
MAGGIE

APN: 336S00020



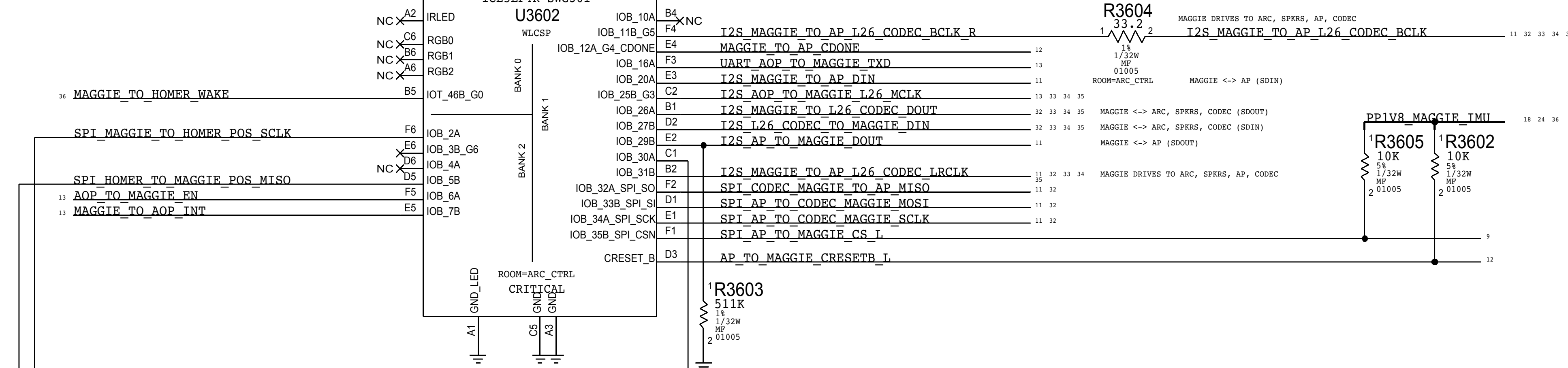
HOMER STM32L0 MICRO

STM32L03 APN: 337S00231



ICB5LP4K-SWG36I

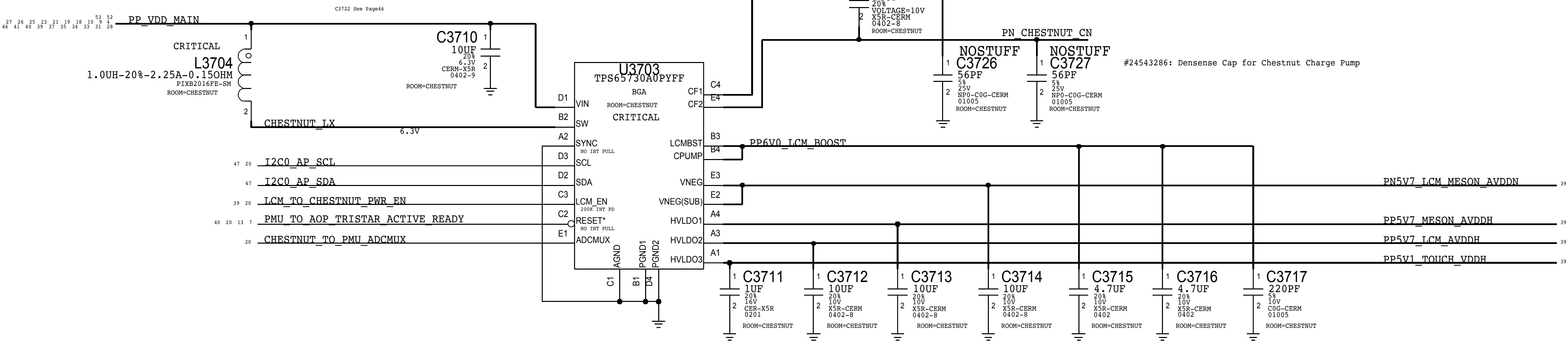
U3602



DISPLAY & TOUCH - POWER SUPPLIES

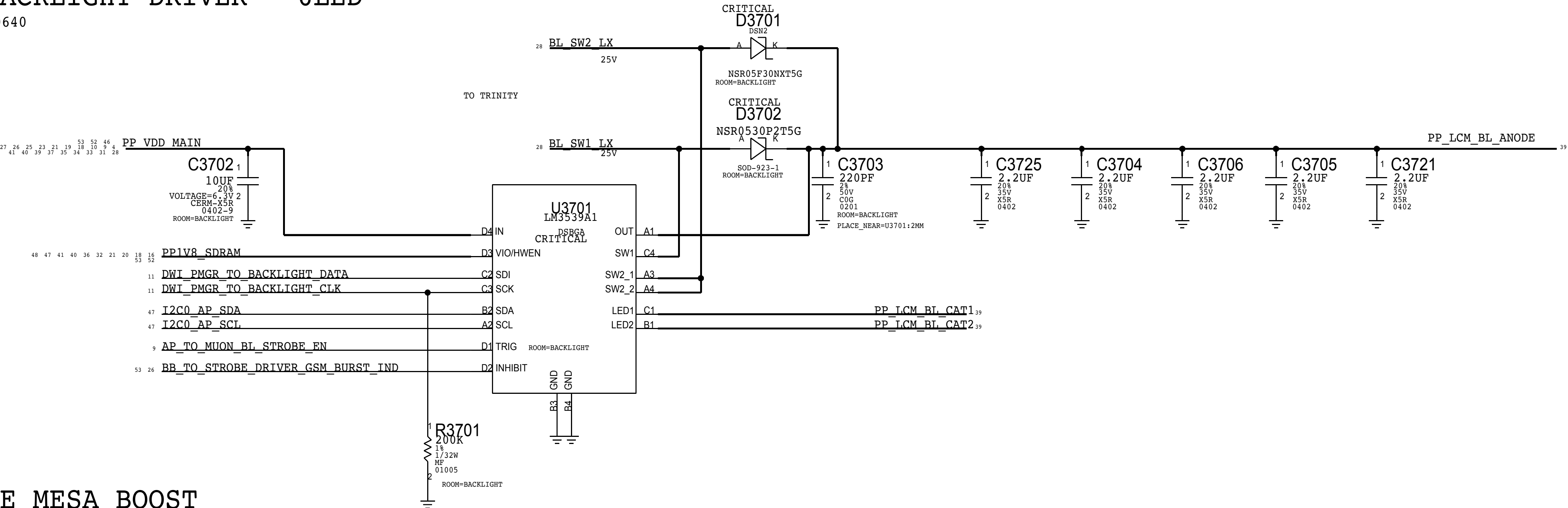
CHESTNUT DISPLAY PMU

APN:338S1172



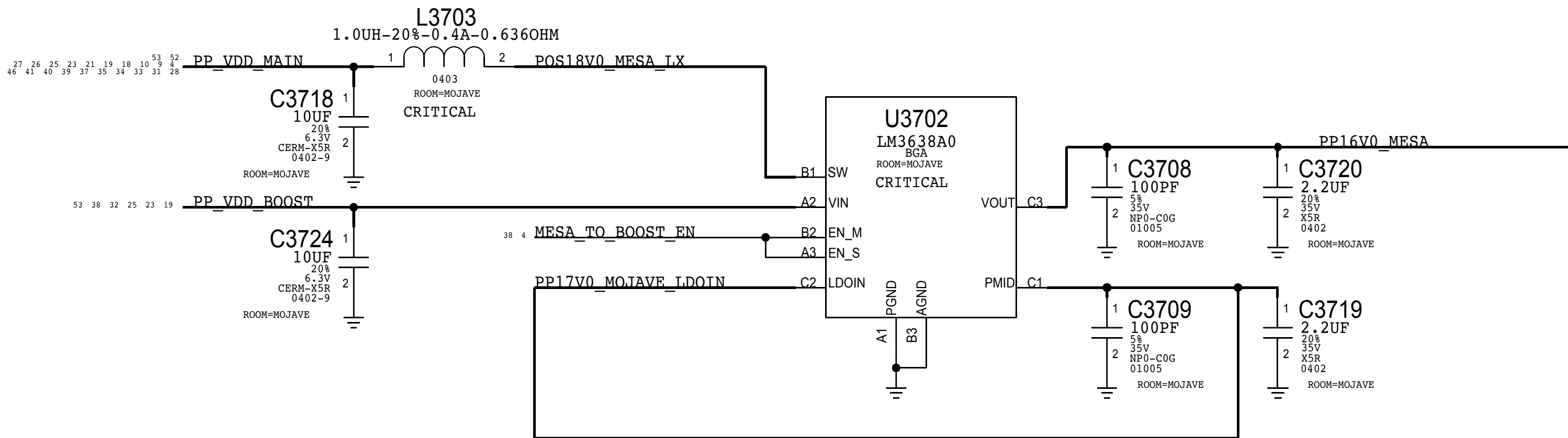
LED BACKLIGHT DRIVER - 6LED

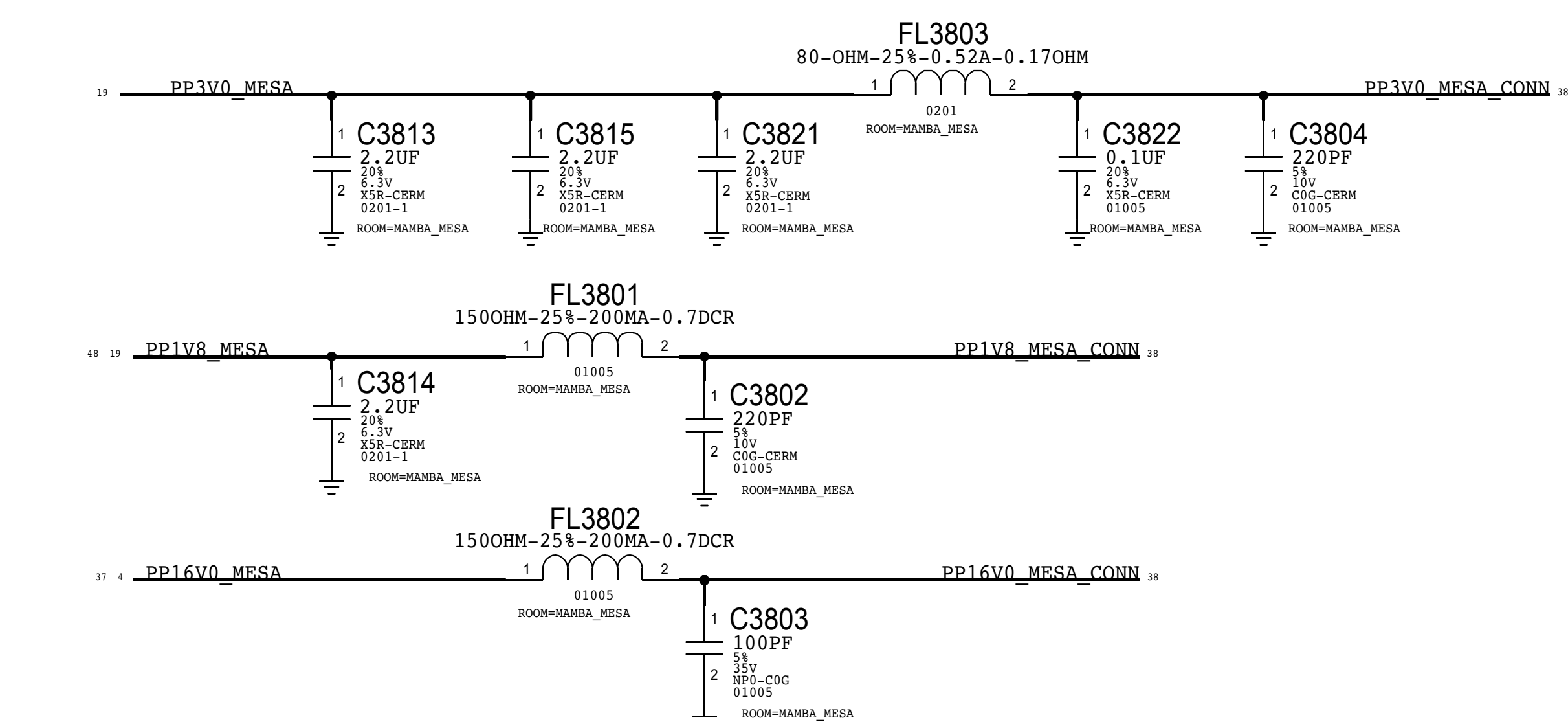
APN:353S00640



MOJAVE MESA BOOST

APN:353S00671

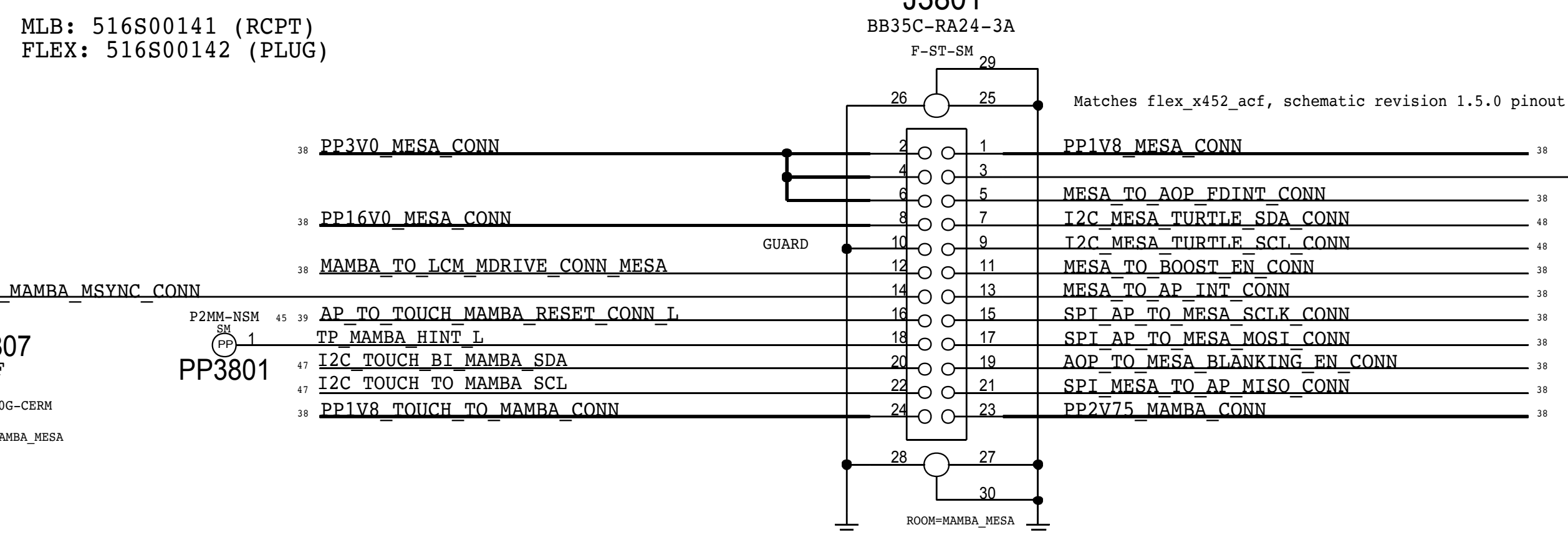




The image displays a PCB layout with five signal traces, each featuring a component footprint and its associated values. The components are arranged in a vertical column, with their values and footprints listed to the right of the trace labels.

- Trace 1:** `SPI_AP_TO_MESA_MOST` to `SPI_AP_TO_MESA_MOST_CONN`. Component: `FL3807`, `1500HM-25%-200MA-0.7DCR`. Footprint: `01005`.
- Trace 2:** `SPI_AP_TO_MESA_SCLK` to `SPI_AP_TO_MESA_SCLK_CONN`. Component: `R3807`, `511K`. Footprint: `1/32W MF 01005`.
- Trace 3:** `SPI_MESA_TO_AP_MISO` to `SPI_MESA_TO_AP_MISO_CONN`. Component: `R3808`, `511K`. Footprint: `1/32W MF 01005`.
- Trace 4:** `MESA_TO_AP_INT` to `MESA_TO_AP_INT_CONN`. Component: `R3809`, `511K`. Footprint: `1/32W MF 01005`.
- Trace 5:** `MESA_TO_BOOST_EN` to `MESA_TO_BOOST_EN_CONN`. Component: `R3810`, `511K`. Footprint: `1/32W MF 01005`.
- Trace 6:** `AOP_TO_MESA_BLANKING_EN` to `AOP_TO_MESA_BLANKING_EN_CONN`. Component: `FL3811`, `1500HM-25%-200MA-0.7DCR`. Footprint: `01005`.

Each component footprint is labeled with its value and footprint code. The footprint code is `01005` for the resistors and `01005` for the inductors.



C3828
 2.2UF
 20%
 50V
 X5R-CERM
 0201-1
 ROOM-MAMBA_MESA
 XW3801
 SHORT-20U-0.05M-SM
 ROOM-FMU
 ONTT
 PP1V8 TOUCH
 47 46 39 38 19

U3801
 LP5907SX-2.75
 X250N
 ROOM-MAMBA_MESA
 EN
 GND EPAD
 4 3 2 1

C3823
 10UF
 20%
 10V
 X5R-CERM
 0402-8
 ROOM-MAMBA_MESA
 1 2

C3812
 220PF
 5%
 50V
 C0G-CERM
 01005
 ROOM-MAMBA_MESA
 1 2

PP2V75_MAMBA_CONN
 38

PP1V8 TOUCH
 47 46 39 38 18

R3805
 0.00
 0%
 1/32W
 0F
 01005
 ROOM-MAMBA_MESA
 1 2

C3824
 2.2UF
 20%
 6.3V
 X5R-CERM
 0201-1
 ROOM-MAMBA_MESA
 1 2

C3811
 220PF
 5%
 50V
 C0G-CERM
 01005
 ROOM-MAMBA_MESA
 1 2

PP1V8_TOUCH_TO_MAMBA_CONN
 38

MAMBA DIGITAL I/O

FL3804
 1500HM-25%-200MA-0.7DCR
 01005
 ROOM-MAMBA_MESA
 1 2

C3805
 56PF
 5%
 25V
 NP0-C0G-CERM
 01005
 ROOM-MAMBA_MESA
 1 2

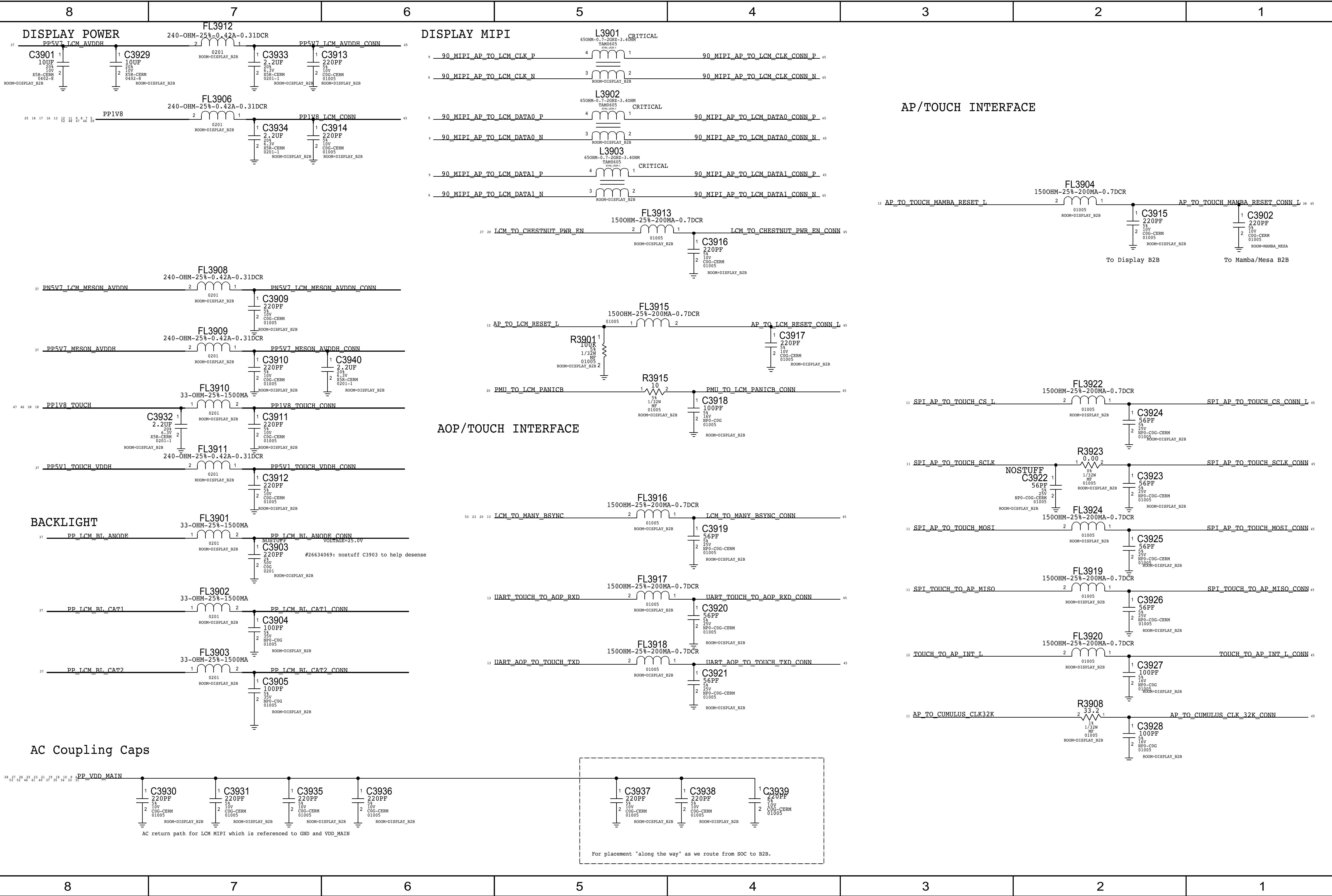
NOSTUFF
 C3806
 56PF
 5%
 25V
 NP0-C0G-CERM
 01005
 ROOM-MAMBA_MESA
 1 2

MAMBA_TO_LCM_MDRIVE
 45

MAMBA_TO_LCM_MDRIVE_CONN_MESA

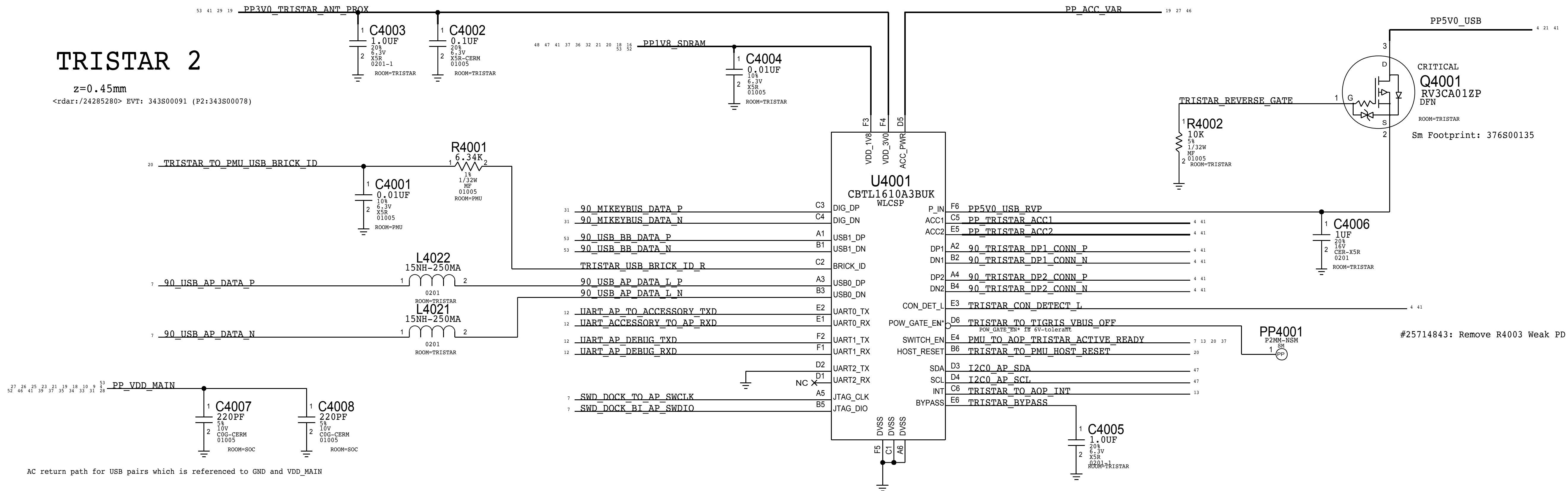
The schematic diagram illustrates the power and ground connections for the MESA board. It features several key components and their connections:

- Top Section:**
 - FL3804:** A 150OHM-25%-200MA-0.7DCR resistor connecting the **MAMBA_TO_ICM_MDRIVE** signal to the **MAMBA_TO_ICM_MDRIVE_CONN_MESA** signal.
 - C3805:** A 56PF, 5V, 25V NP0-C0G-CERM capacitor with footprint 01005, connected between the MAMBA_TO_ICM_MDRIVE line and ground (ROOM=MAMBA_MESA).
 - C3806:** A 56PF, 5V, 25V NP0-C0G-CERM capacitor with footprint 01005, connected between the MAMBA_TO_ICM_MDRIVE_CONN_MESA line and ground (ROOM=MAMBA_MESA).
- Bottom Section:**
 - FL3806:** A 150OHM-25%-200MA-0.7DCR resistor connecting the **MESA_TO_AOP_FDINT** signal to the **MESA_TO_AOP_FDINT_CONN** signal.
 - C3826:** A 100PF, 16V, 16V NP0-C0G capacitor with footprint 01005, connected between the MESA_TO_AOP_FDINT_CONN line and ground (ROOM=MAMBA_MESA). A note specifies: #24543342: stuff 100pF.



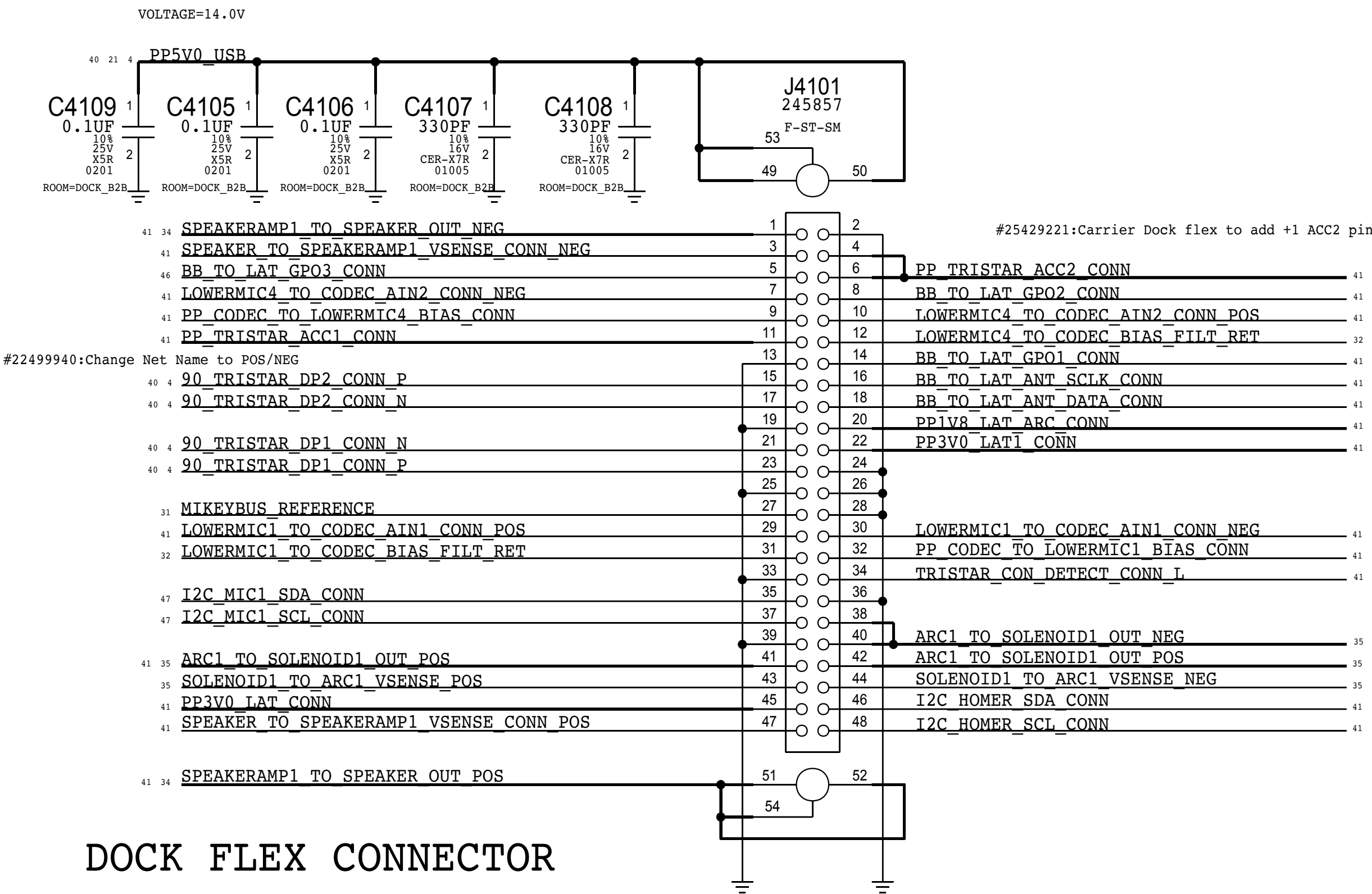
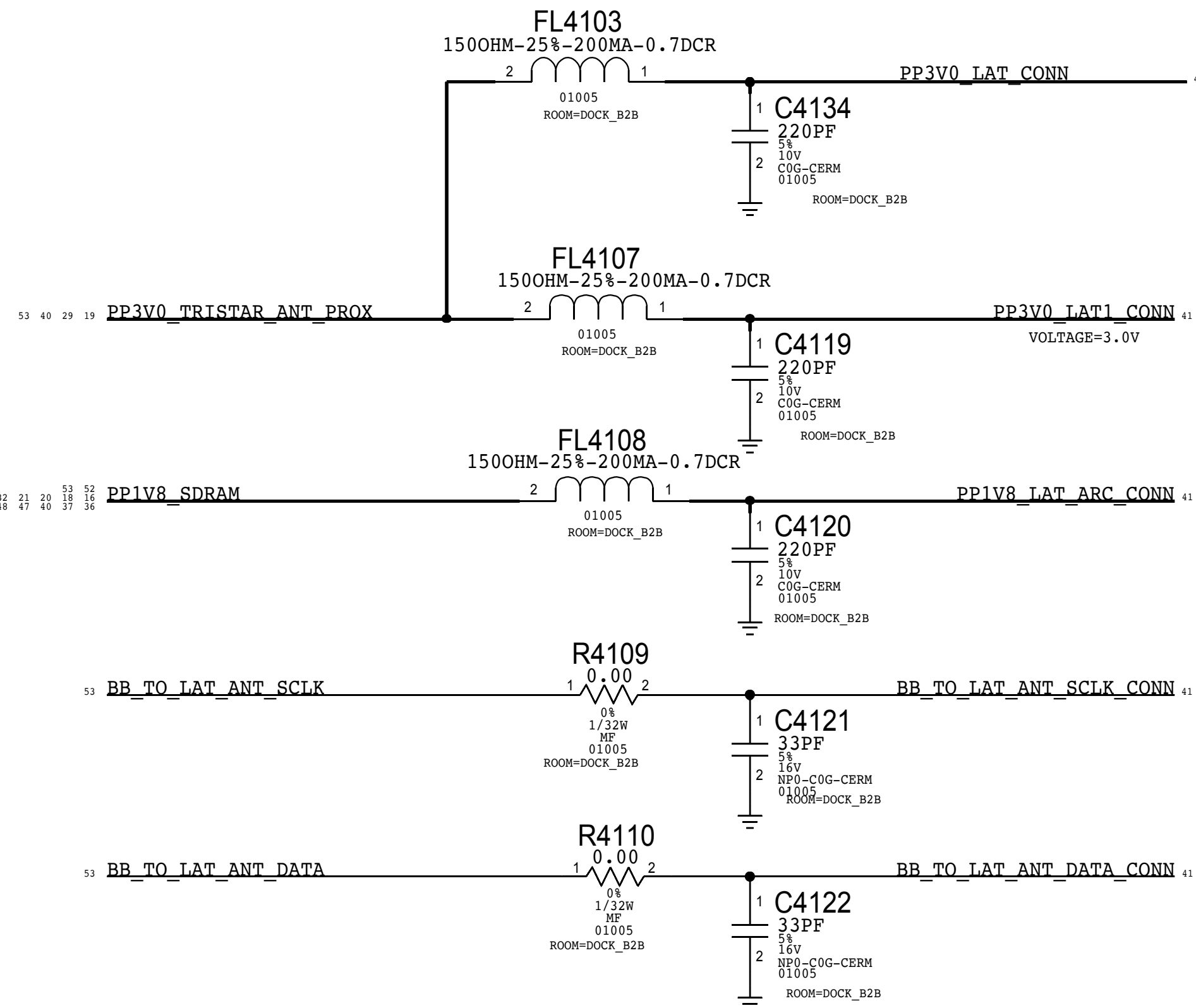
TRISTAR 2

z=0.45mm
<rdar://24285280> EVT: 343S00091 (P2:343S00078)



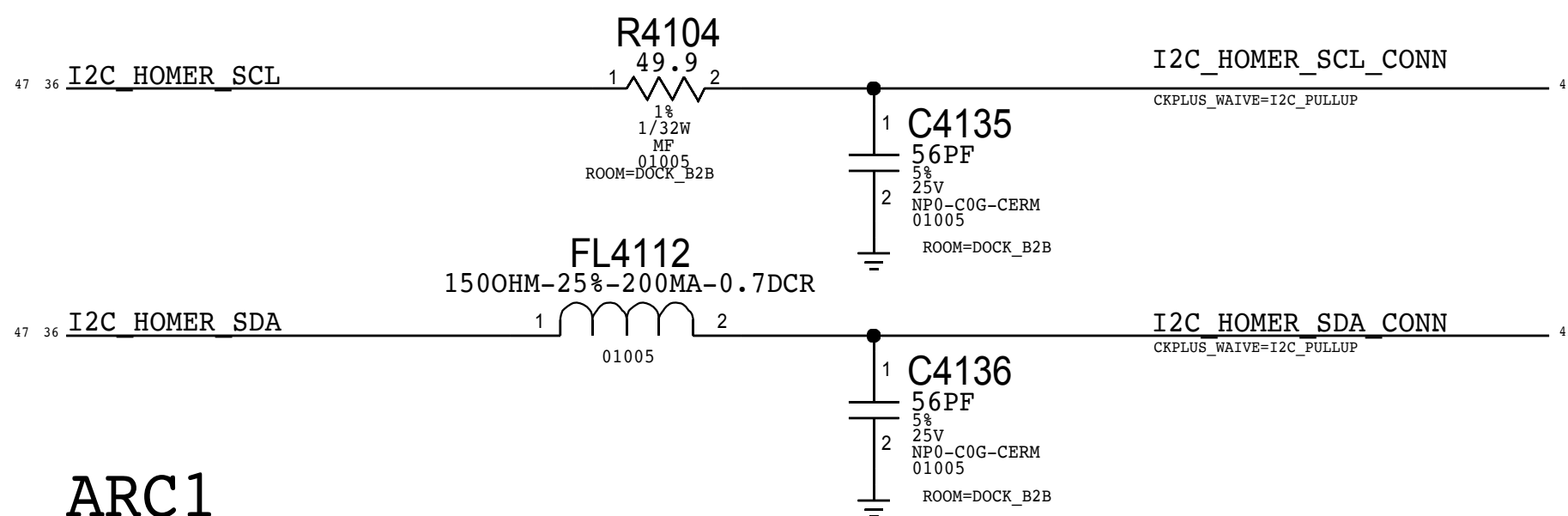
ANTENNA

Please loop in Matt Mow (Antenna Team)
when changing these components!

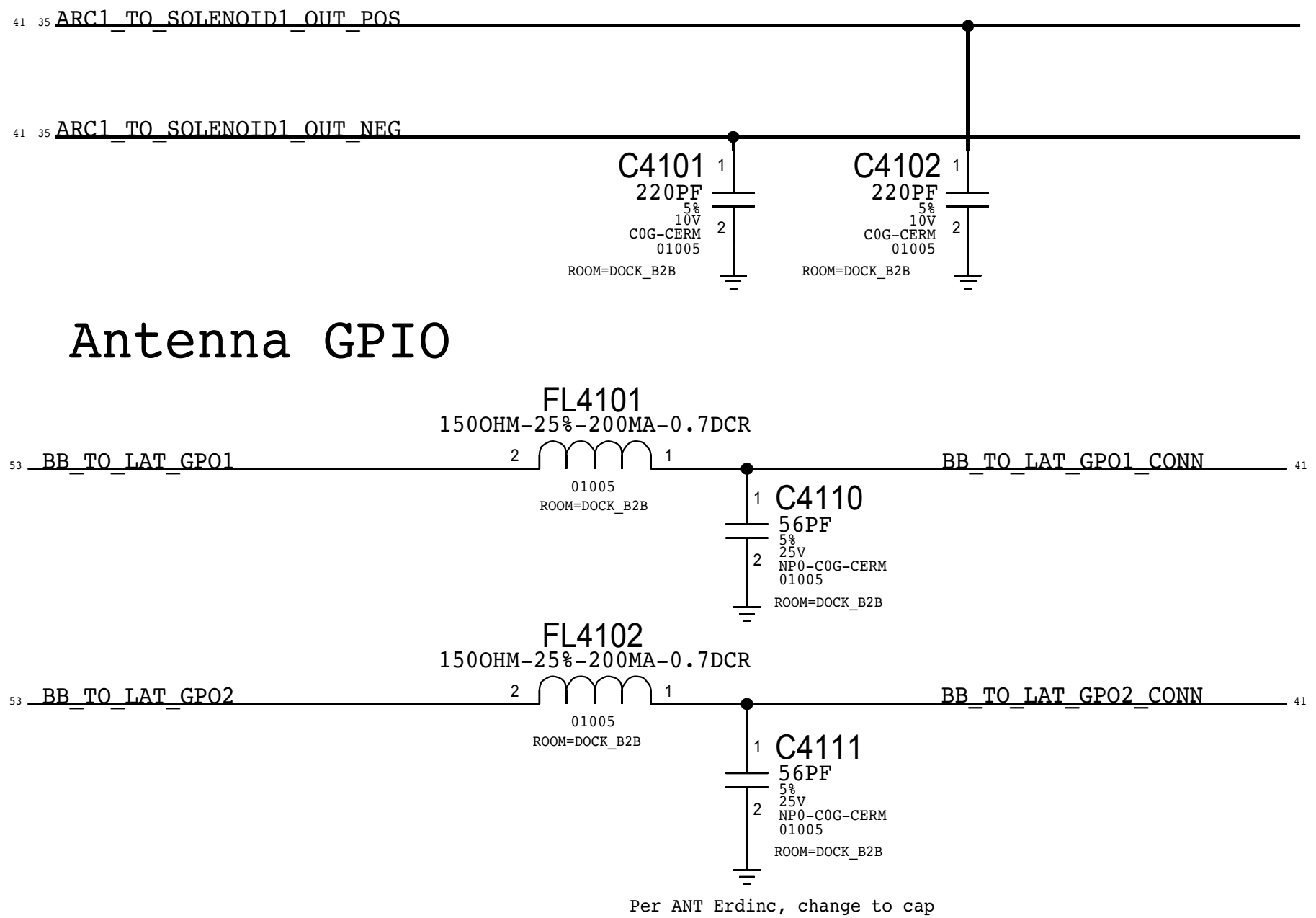


ARC CONTROL

#26118161: Update FL4104 to 49.9ohm

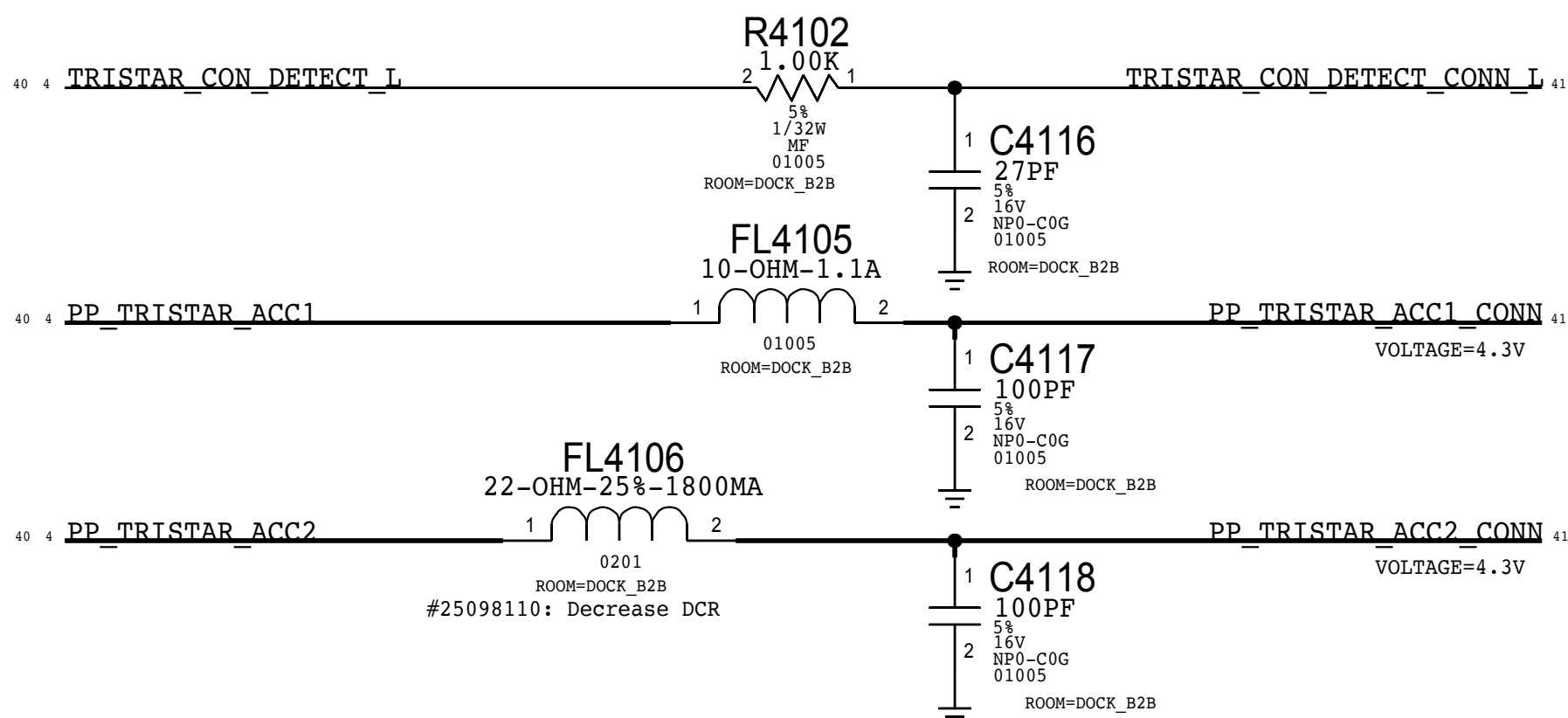


ARC1

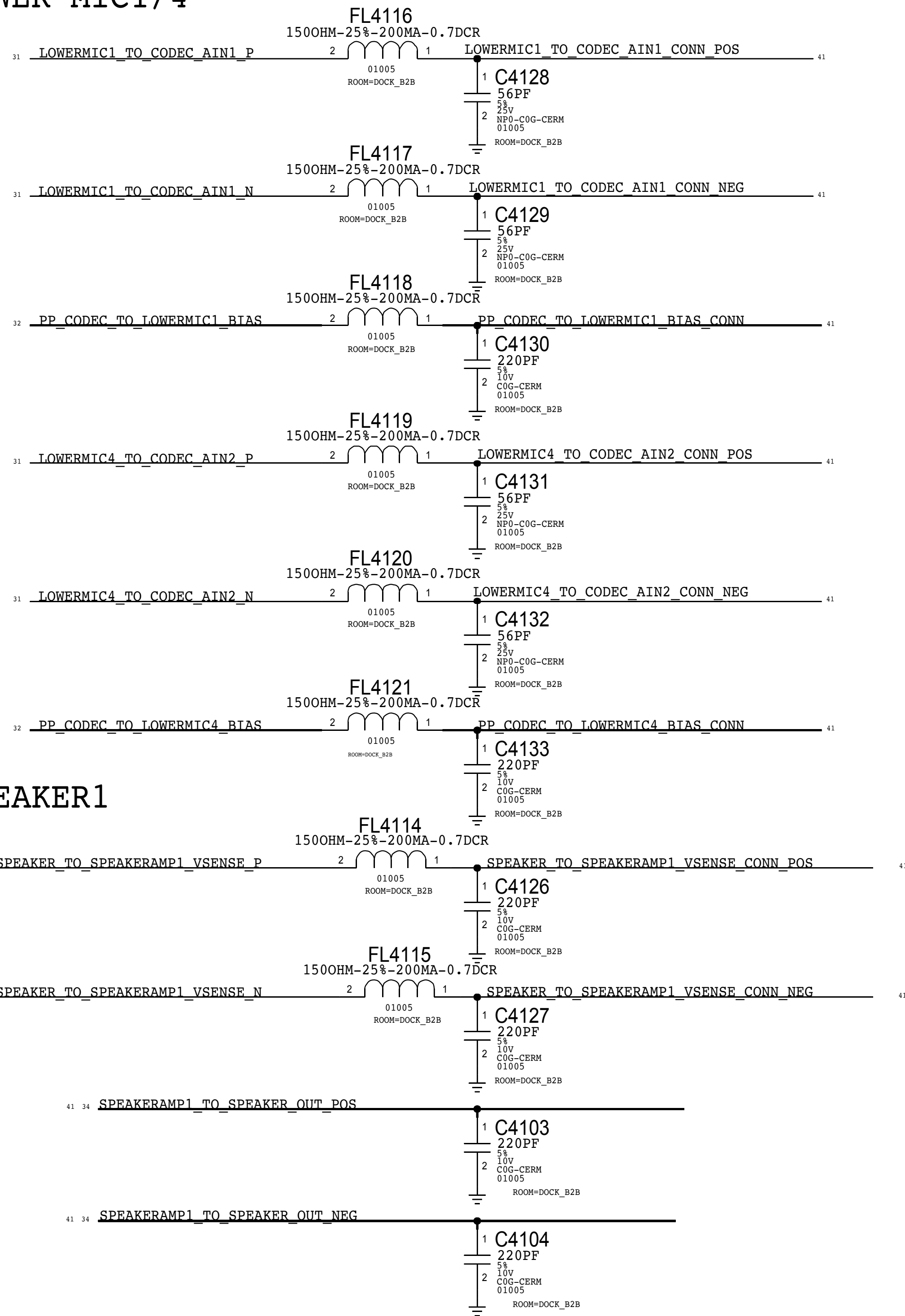


Antenna GPIO

TRISTAR



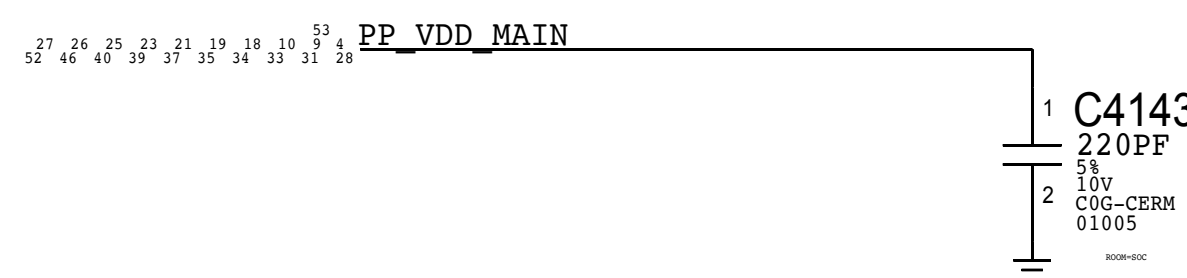
LOWER MIC1/4



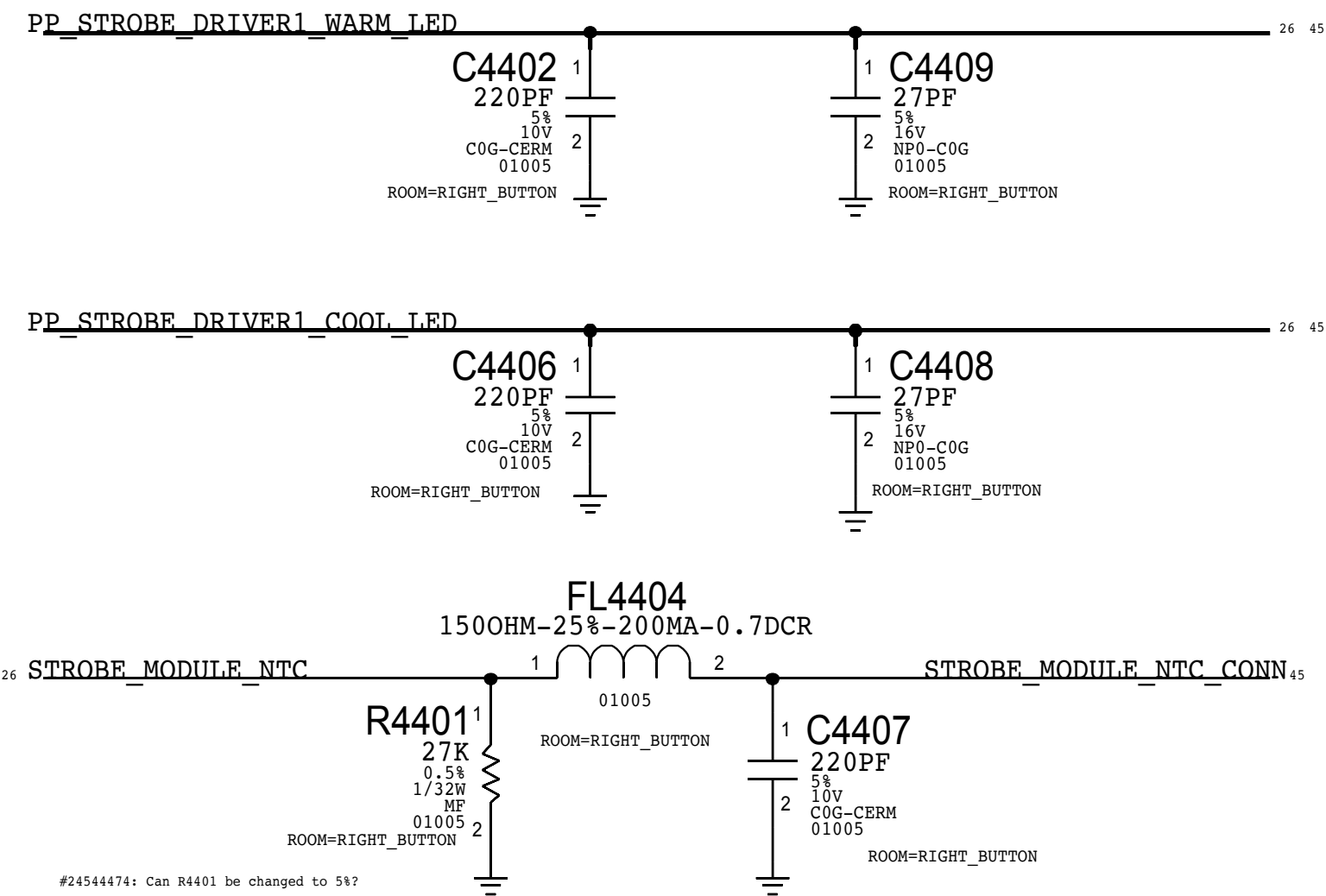
SPEAKER1

USB AC Coupling

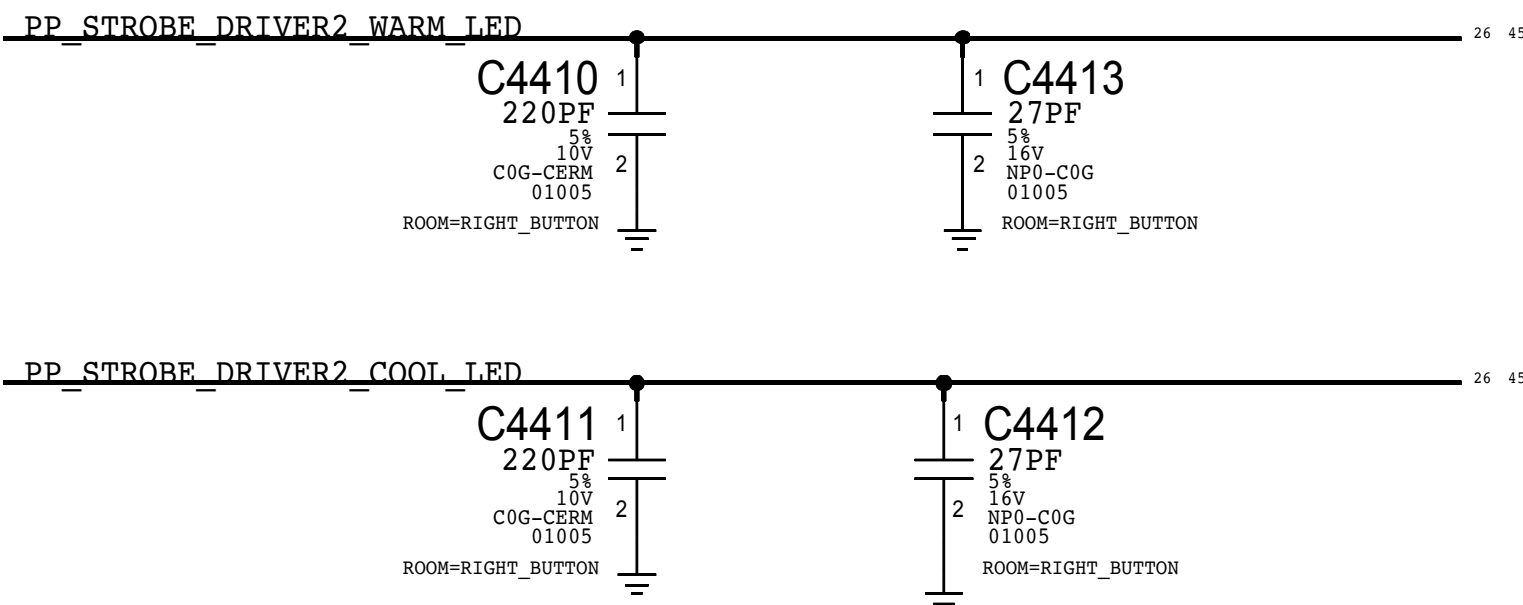
AC return path for USB pairs which is referenced to GND and VDD_MAIN



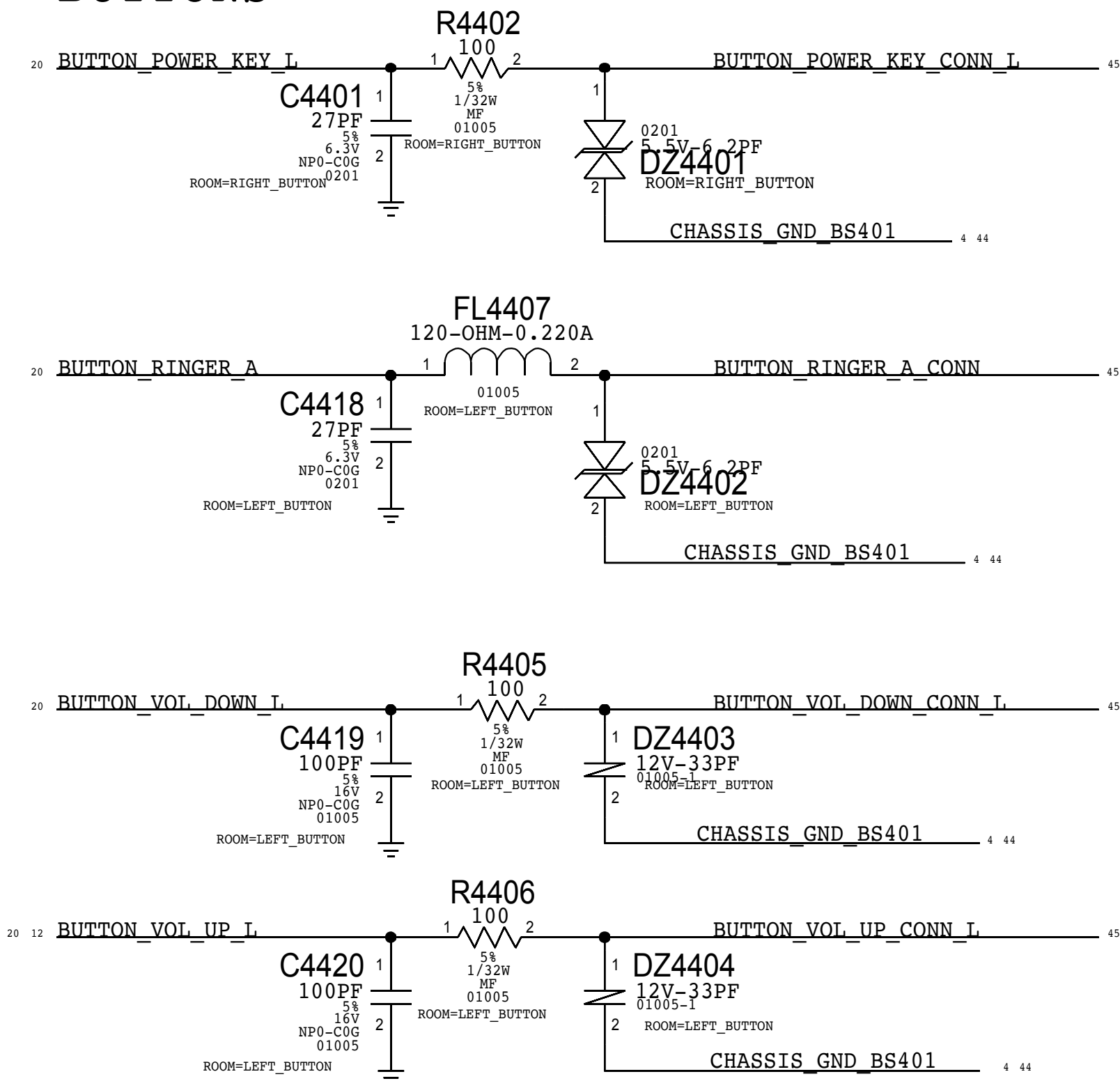
STROBE1



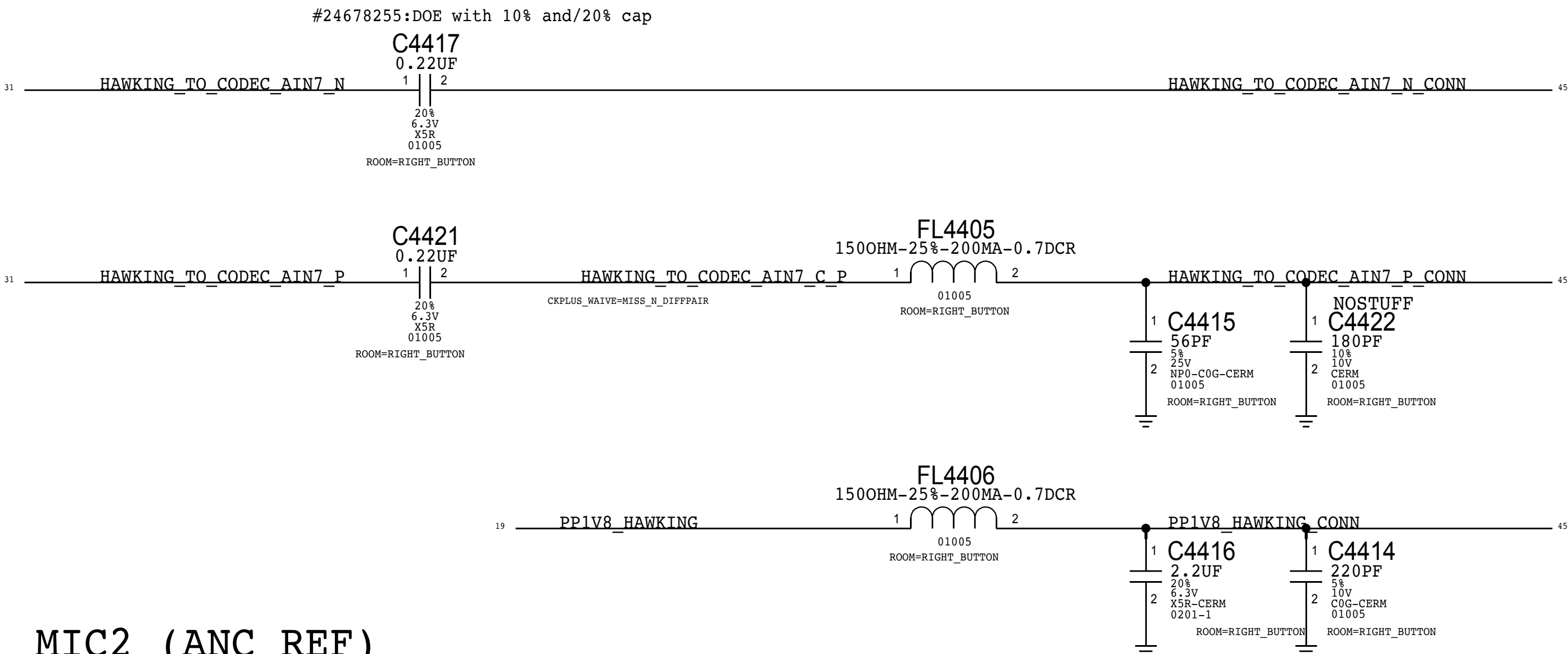
STROBE2



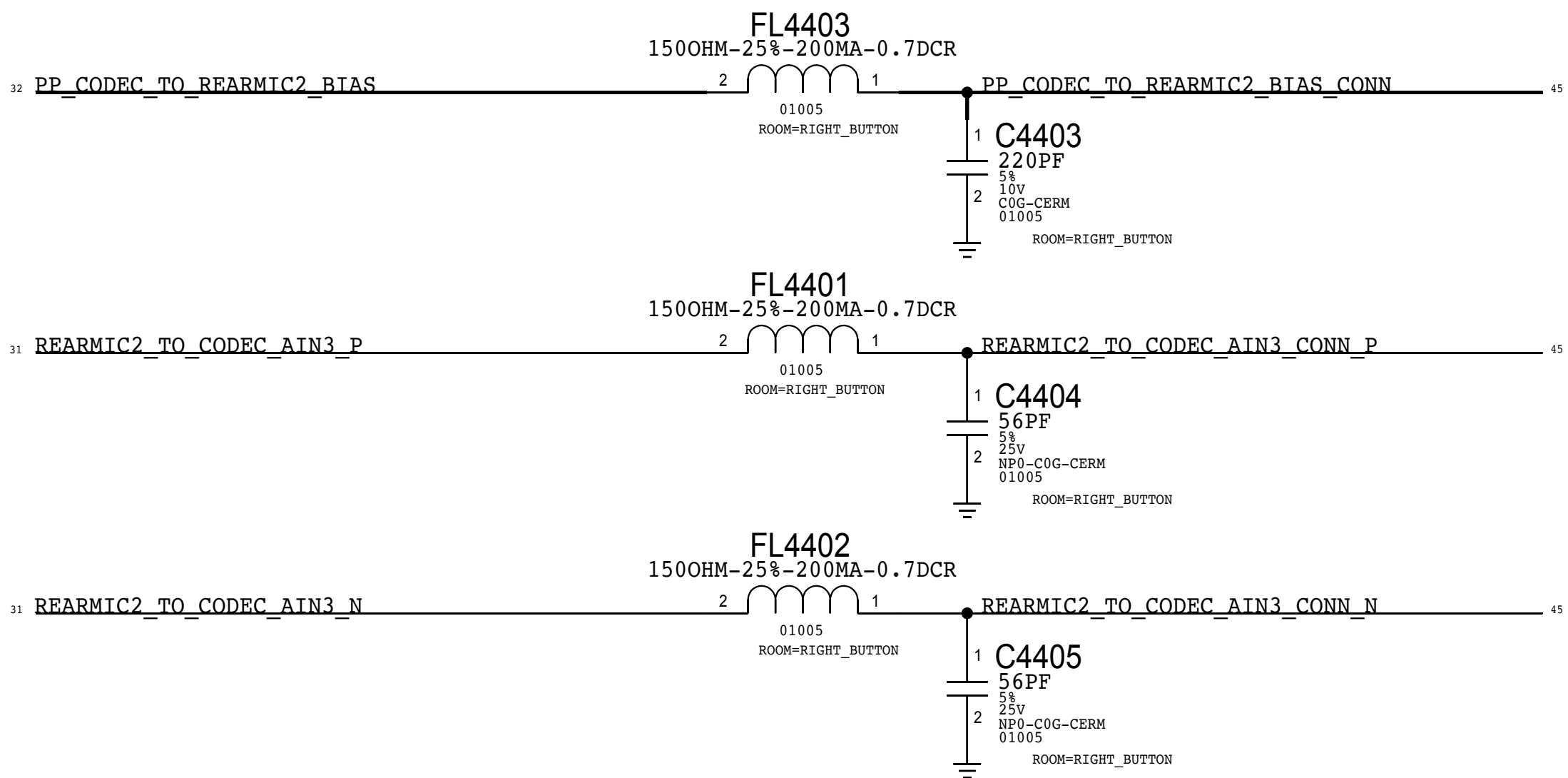
BUTTONS



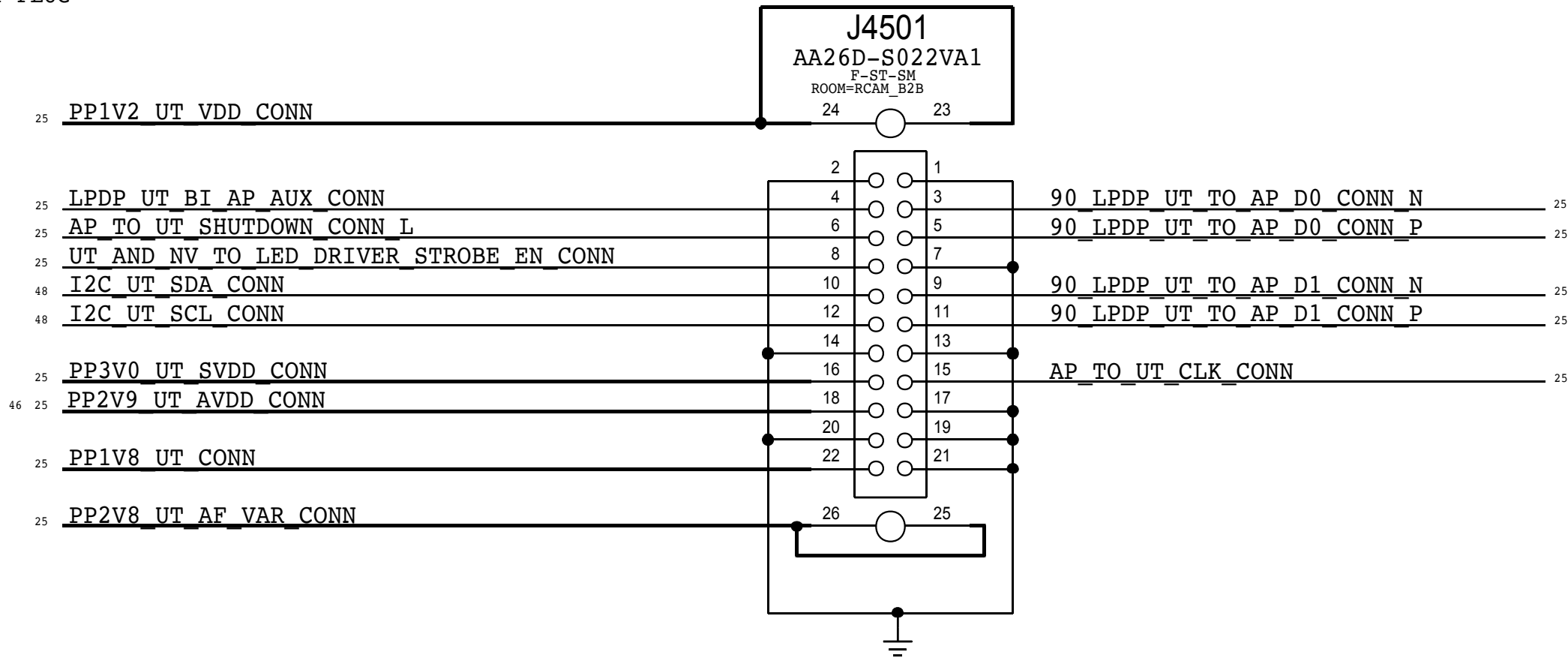
HAWKING



MIC2 (ANC REF)

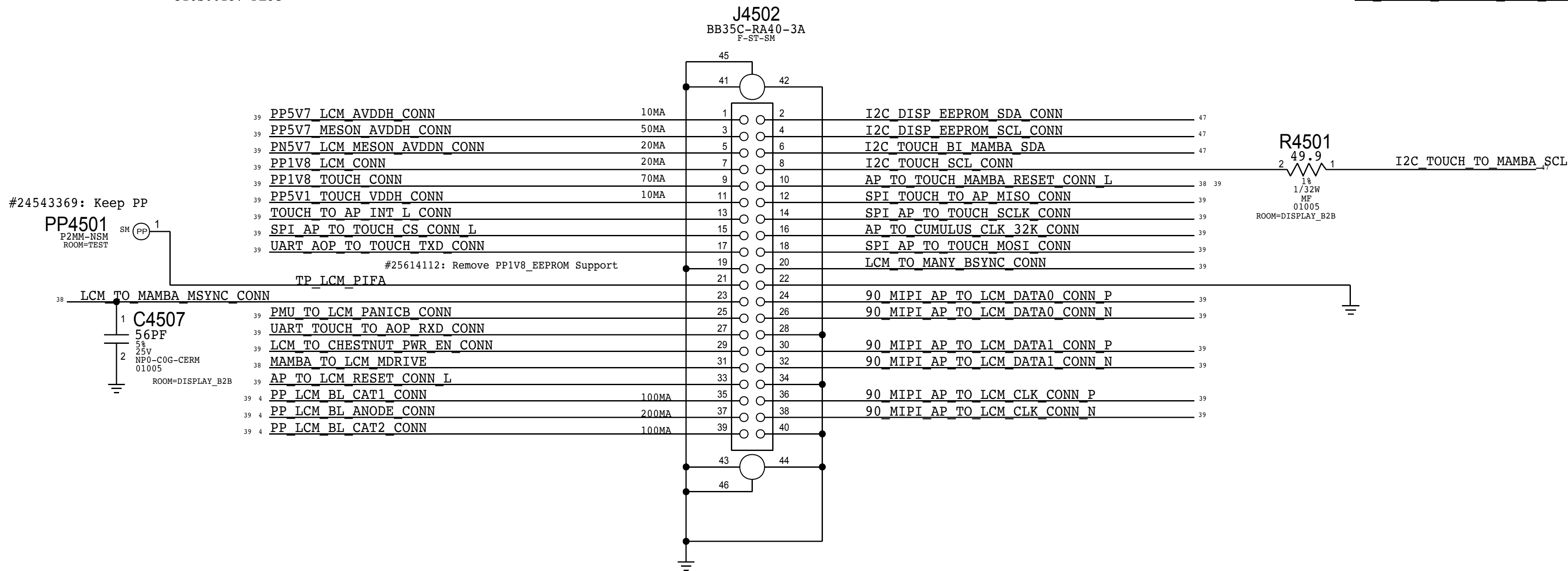


THIS ONE ----> 516S00152 RCPT (USED ON MLB)
516S00151 PLUG



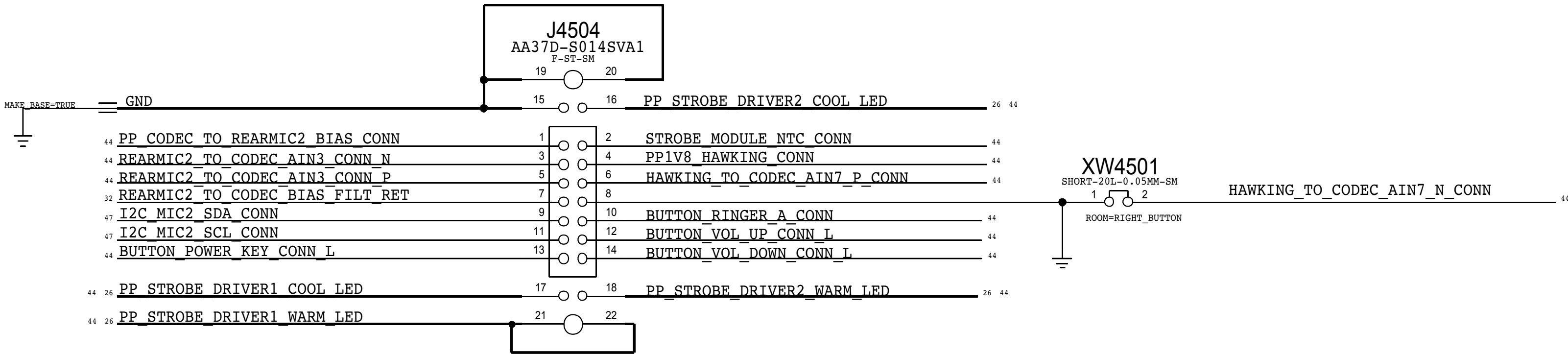
DISPLAY / TOUCH FLEX CONNECTOR

THIS ONE ----> 516S00138 RCPT (USED ON MLB)
516S00137 PLUG



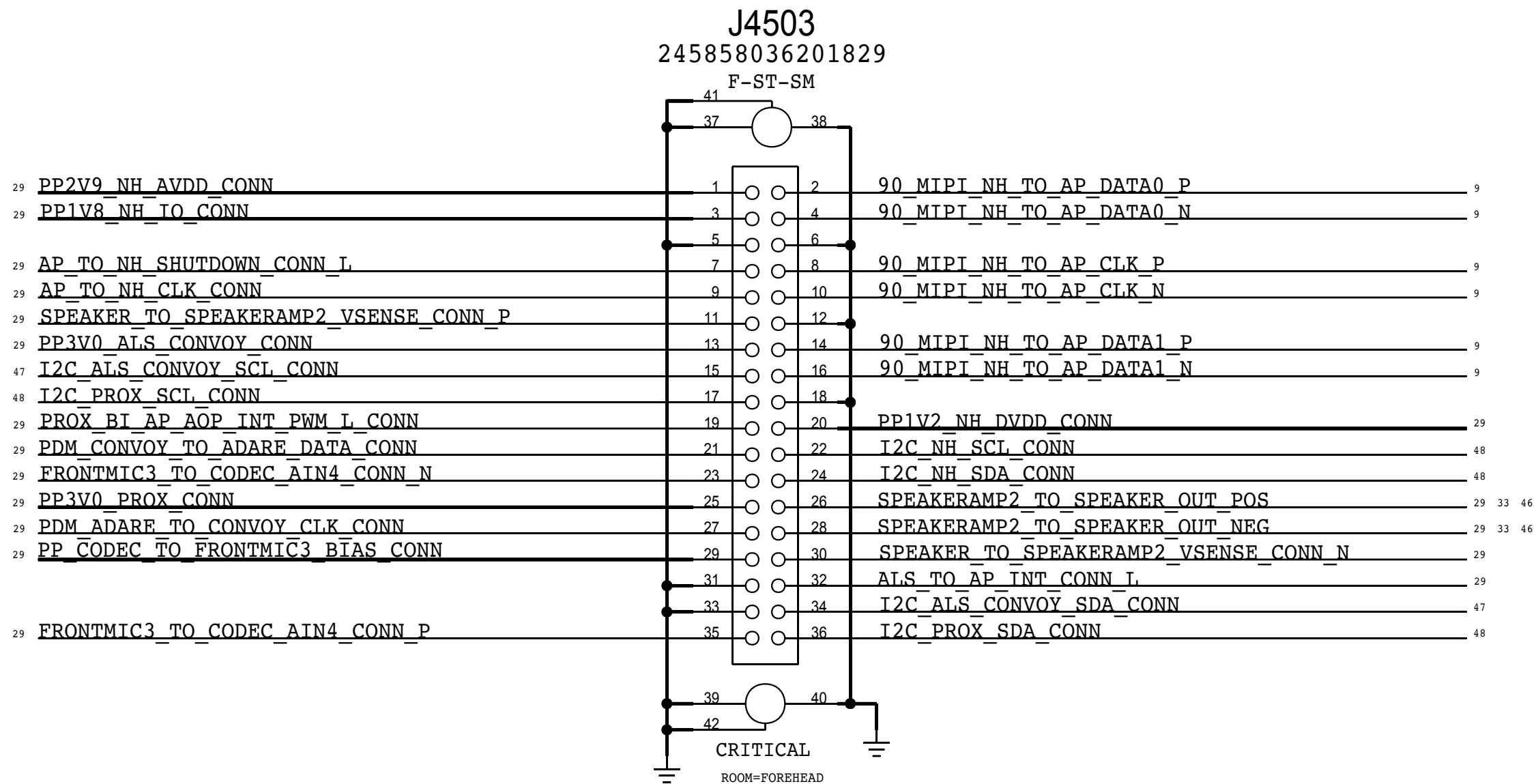
COMBINED BUTTON FLEX CONNECTOR

MLB APN: 516S00150
FLEX APN: 516S00149

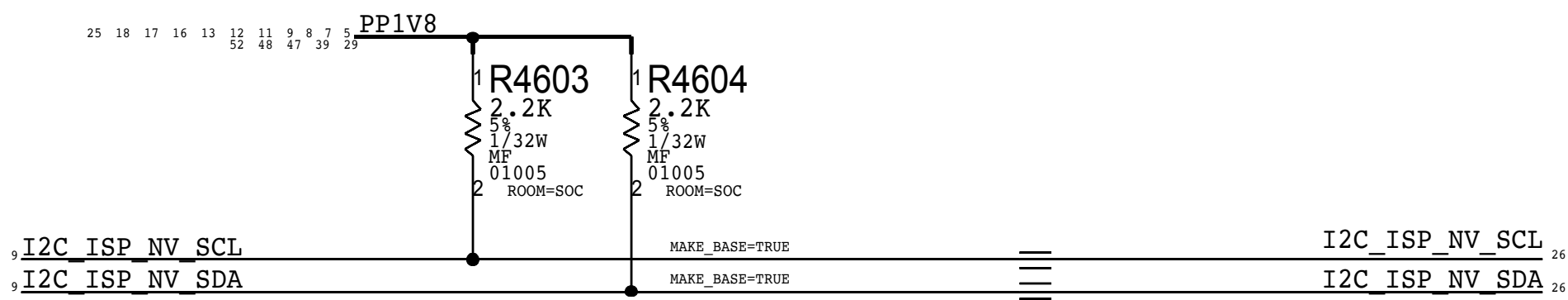


FOREHEAD FLEX CONNECTOR

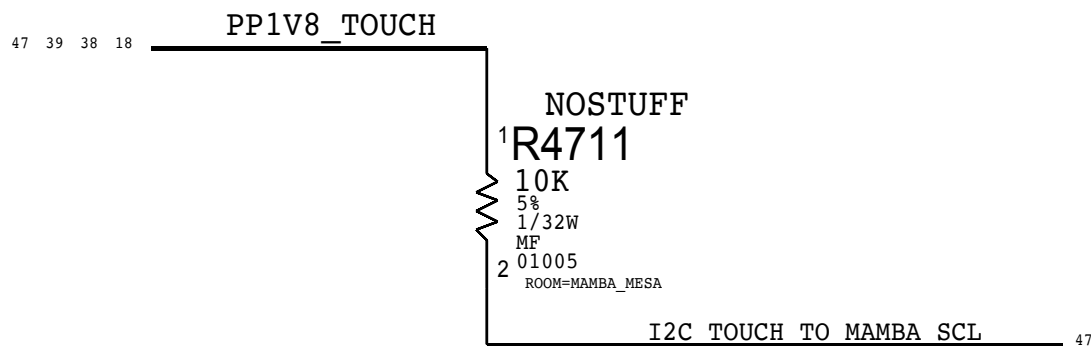
THIS ONE ----> 516S00146 RCPT (USED ON MLB)
516S00145 PLUG



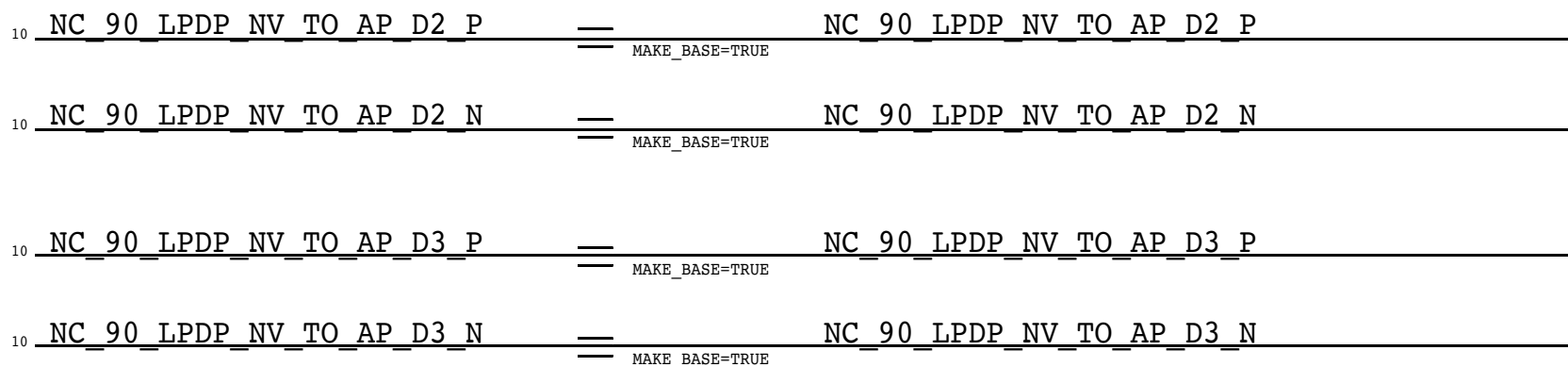
ISP I2C1



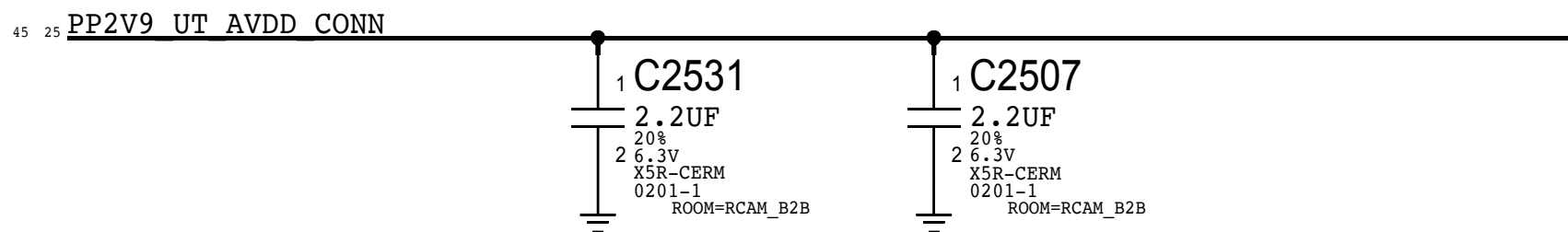
#26682438:Move to Page 46
TOUCH I2C



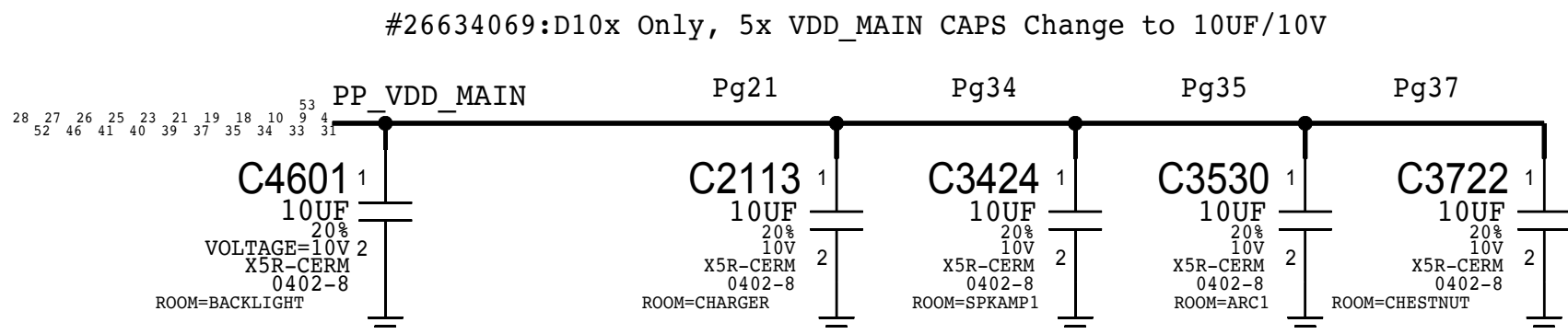
NC Nets in Small FF



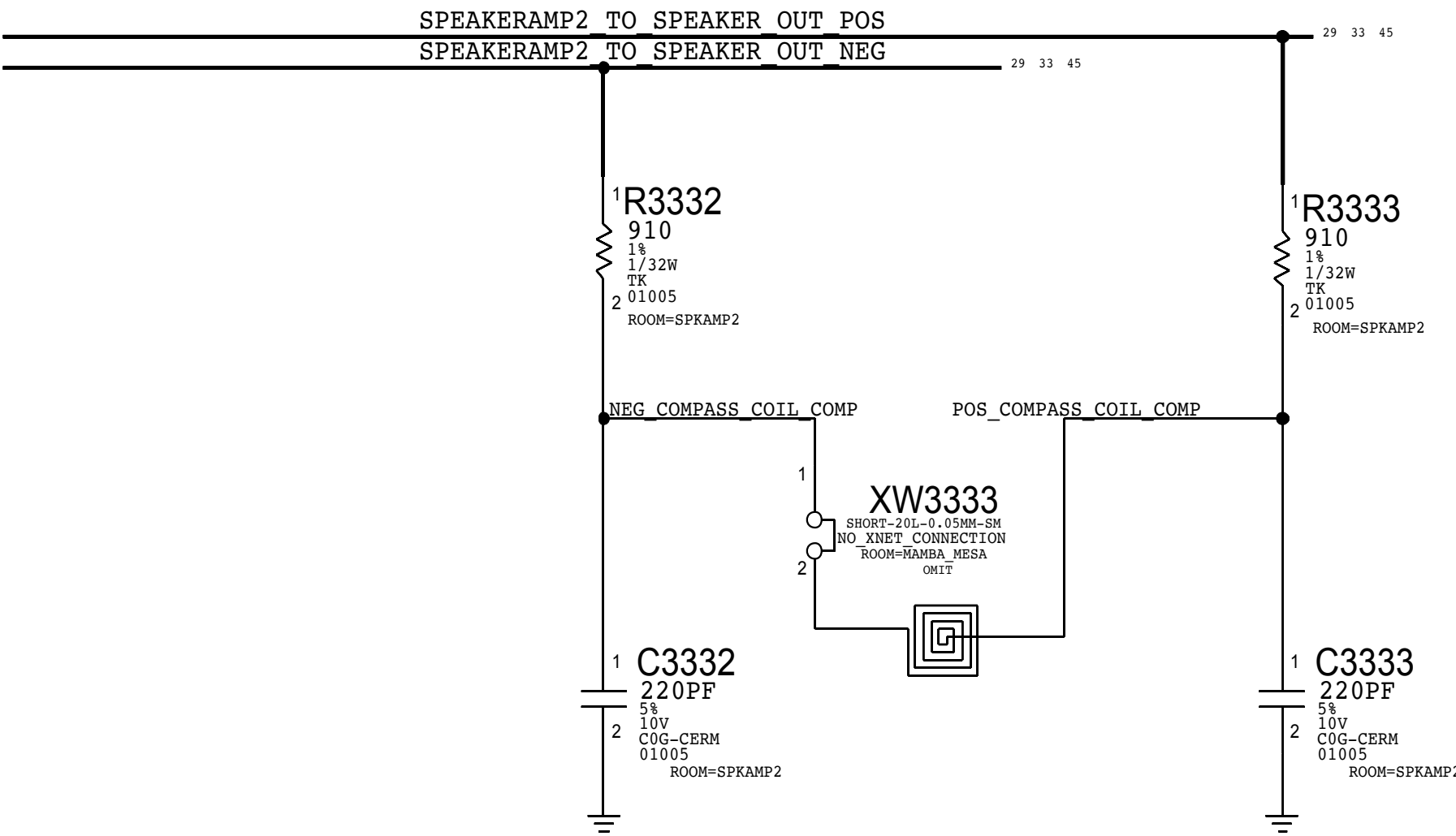
UT B2B



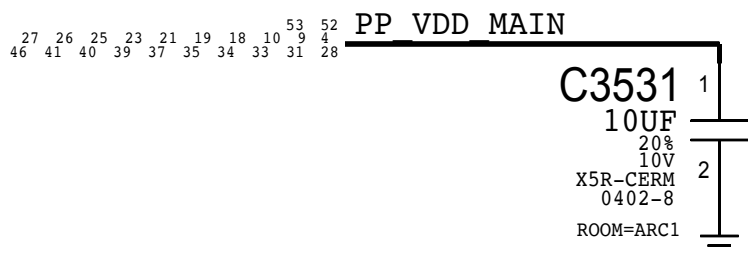
VDD_MAIN Cap



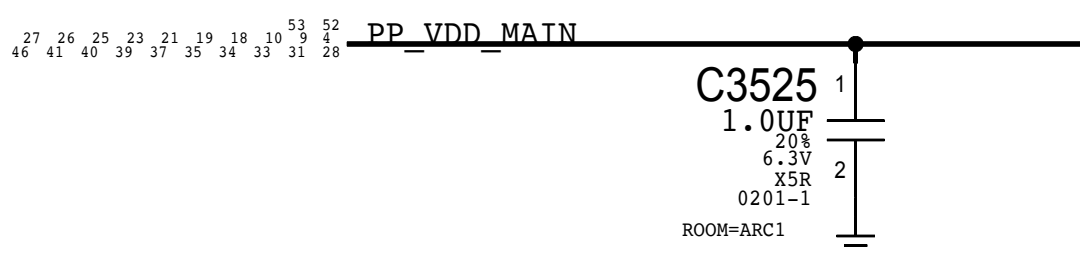
Top Speaker Compass Coil



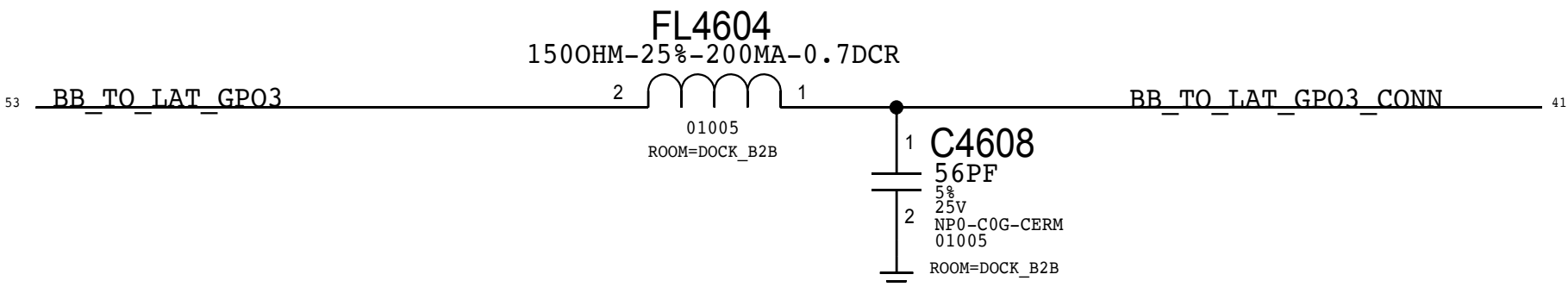
#25742582,Add back C3531 in layout at ARC



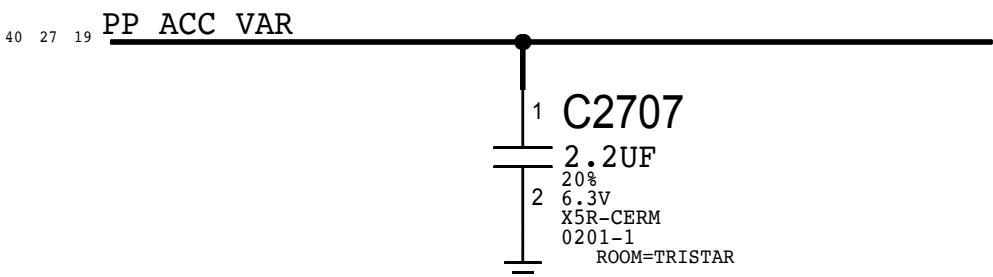
#26104509:C3525 Change to 1UF 0201 in DVT



Dock B2B (Pg 41)

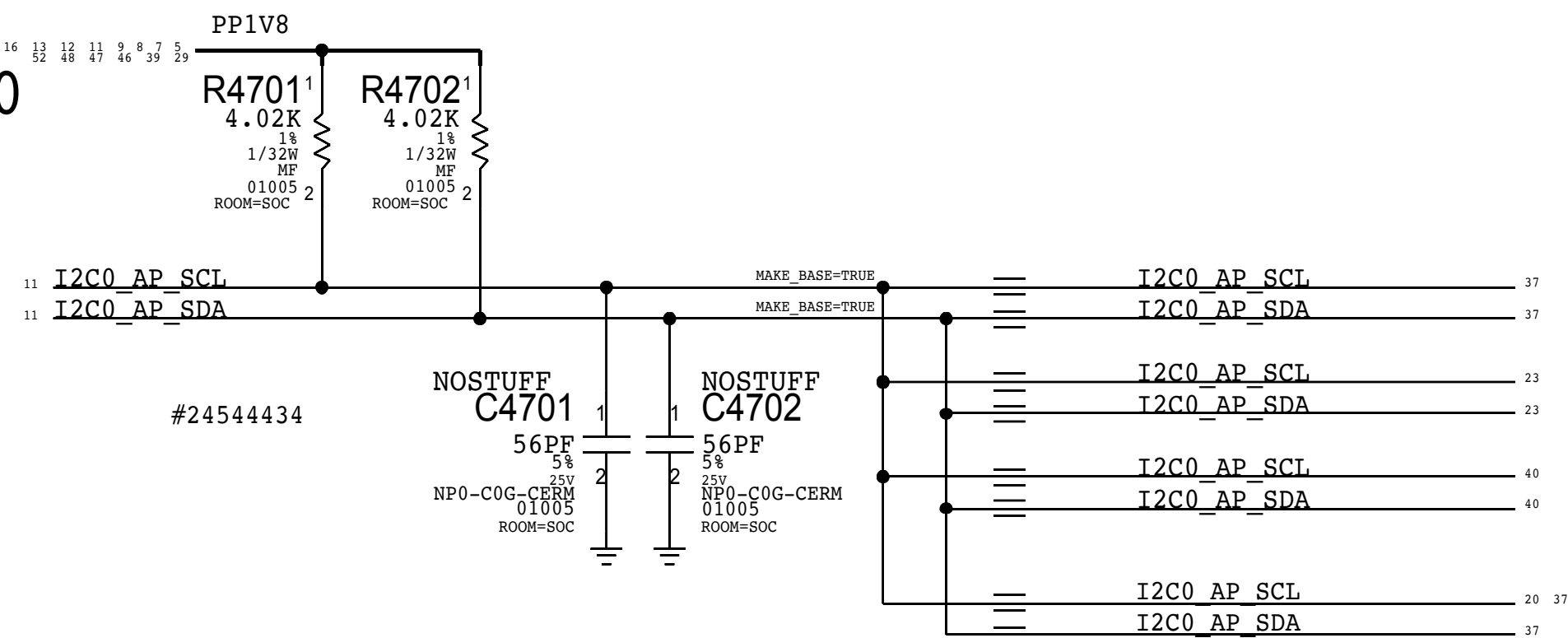


ACC Buck Caps

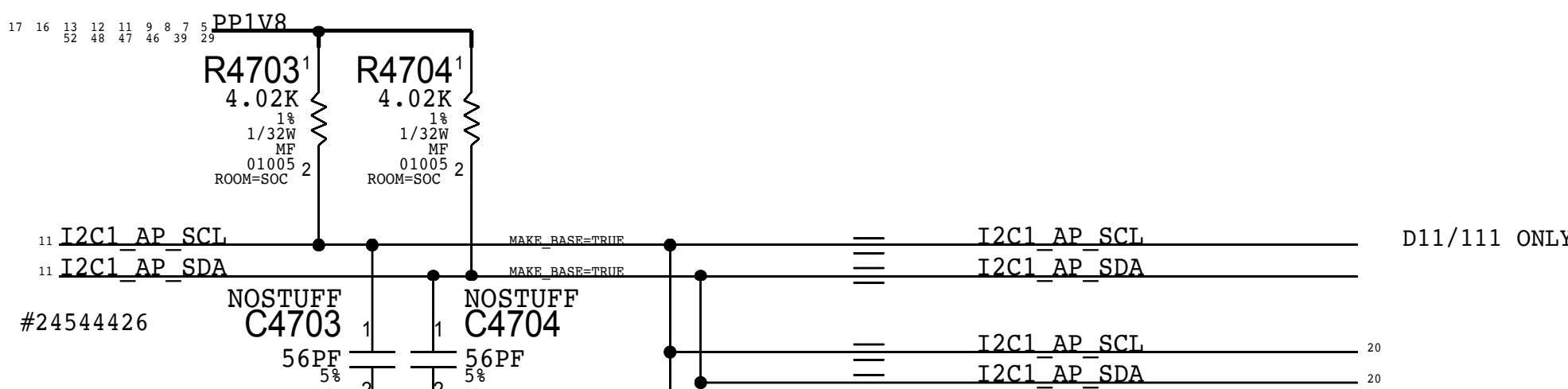


AP

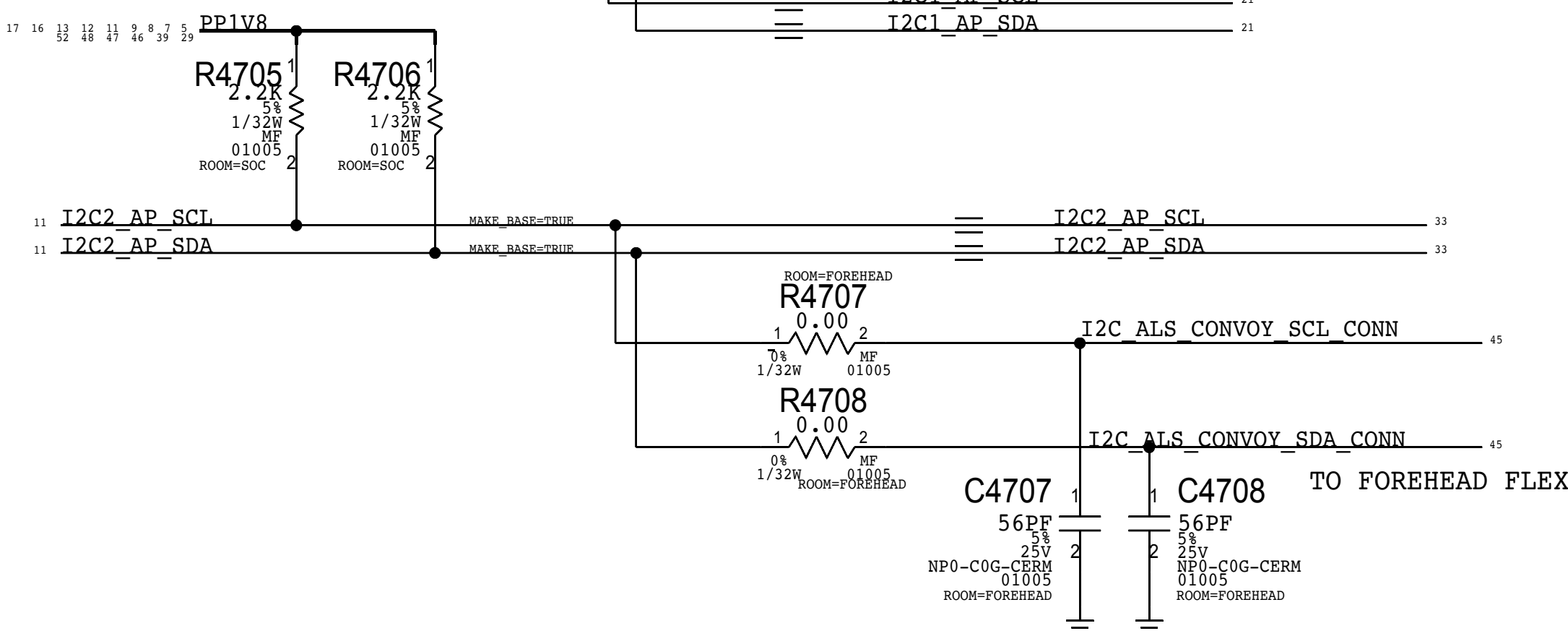
I2C0



I2C1

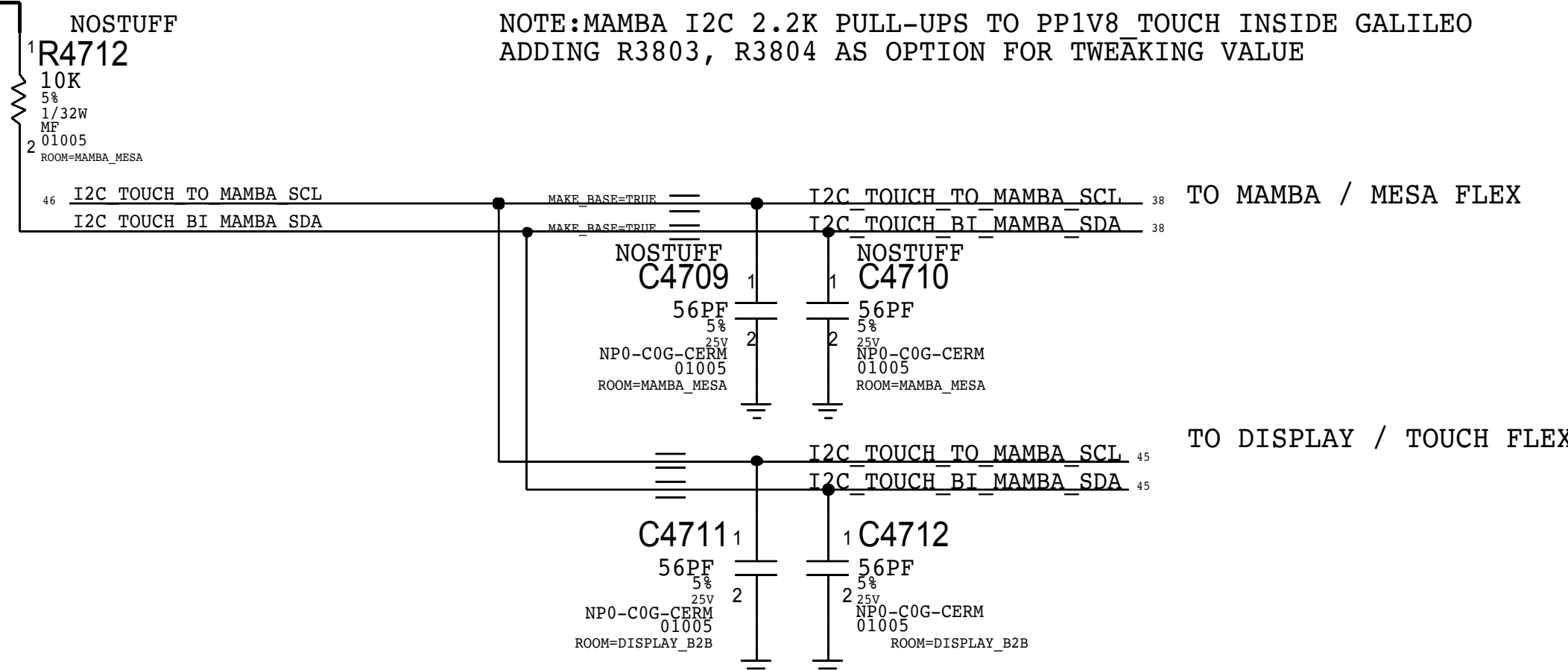


I2C2

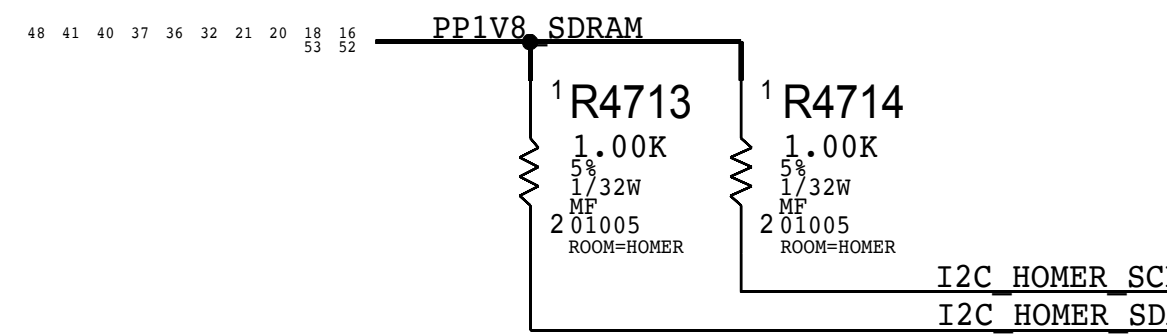


TOUCH

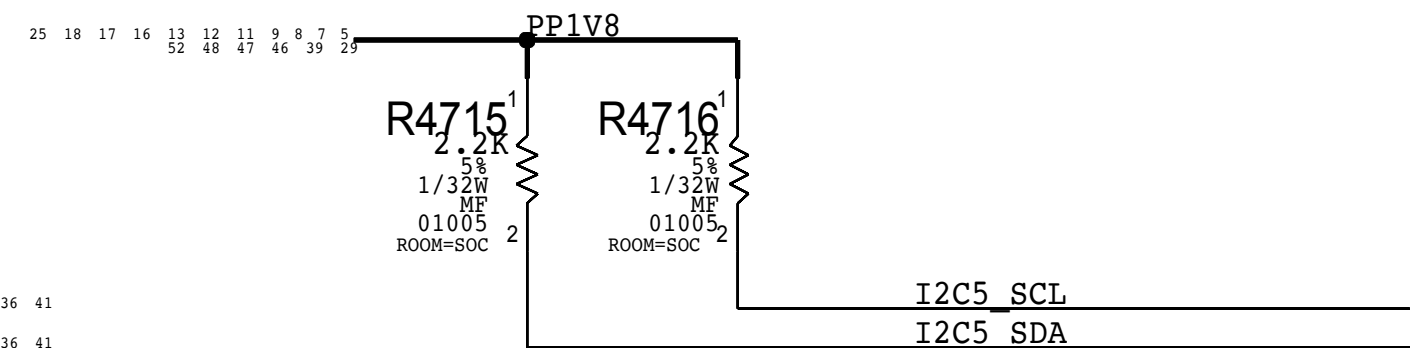
#26682438:Move to Page 46



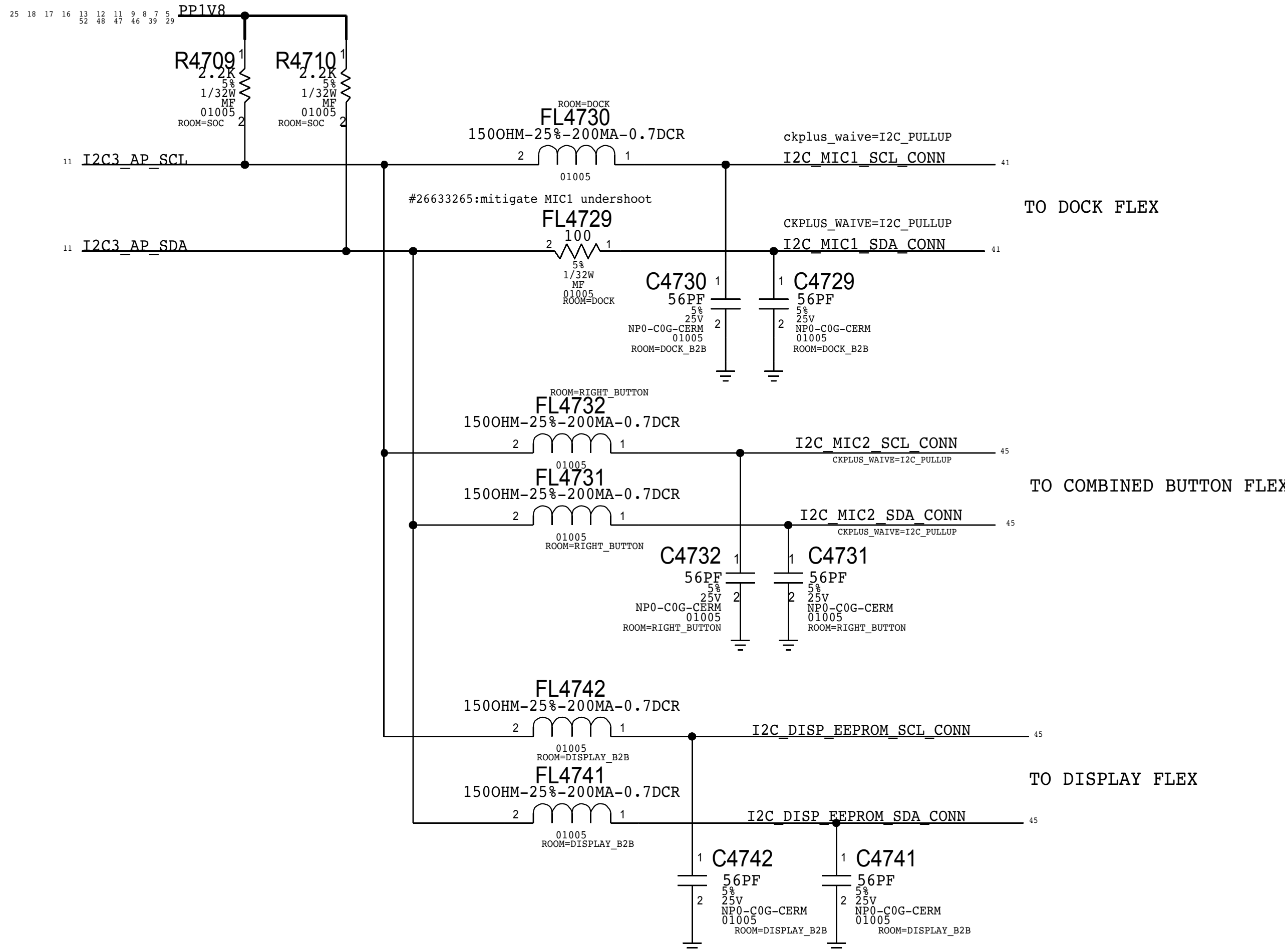
HOMER

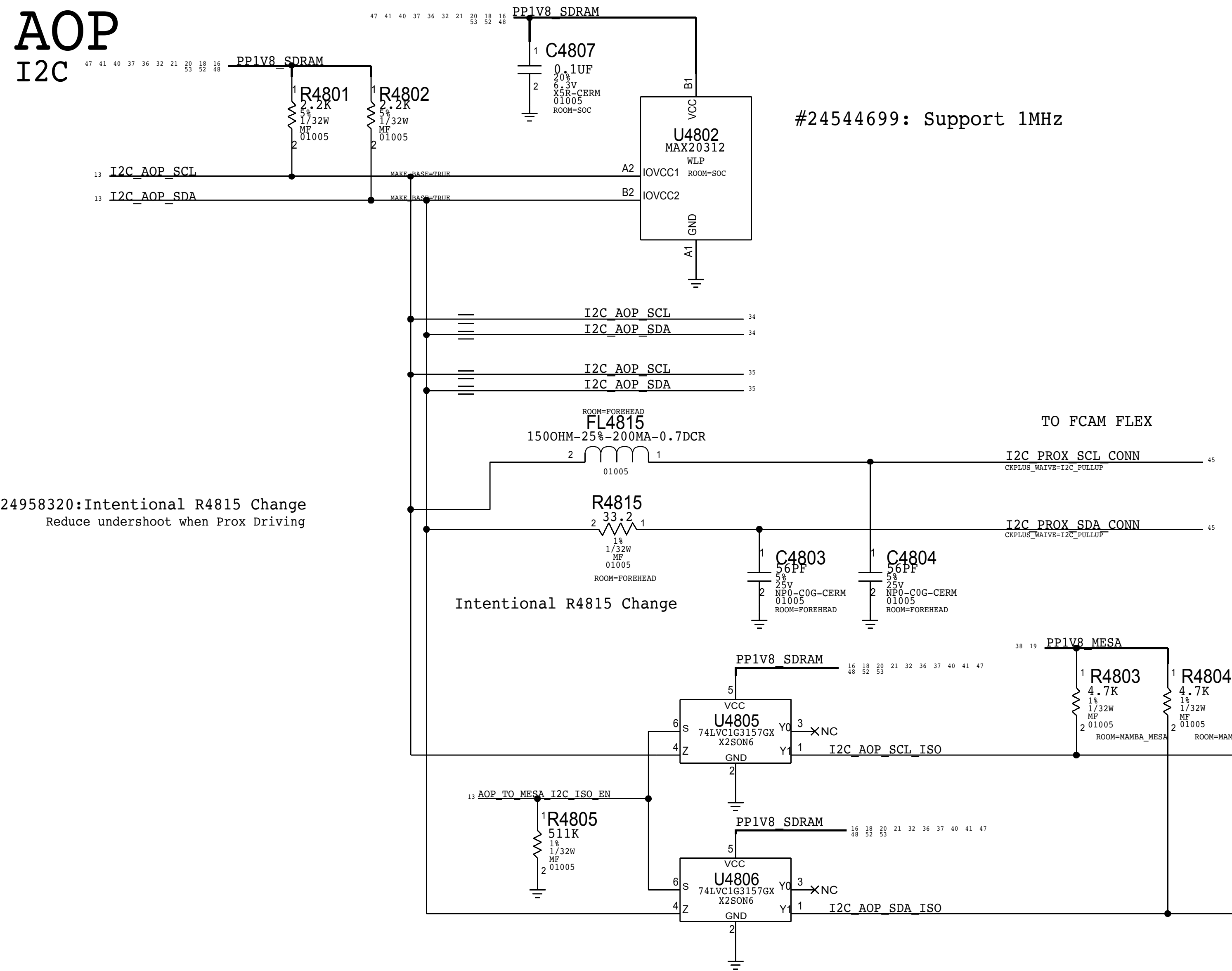


I2C5



I2C3

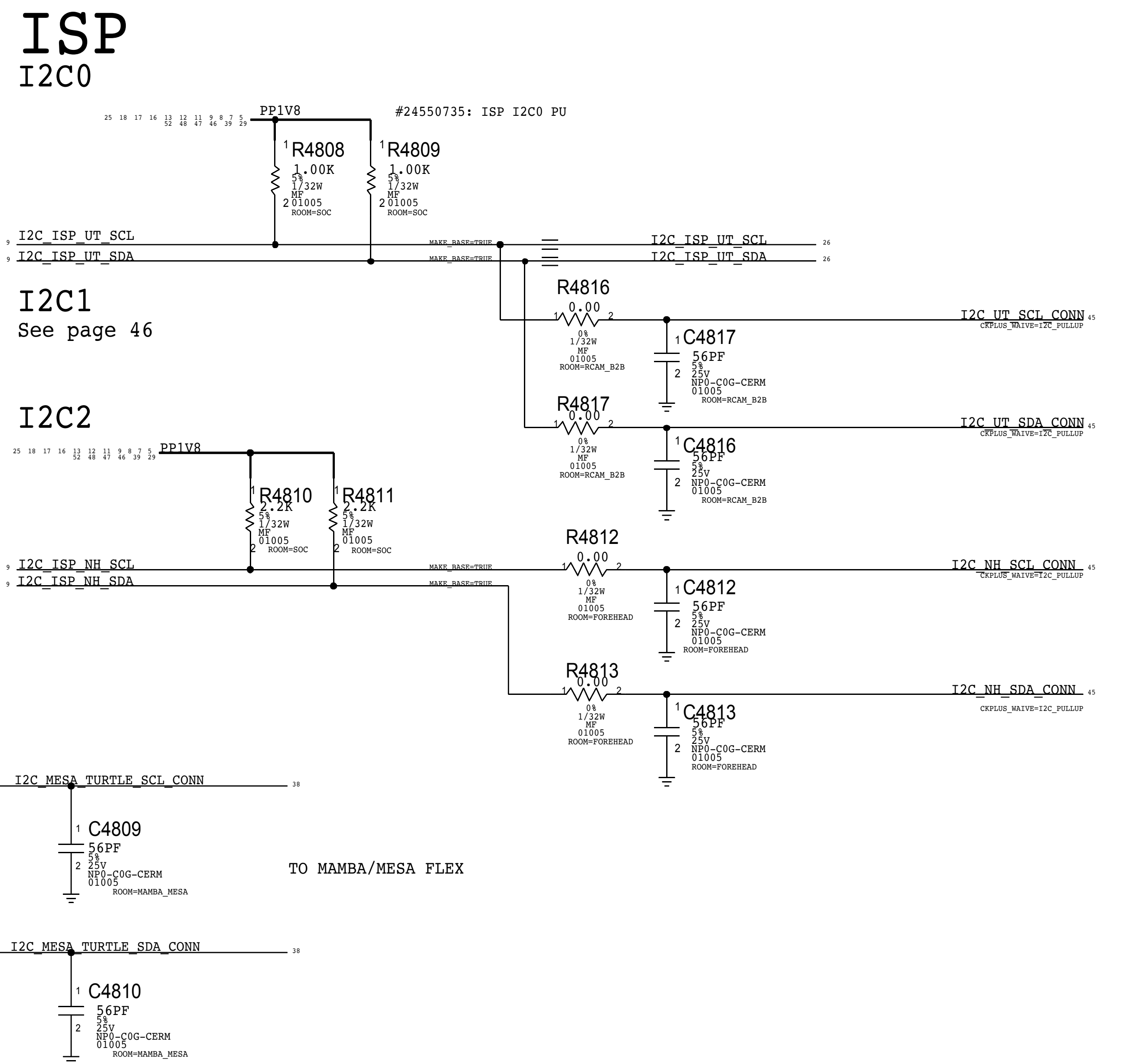




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#24544699: Support 1MHz
```

```
#24958320: Intentional R4815 Change
      Reduce undershoot when Prox Driving
```

Intentional R4815 Change

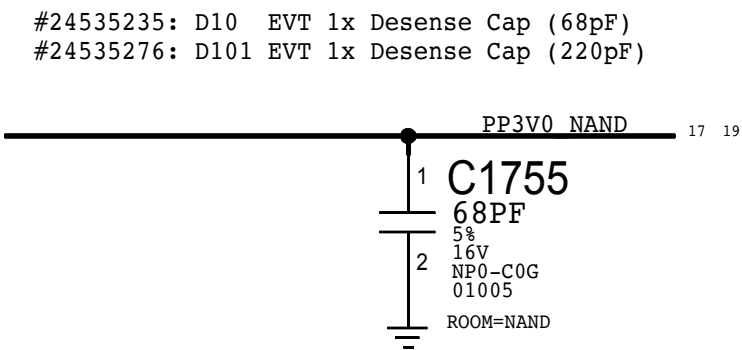
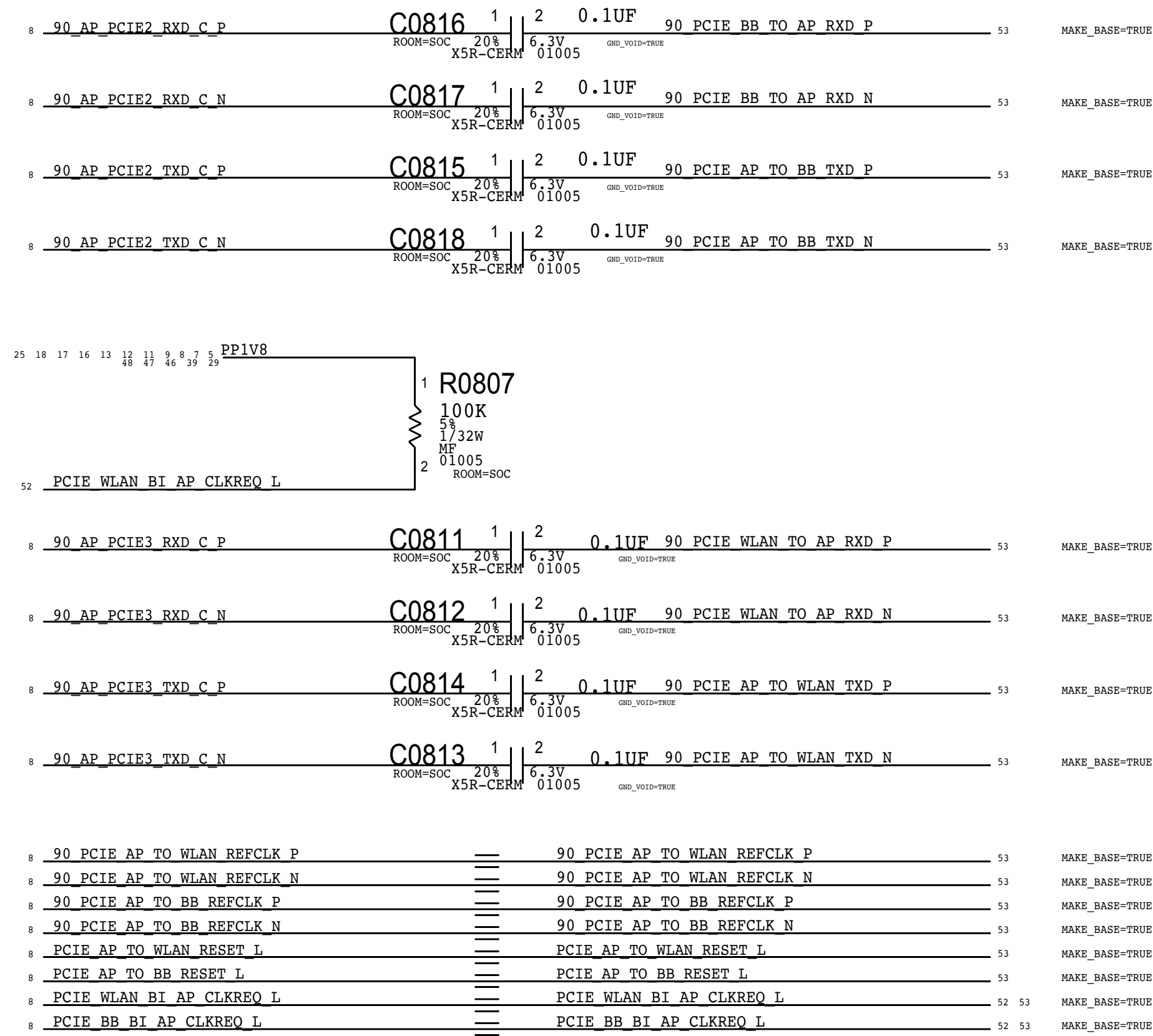
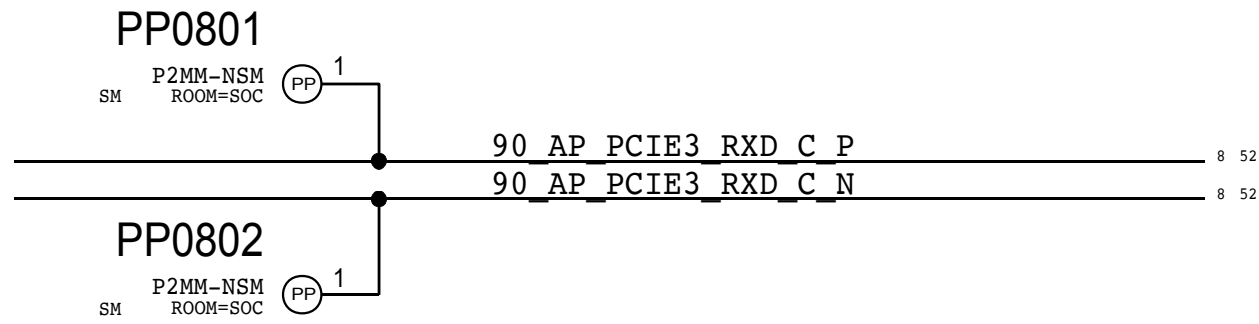


D1x I2C Table

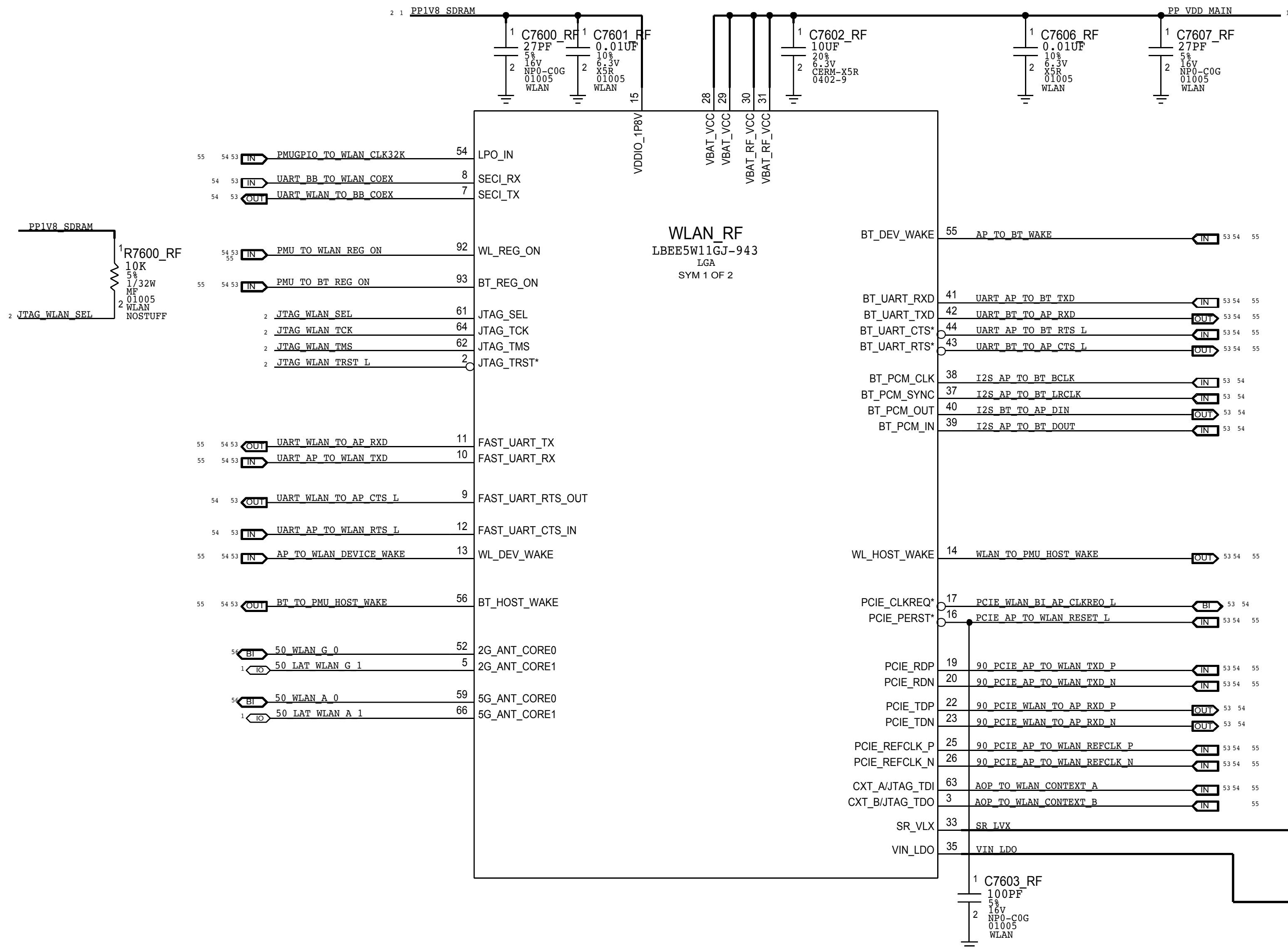
Bus	Device	Binary	7-bit Address	8-bit Address	Max Speed
ISP I2C0 1.5Mhz	SALT LAKE (PRIMARY)	0010000X	0x10	0x20	1.5Mhz
	SALT LAKE (SECONDARY)	0110000X	0x30	0x60	1.5Mhz
	GRUNBERG (STANDARD)	0011100X	0x10	0x08	1.5Mhz
	GRUNBERG (SPECIAL ABSOLUTE RW)	0011101X	0x10	0x0A	1.5Mhz
	GRUNBERG (SPECIAL DELTA READ)	0011110X	0x1E	0x3C	1.5Mhz
	STROBE 1	1100111X	0x63	0xC6	1.5Mhz
ISP I2C1 1.5Mhz	LAS VEGAS (PRIMARY)	0100000X	0x20	0x40	1.5Mhz
	LAS VEGAS (SECONDARY)	0110000X	0x30	0x60	1.5Mhz
	EDWIN	1100011X	0x63	0xC6	400 kHz
	STROBE 2	0001110X	0x06	0x0C	1.5Mhz
ISP I2C2 400kHz	CONCORD	0010000X	0x10	0x20	1.5Mhz
ADP I2C 1.5Mhz	ARC DRIVER 1	100000YX	0x41	0x82	1.5Mhz
	PROX	1011000X	0x58	0x80	1.5Mhz
	SPKAMP1 (L2B)	100000YX	0x40	0x80	1.5Mhz
	TURTLE	0101100X	0x2c	0x58	400 kHz
	MESA	1100001X	0x81	0xC2	400 kHz
Touch I2C 400kHz	MAWBA	1100000X	0x60	0xC0	1.5Mhz
	MESON	1000000X	0x40	0x80	510 kHz
EEPROM I2C	EEPROM	1010001X	0x51	0xA2	400 kHz
AP I2C0 400kHz	BOOST	1110101X	0x75	0xEA	3.4 Mhz
	TRISTAR	0011010X	0x1A	0x34	400 kHz
	BACKLIGHT 1	1100110X	0x62	0xC4	1.5Mhz
	CHESTNUT	0101111X	0x27	0x4E	400 kHz
AP I2C1 400kHz	ADELYN	1110100X	0x74	0xEB	400 kHz
	TIGRIS	1110101X	0x75	0xEA	400 kHz
	BACKLIGHT 2	1100110X	0x62	0xC4	1.5Mhz
AP I2C2 400kHz	ALS	0101001X	0x29	0x52	400 kHz
	CONVOY	0100001X	0x21	0x42	3.4 Mhz
		0100010X	0x22	0x44	400 kHz
	SPKAMP 2 (L2B)	100002YX	0x40	0x80	400 MHz
AP I2C3 400kHz	DOCK FLEX (MIC1)	1010100X	0x54	0xA8	1.5Mhz
	COMBINED BUTTON FLEX (MIC2)	1010100X	0x56	0xAC	1.5Mhz
	DISP EEPROM	1010001X	0x51	0xA2	400kHz
Homer I2C 1.5Mhz	SCHRODINGER (STANDARD)	0001110X	0x1E	0x3C	1.5Mhz
	SCHRODINGER (CAL REG'S)	0001111X	0x1F	0x3D	1.5Mhz
	EEPROM (STANDARD)	1010000X	0x50	0xA0	
	EEPROM (CONFIG)	1010000X	0x50	0xA0	

This page contains items which differ accross all MLB designs

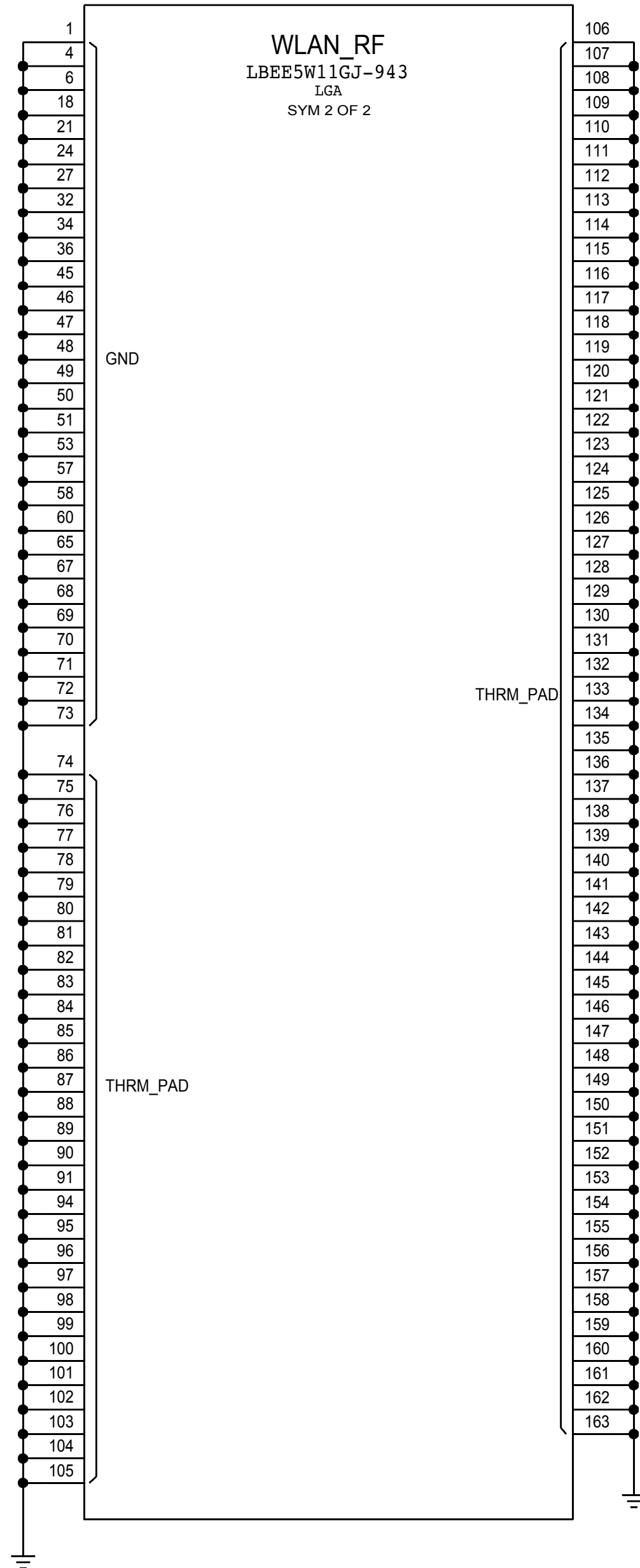
PCIe lanes



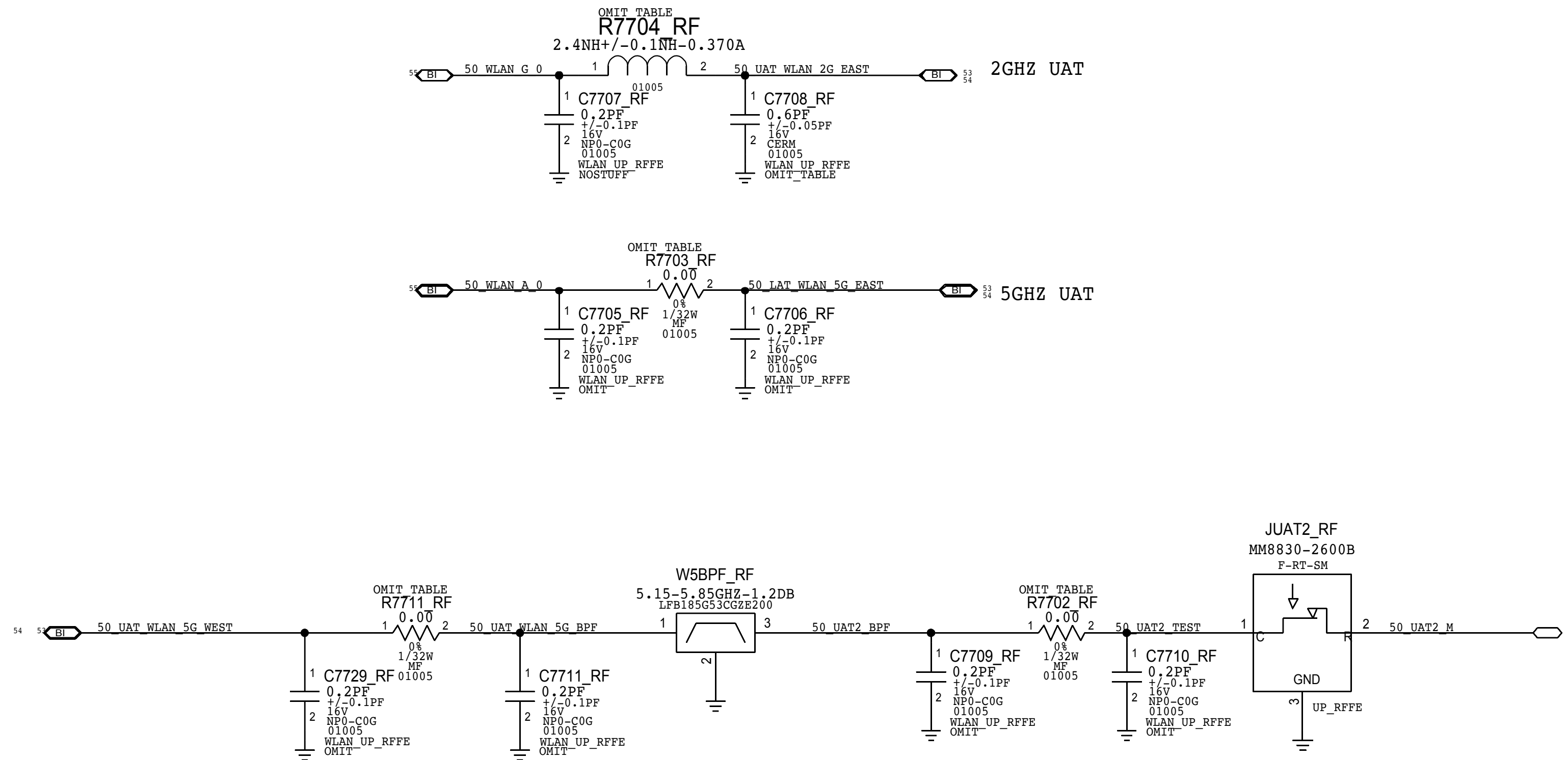
WIFI/BT



PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
339800201	339800199	ALTERNATE	WLAN_RF	ALT WIFI/BT MODULE



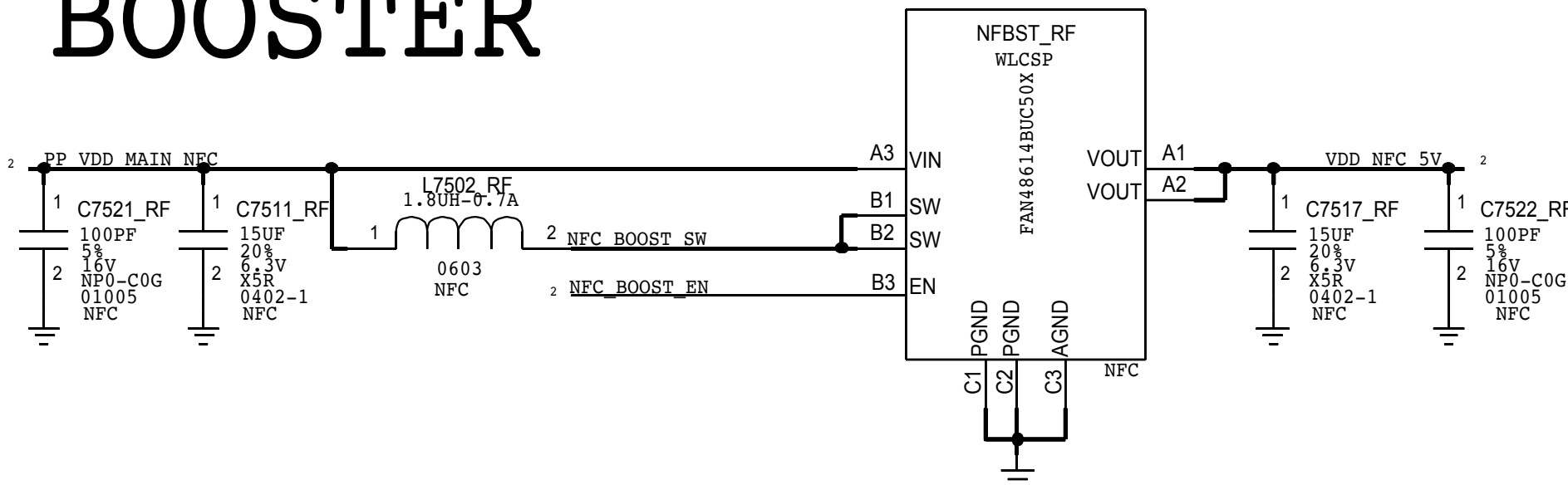
WIFI UPPER ANTENNA FEEDS



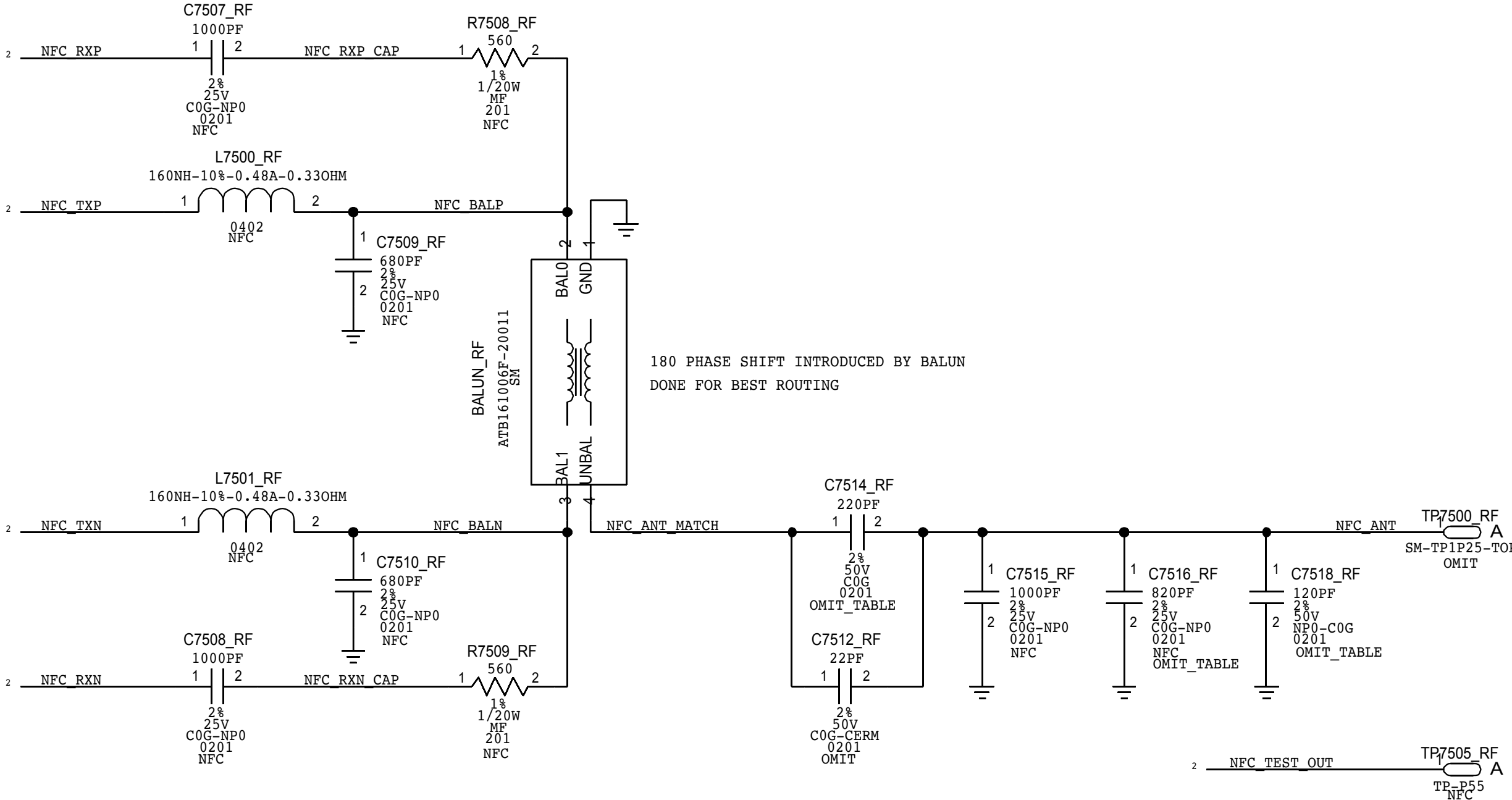
STOCKHOLM

NFC CONTROLLER

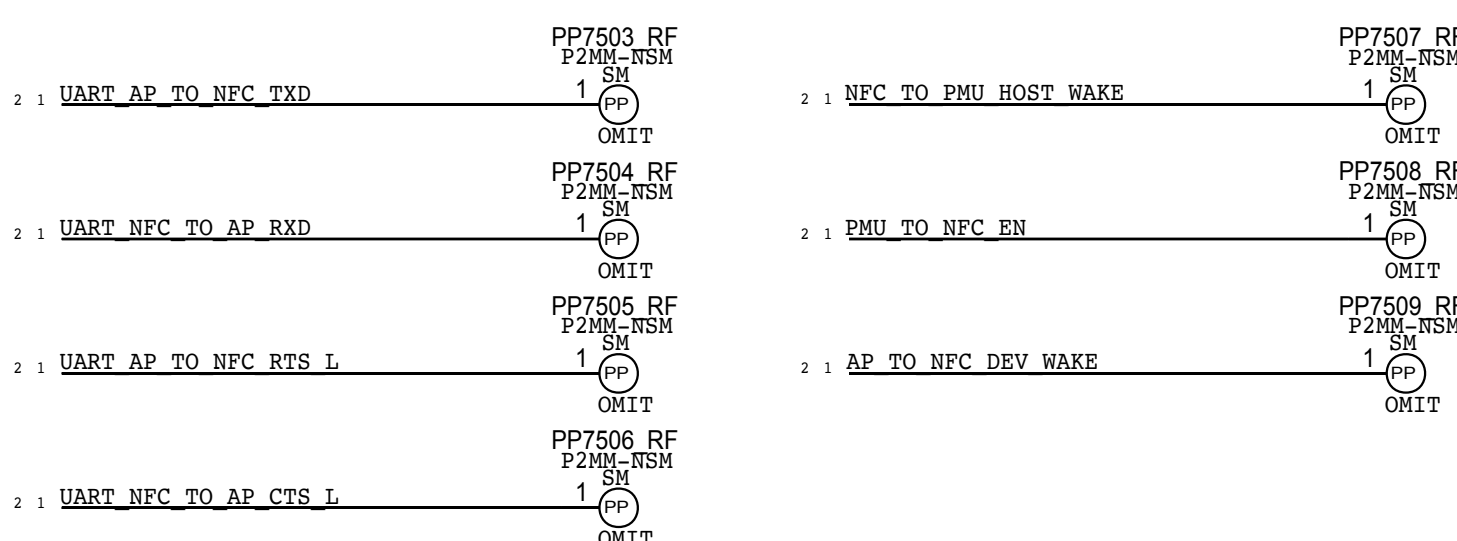
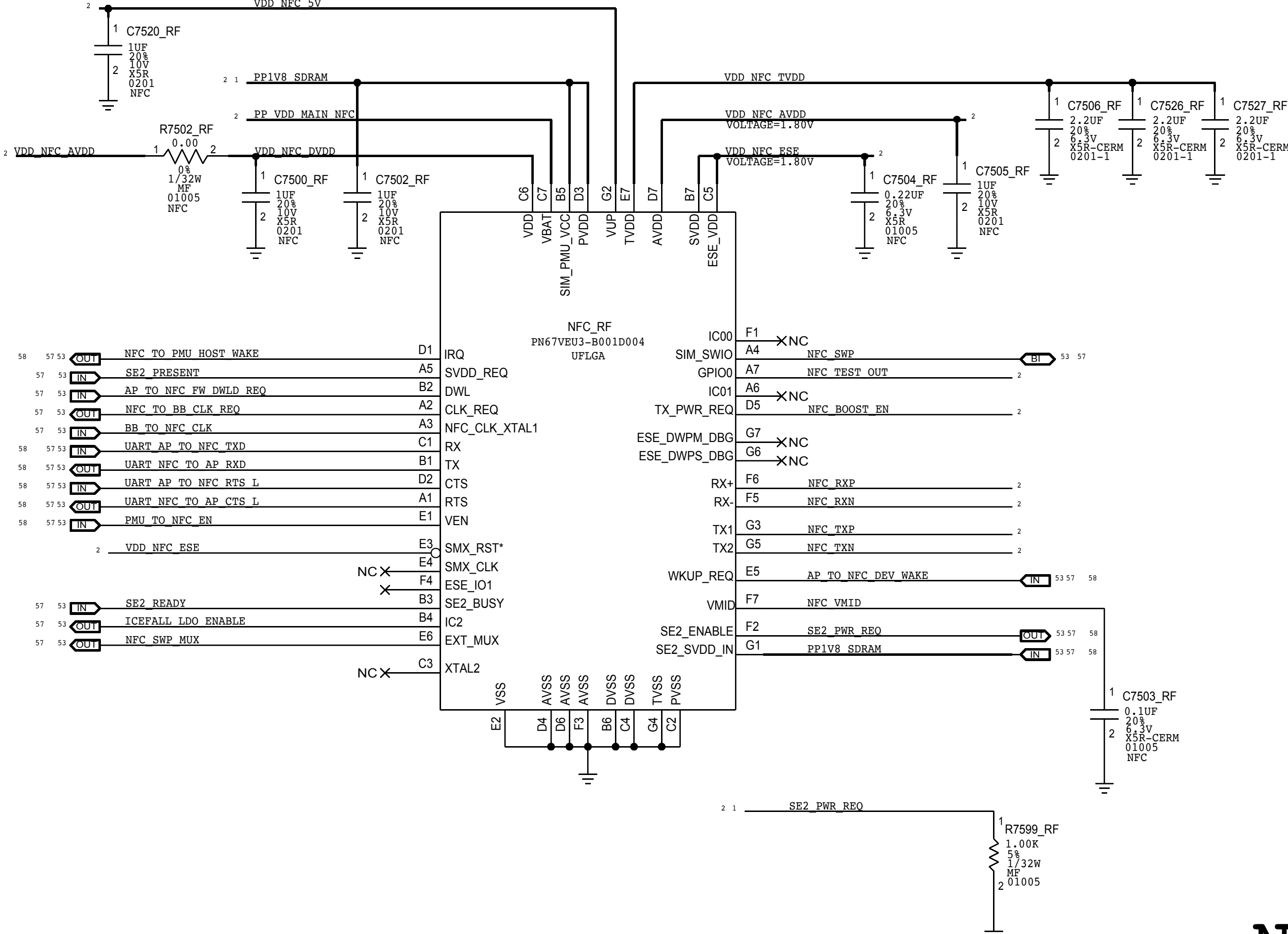
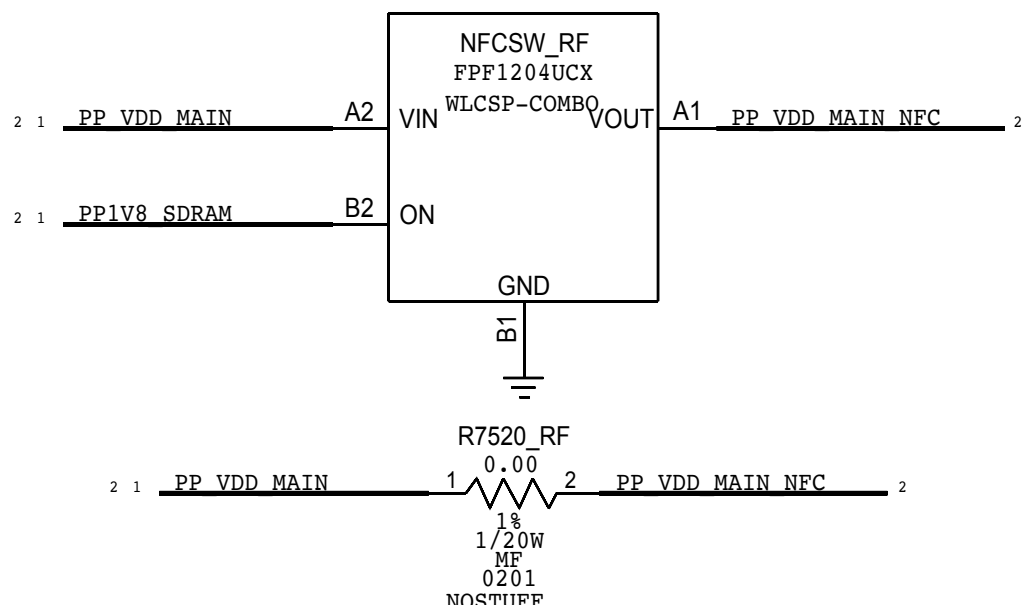
5V BOOSTER



NFC FRONT END



NFC LOAD SWITCH



BASEBAND

D

D

C

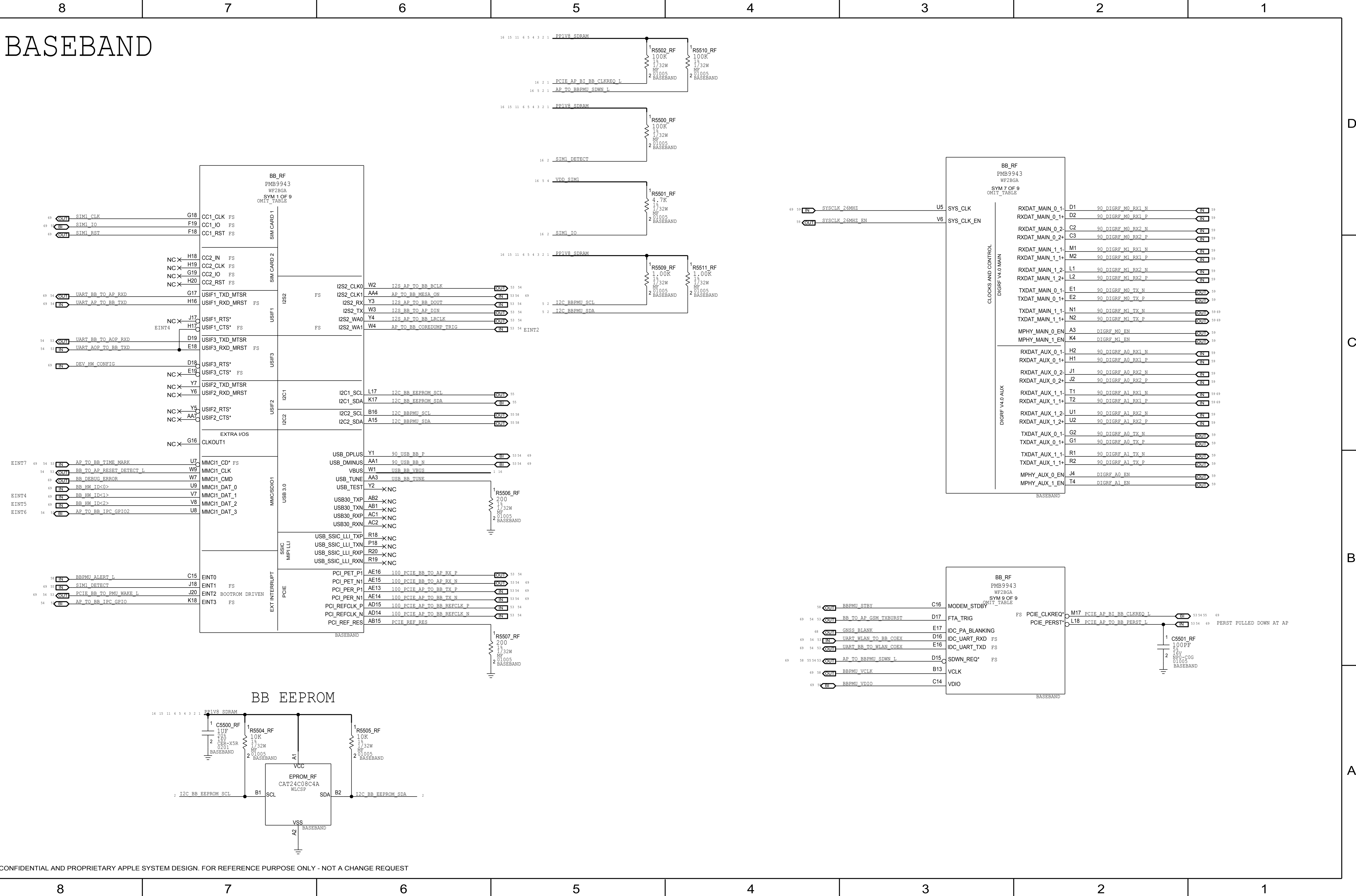
C

B

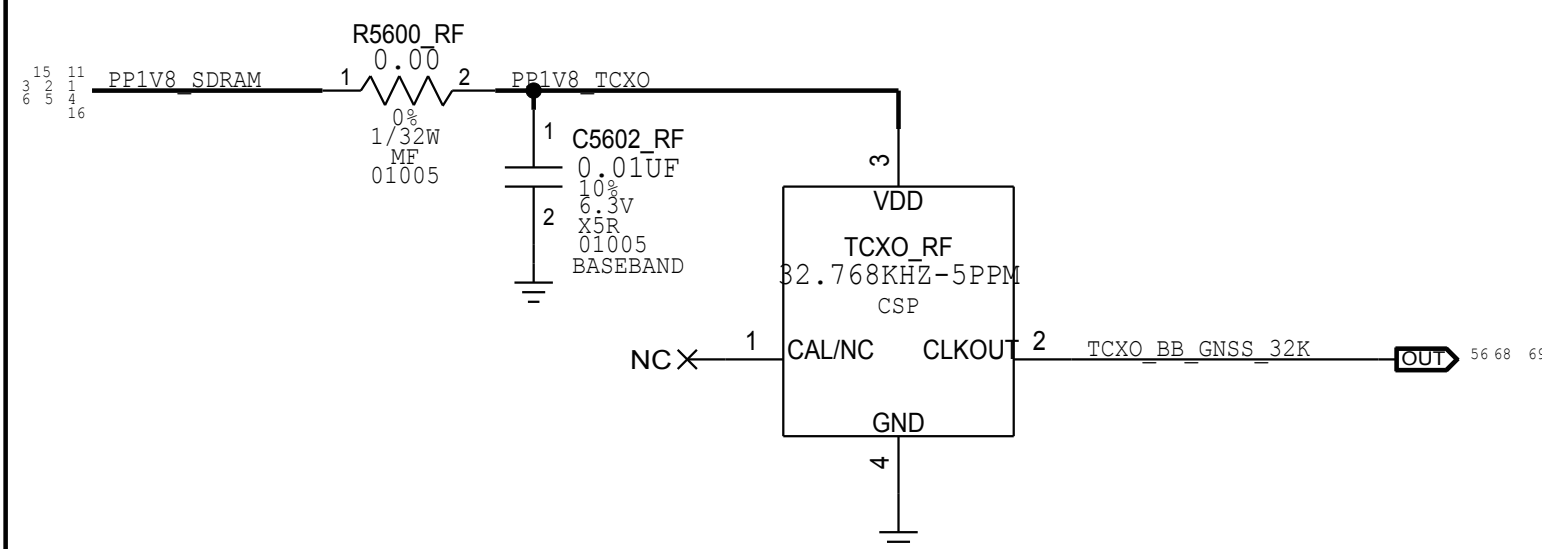
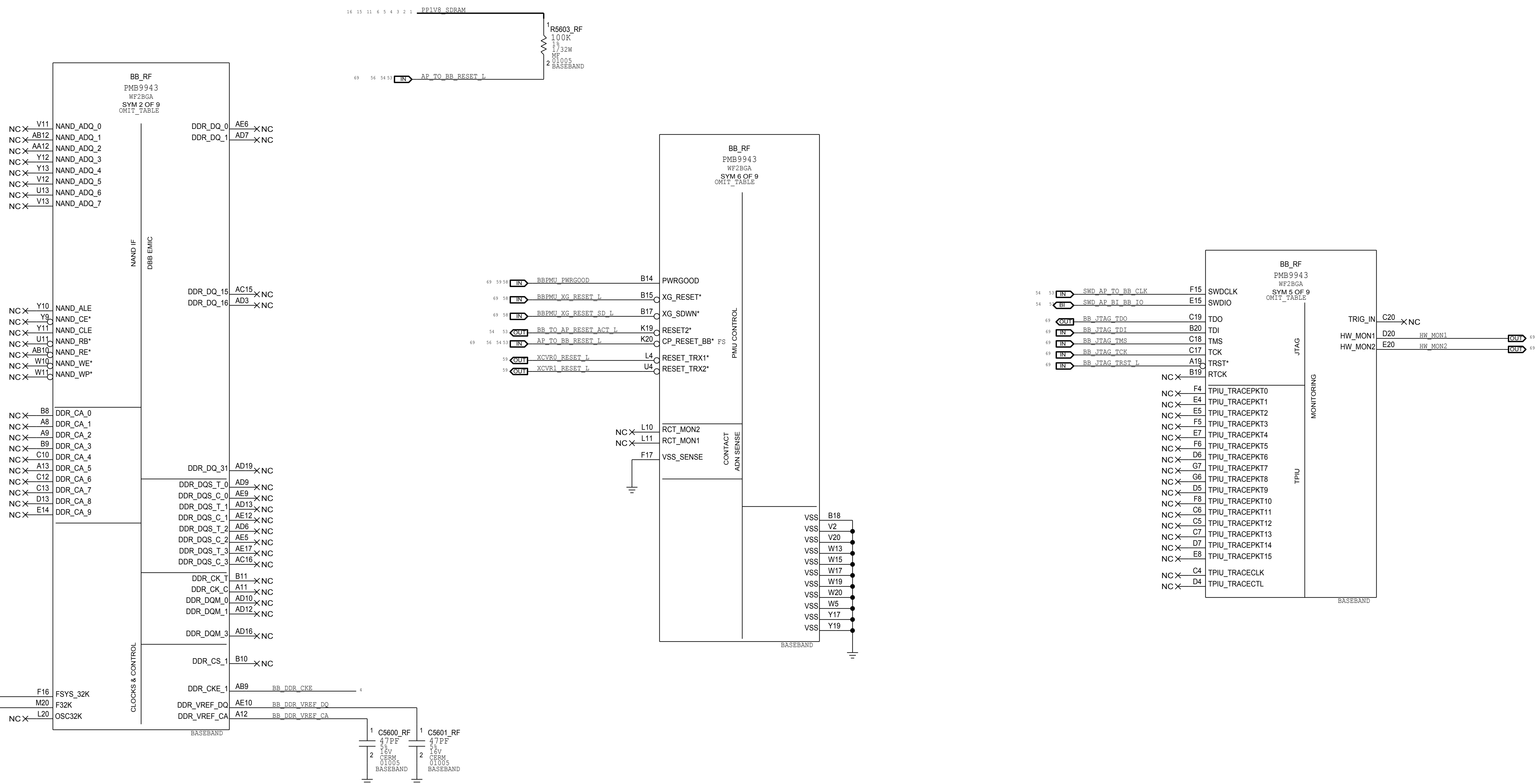
B

A

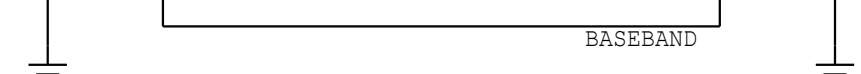
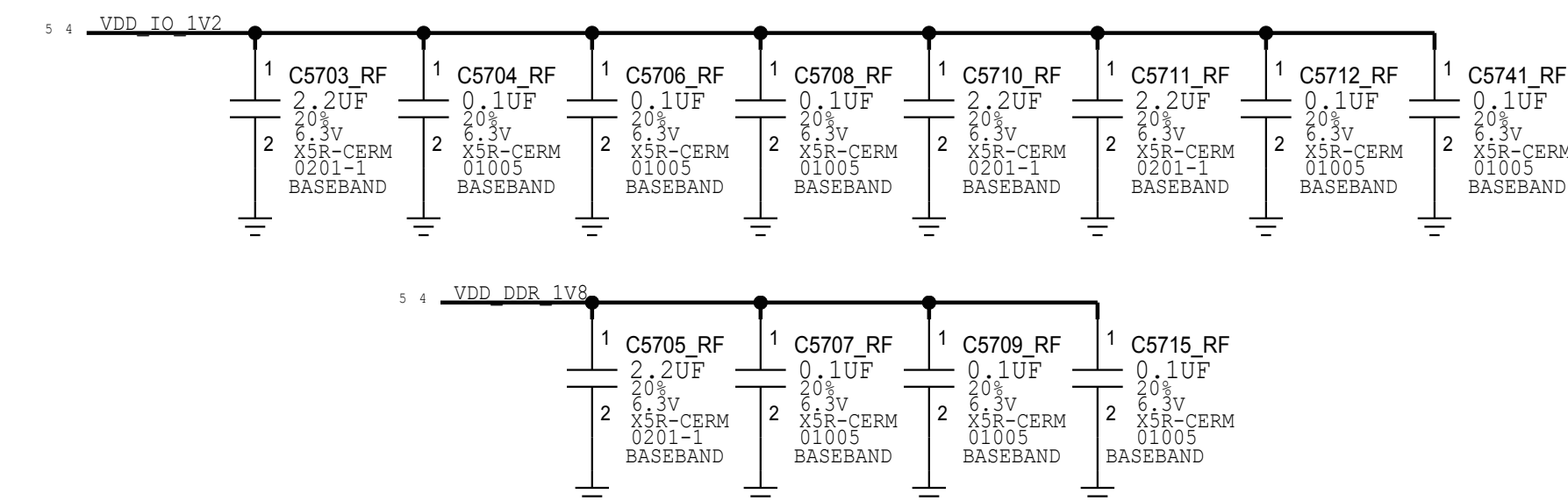
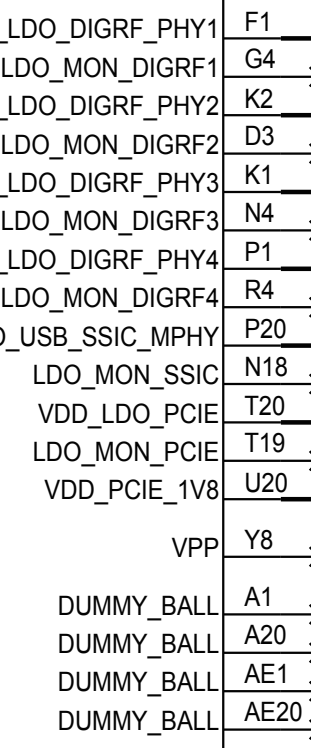
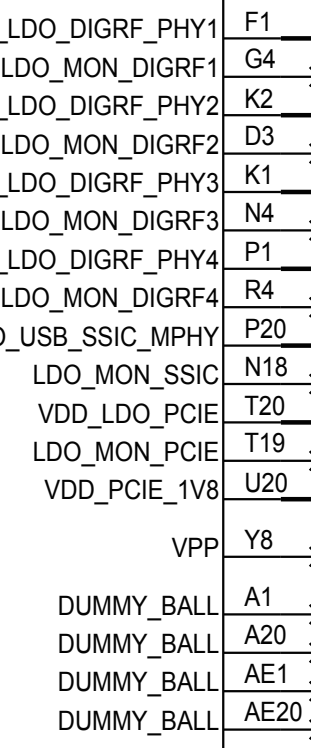
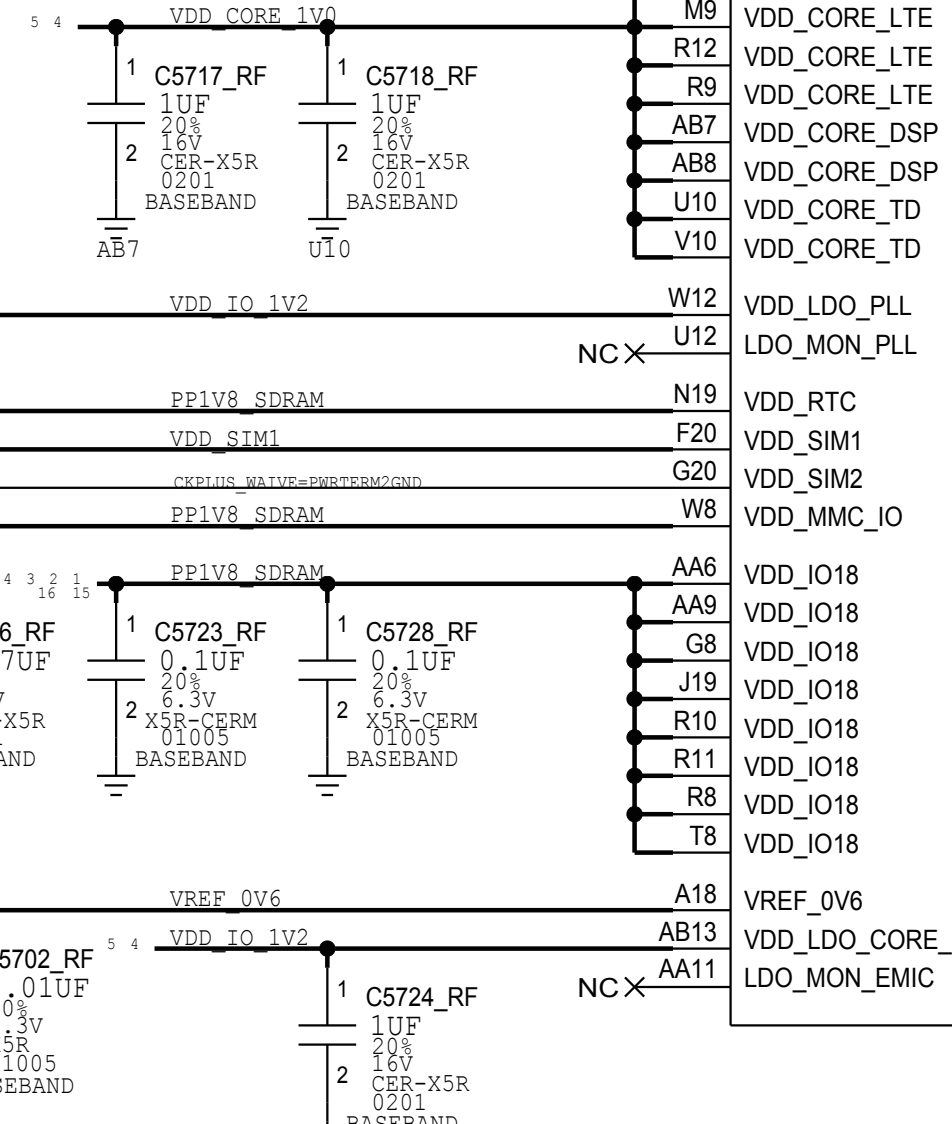
A



BASEBAND MEMORY/DEBUG

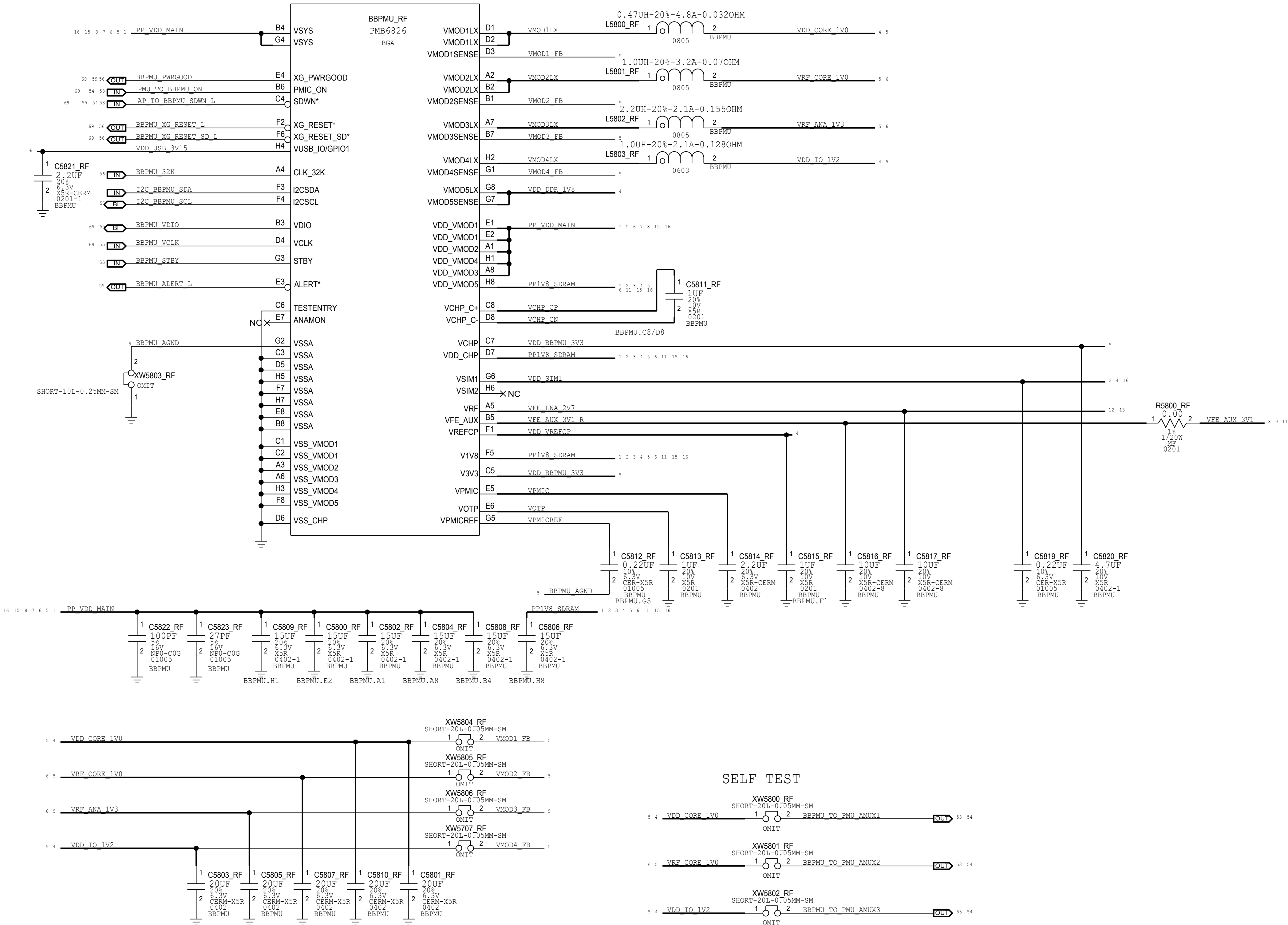


CONFIDENTIAL AND PROPRIETARY APPLE SYSTEM DESIGN. FOR REFERENCE PURPOSE ONLY - NOT A CHANGE REQUEST



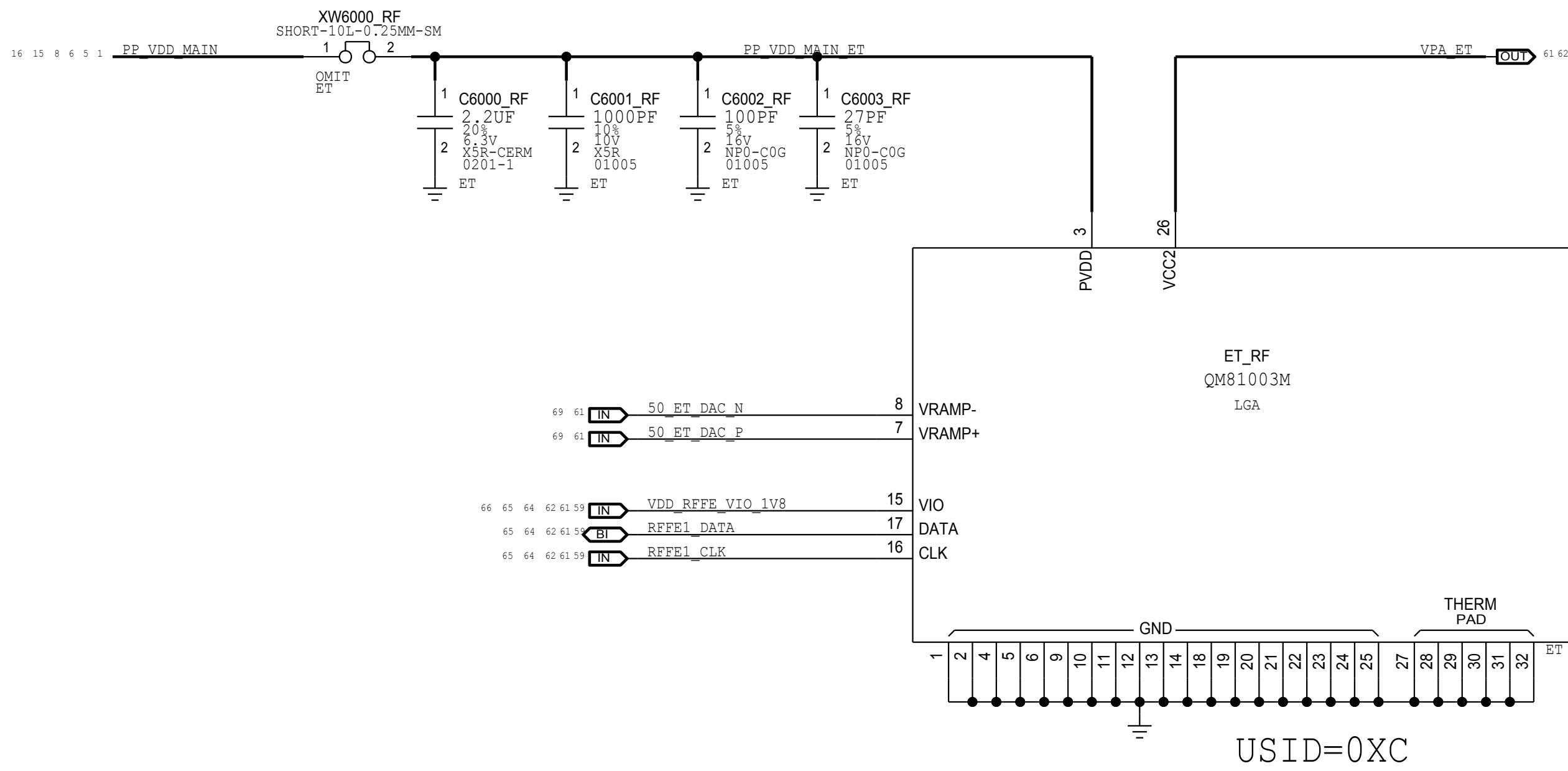
1

BASEBAND PMIC



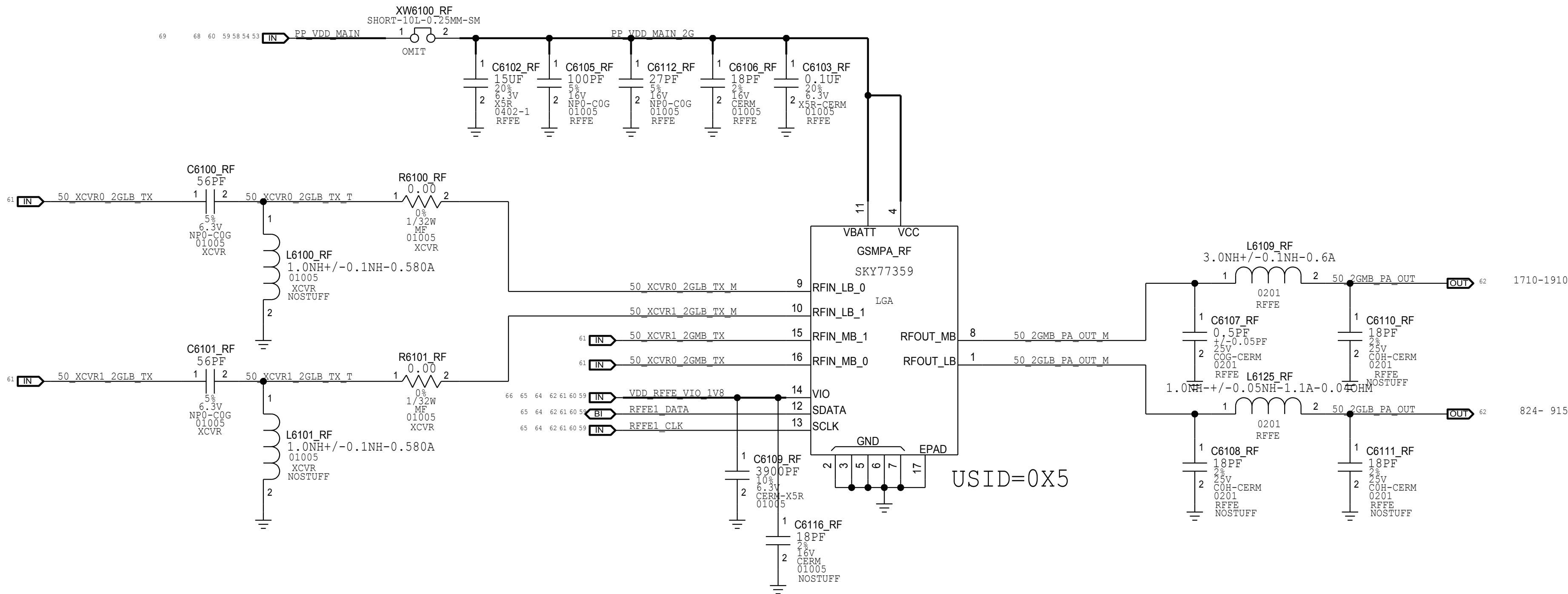
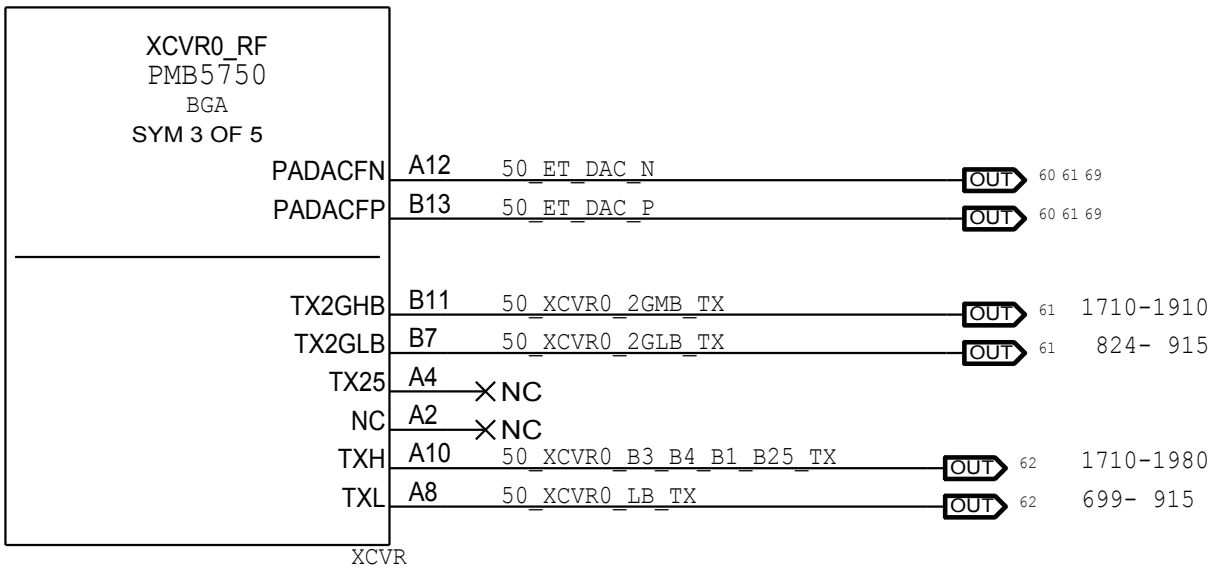
ET MODULATOR

ALPES II ES2

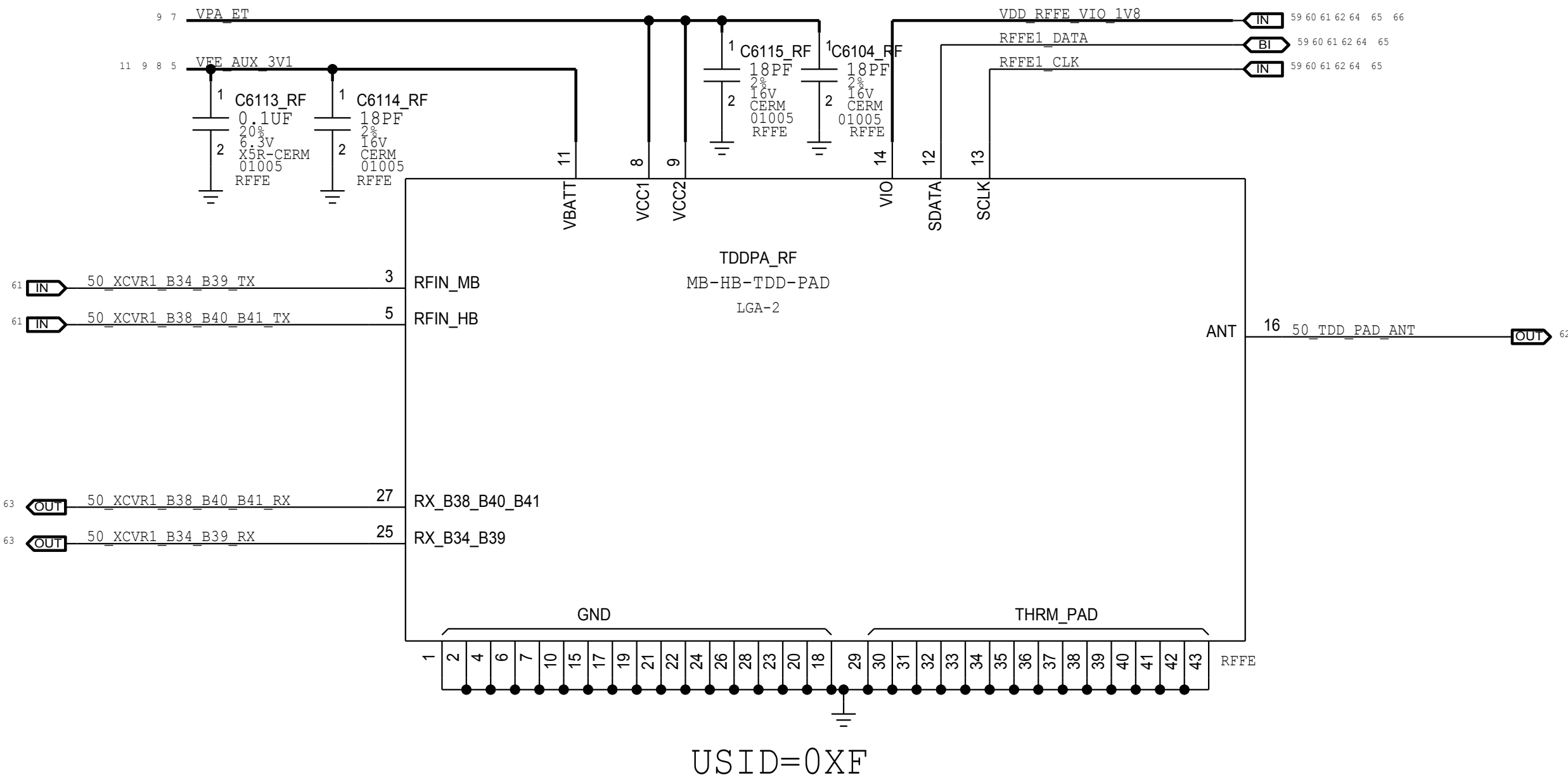
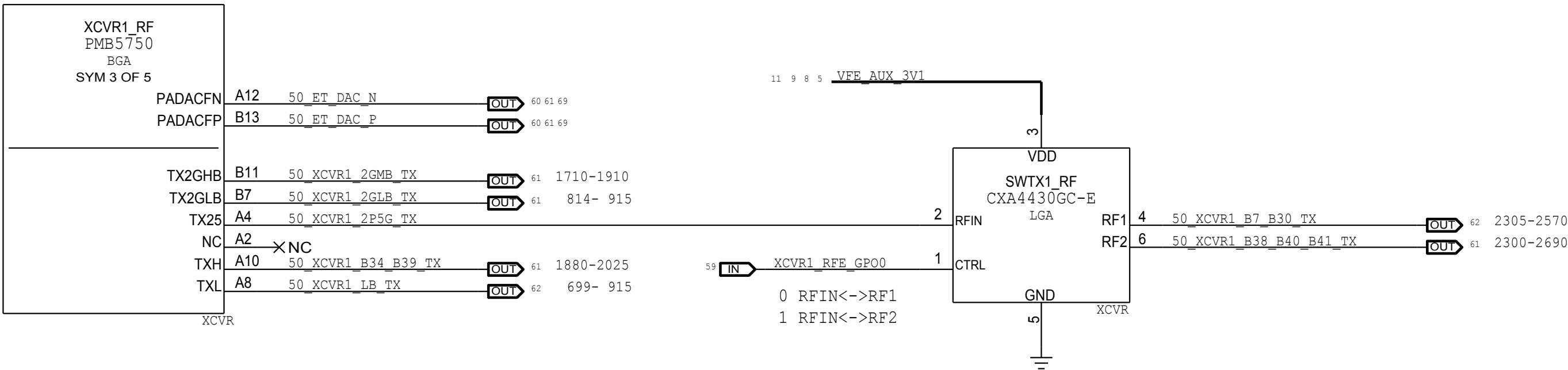


TDD TRANSMIT

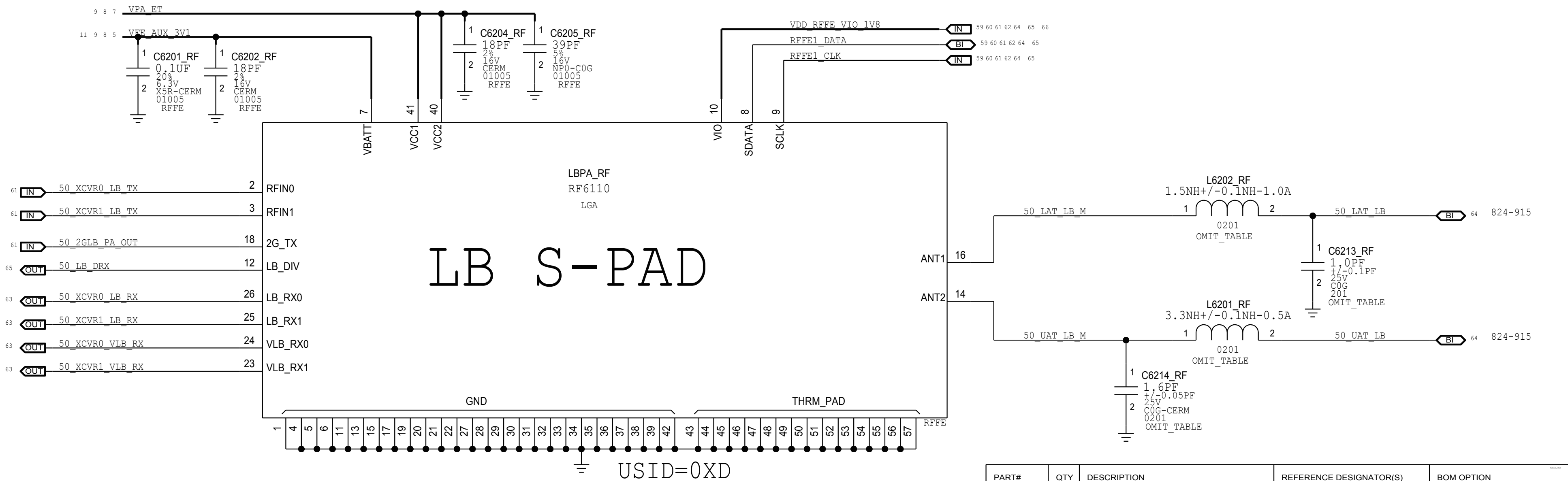
2G PA



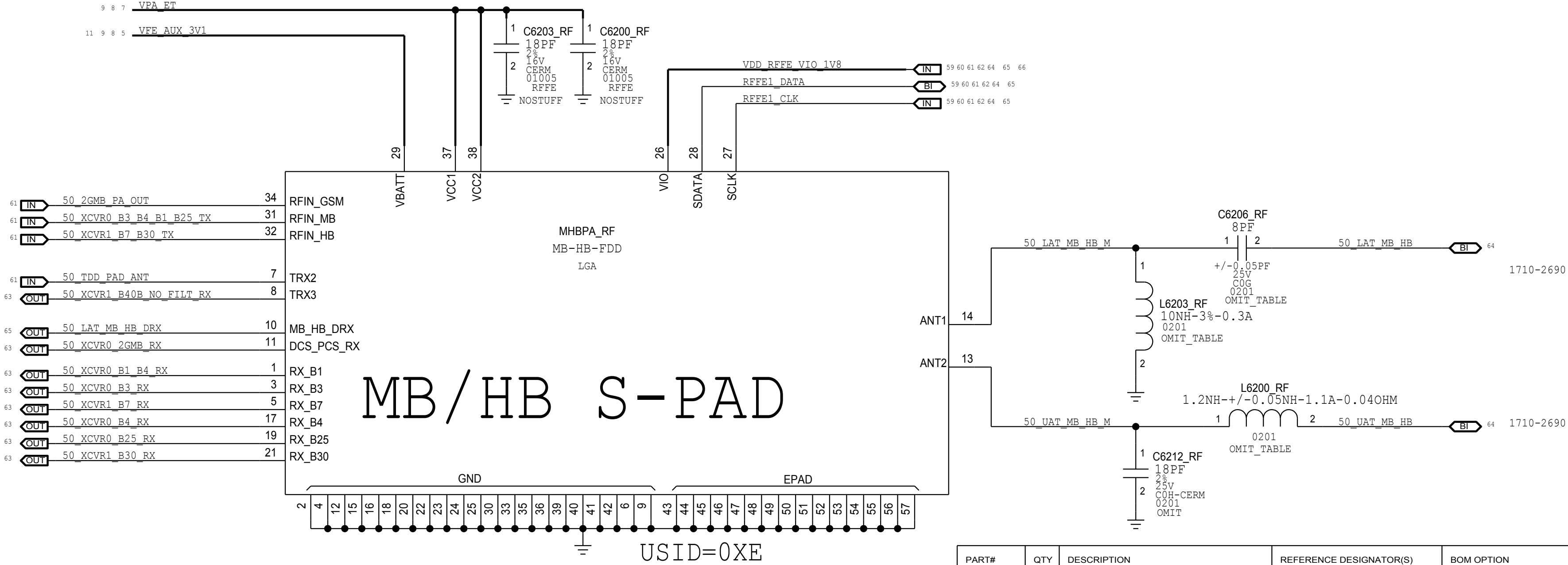
MB HB TDD S-PAD



FDD TRANSMIT



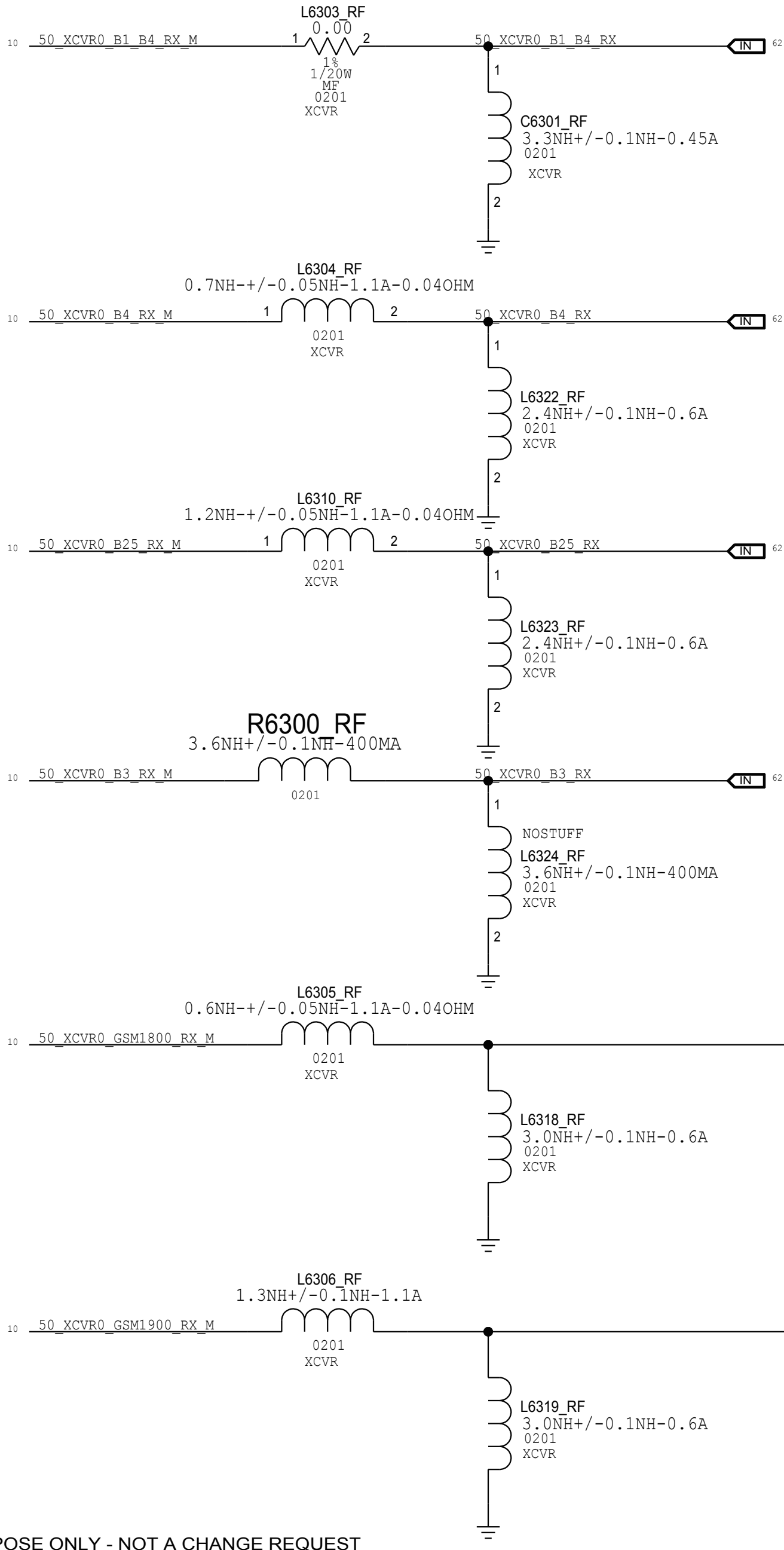
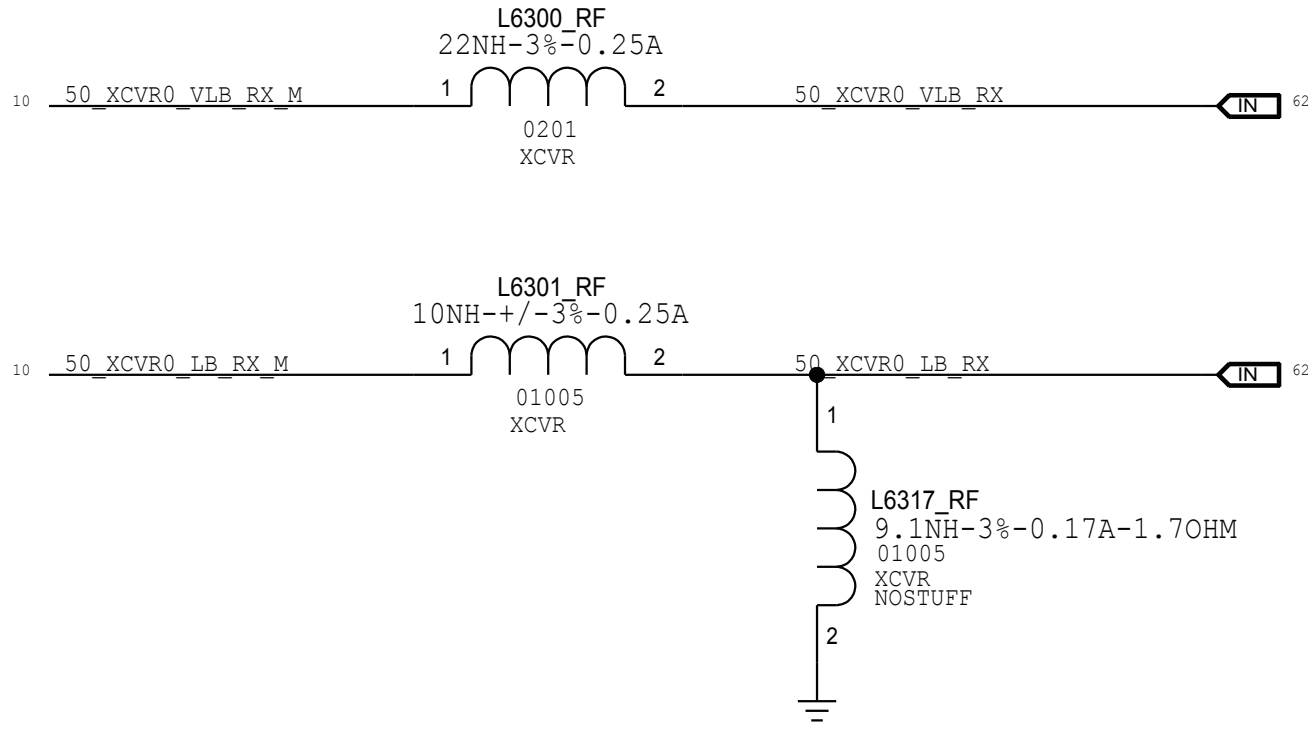
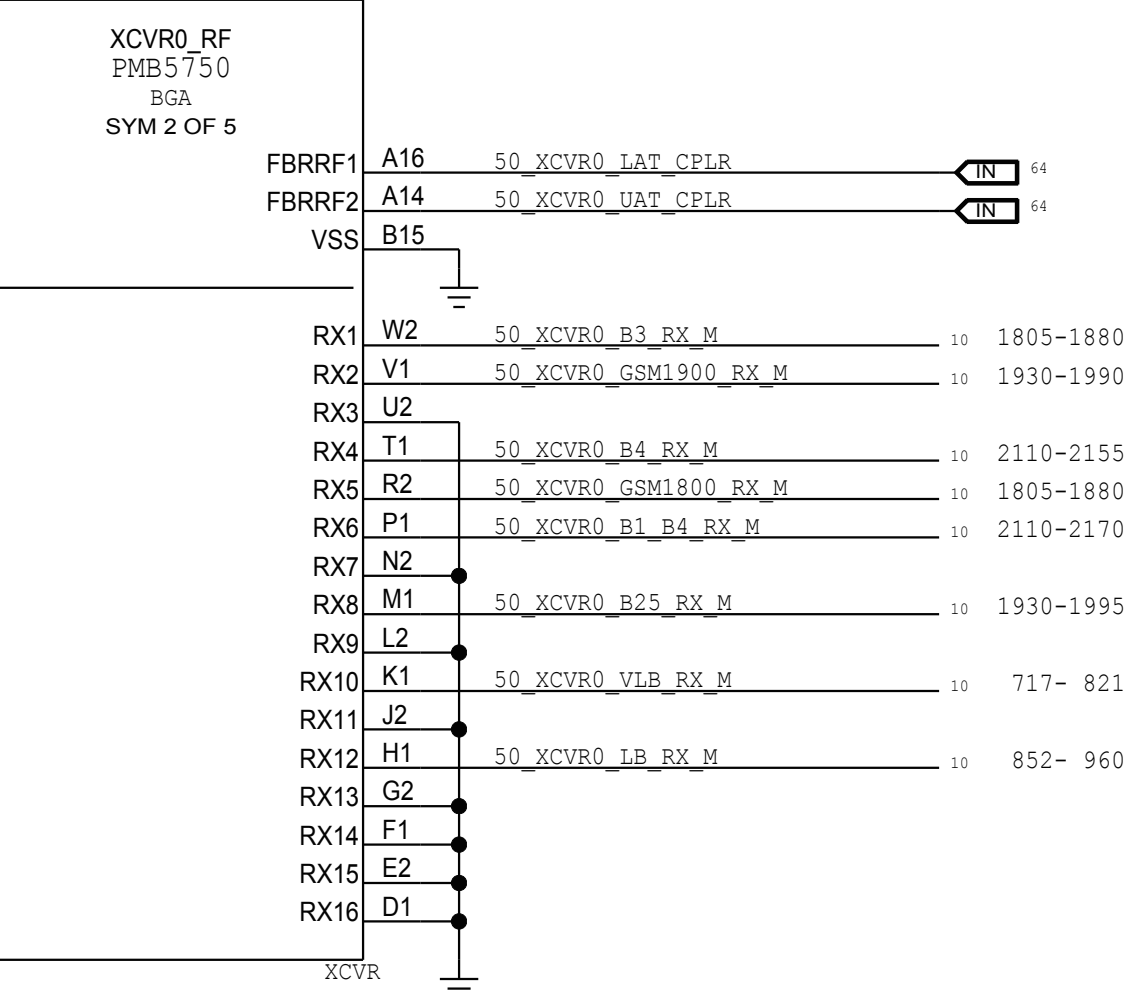
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152S2021	1	IND, 1.5NH, 0201, UH-Q	L6202_RF	D101
131S0278	1	CAP, 1.0PF, 0201, HQ	C6213_RF	D101
152S2004	1	IND, 3.3NH, 0201, UH-Q	L6201_RF	D101
131S0338	1	CAP, 1.6PF, 0201, HF	C6214_RF	D101
152S00151	1	IND, 0.8NH, 0201, UH-Q	L6202_RF	D111
131S0312	1	CAP, 1.2PF, 0201, HQ	C6213_RF	D111
152S2002	1	IND, 2.7NH, 0201, UH-Q	L6201_RF	D111
131S0338	1	CAP, 1.6PF, 0201, HF	C6214_RF	D111



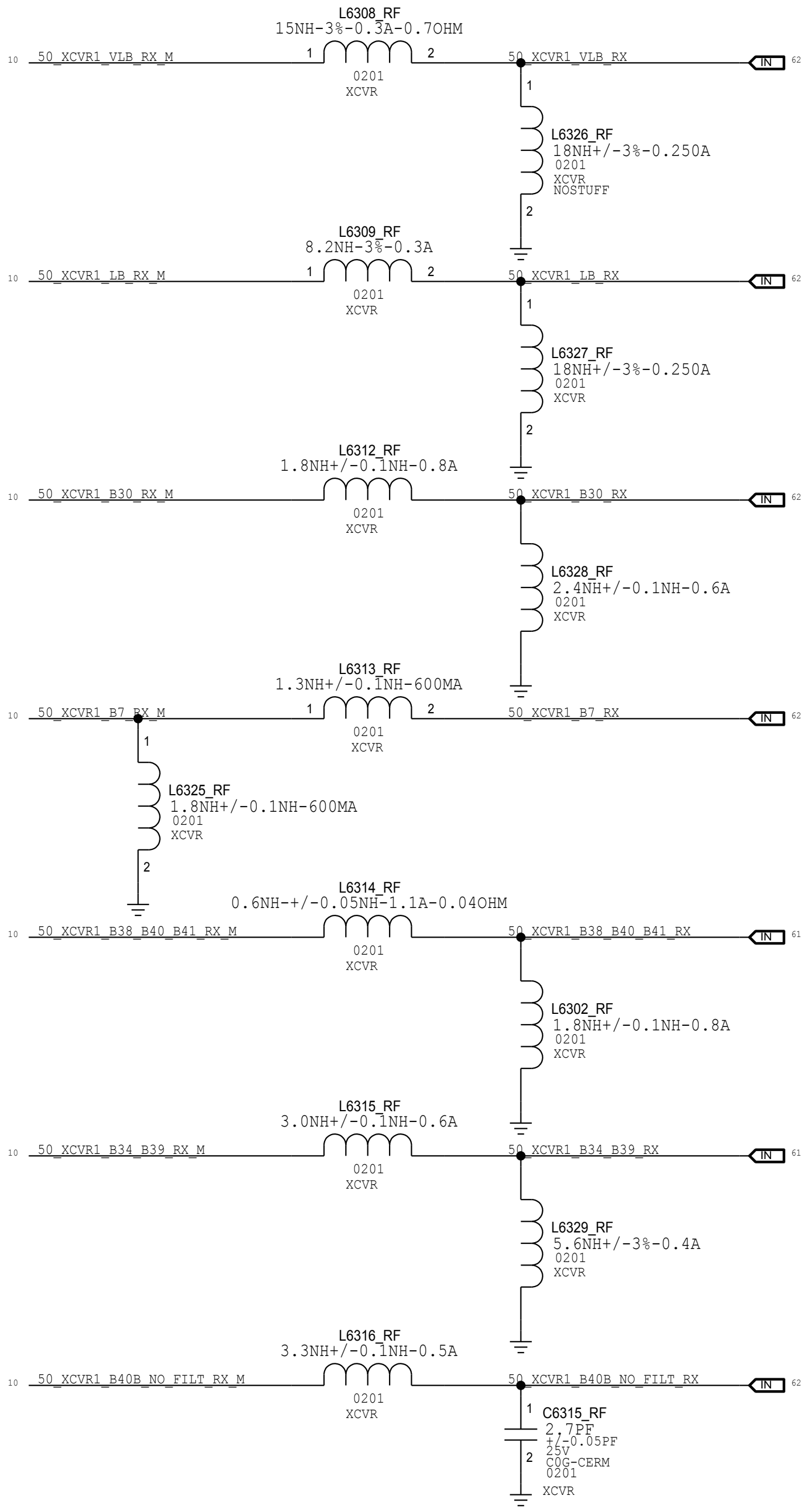
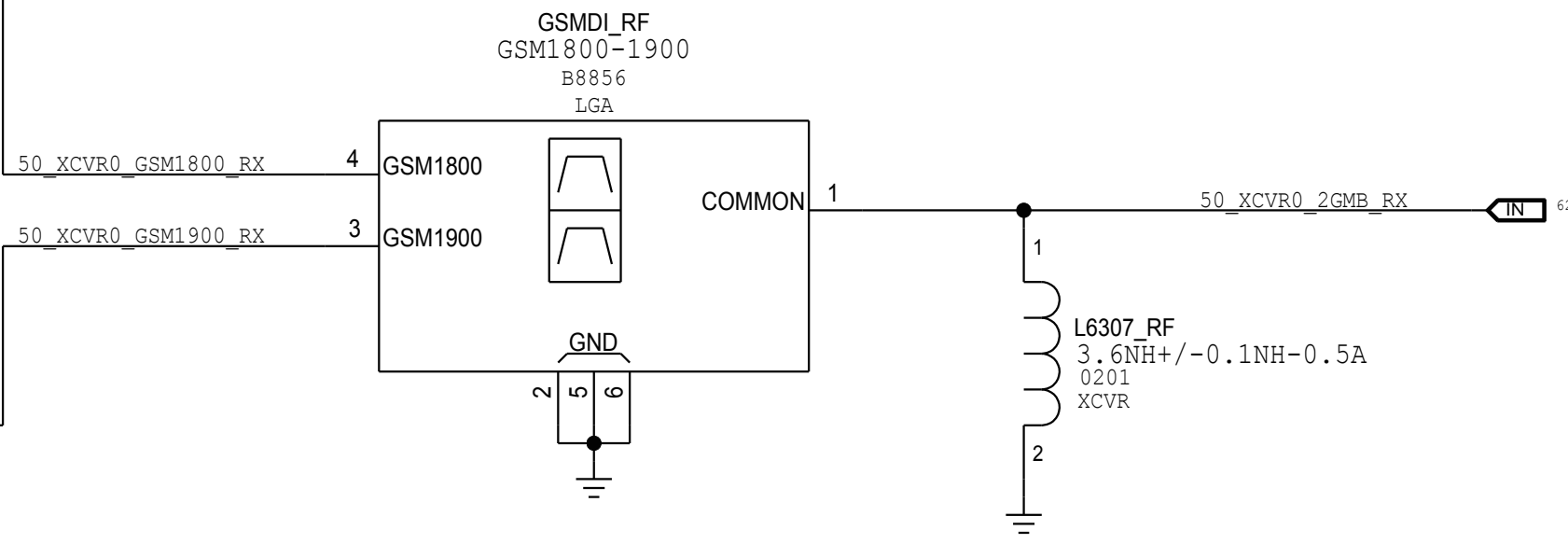
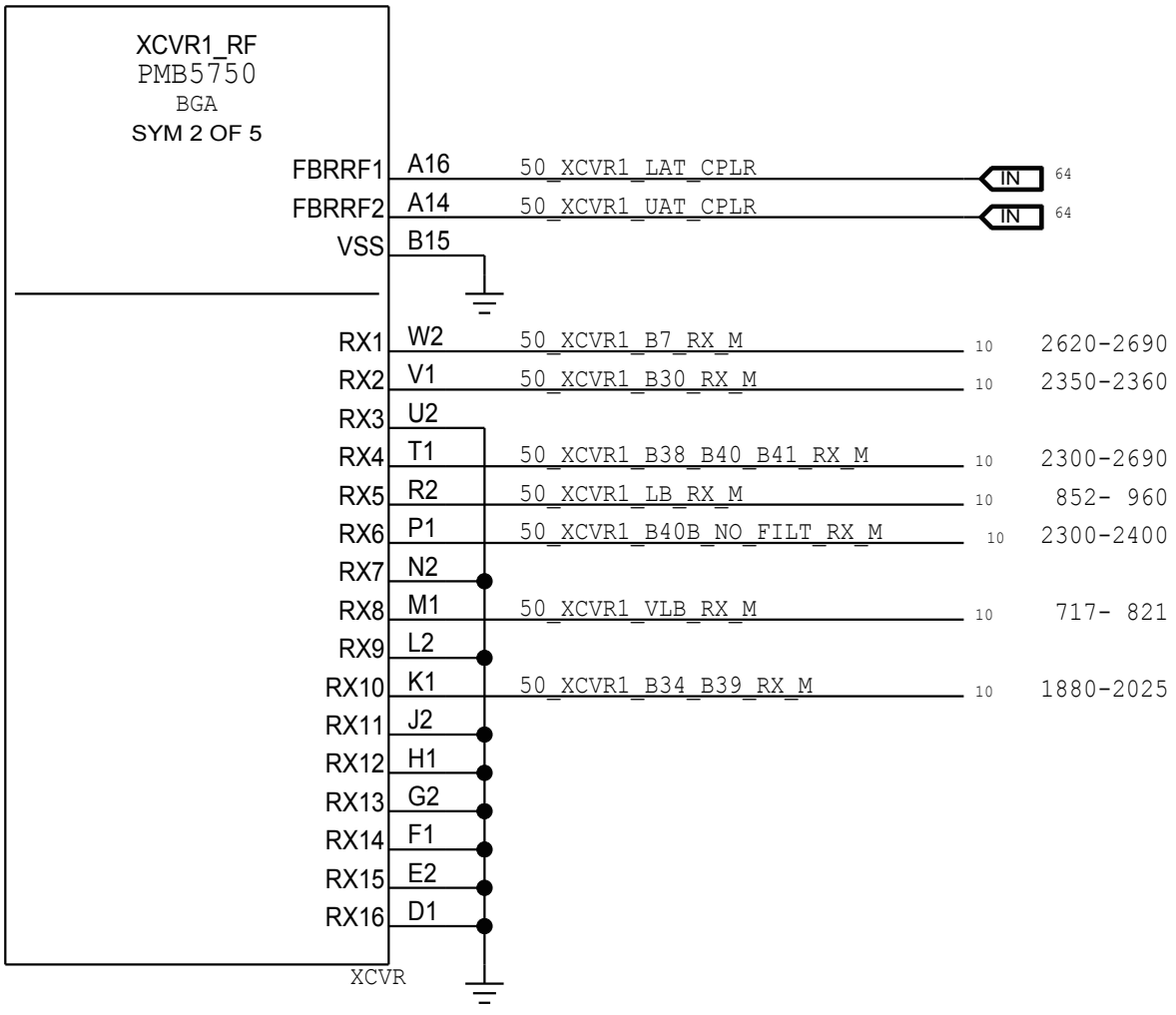
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152S00157	1	IND, 1.2NH, 0201, UH-Q	L6200_RF	D101
		NOSTUFF	C6212_RF	D101
131S00068	1	CAP, 8.0PF, 0201, HQ	C6206_RF	D111
152S00020	1	IND, 12NH, 0201, UH-Q	L6203_RF	D111
152S2021	1	IND, 1.5NH, 0201, UH-Q	L6200_RF	D111
		NOSTUFF	C6212_RF	D111

PRIMARY RECIEVE

XCVR PRX

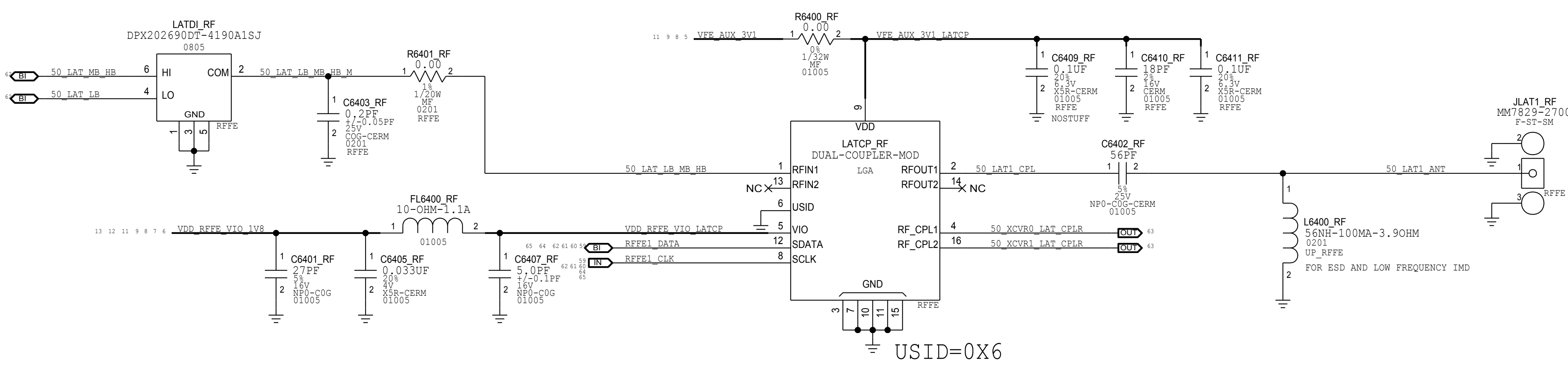


XCVR PRX

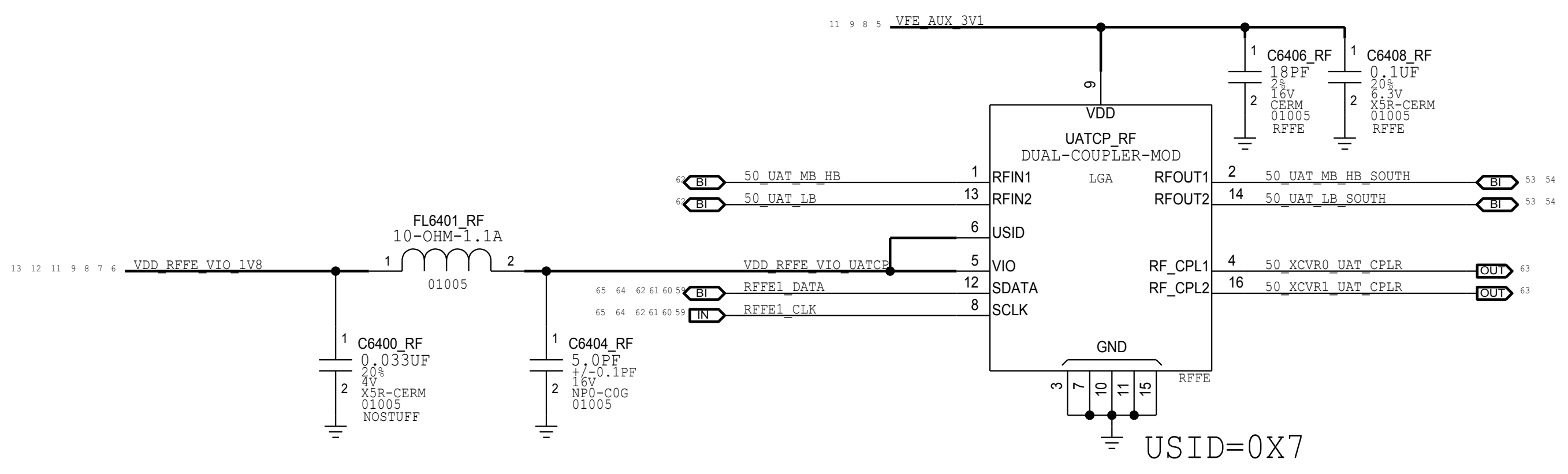


LOWER ANTENNA AND COUPLER

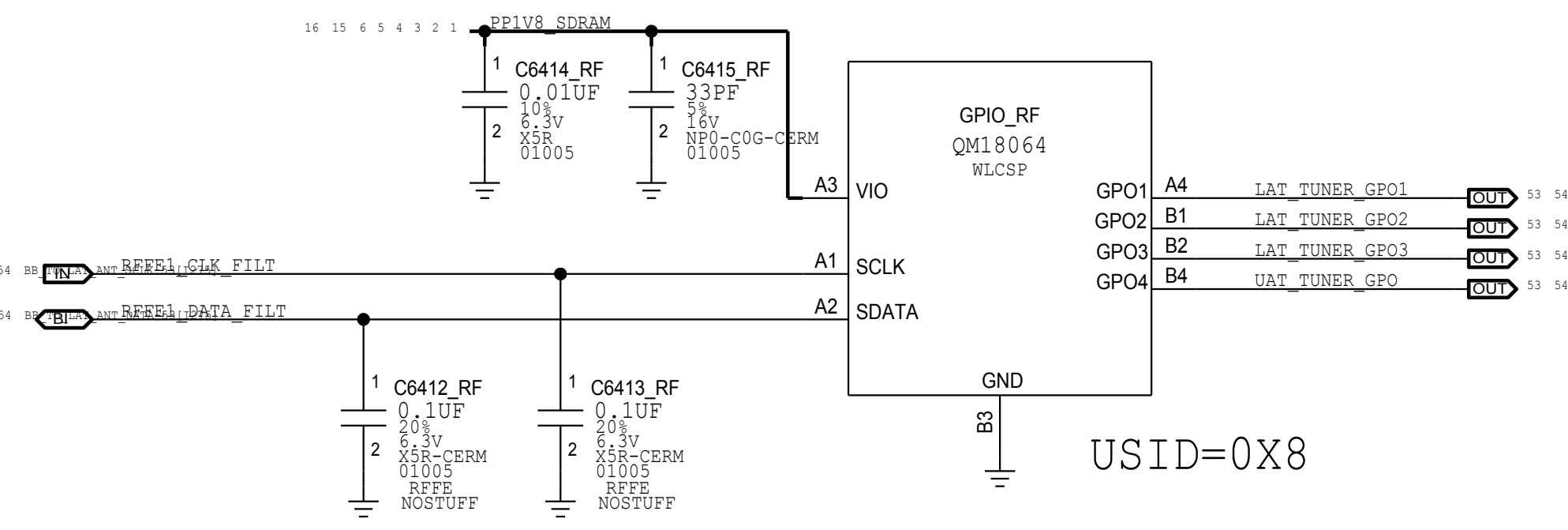
LOWER ANTENNA COUPLER



UPPER ANTENNA COUPLER

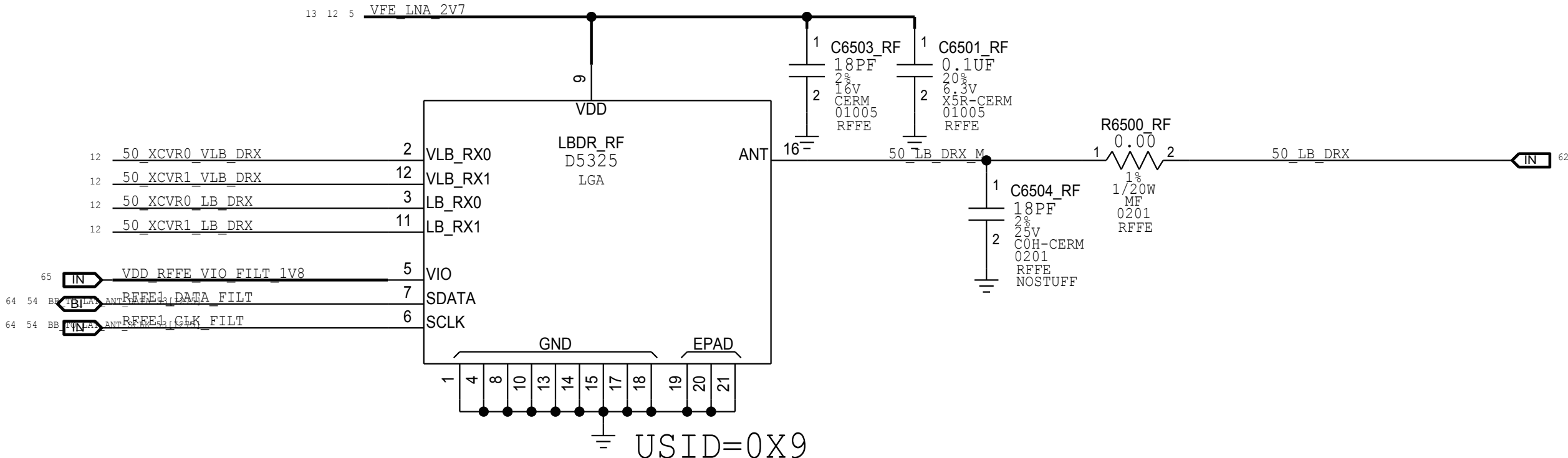
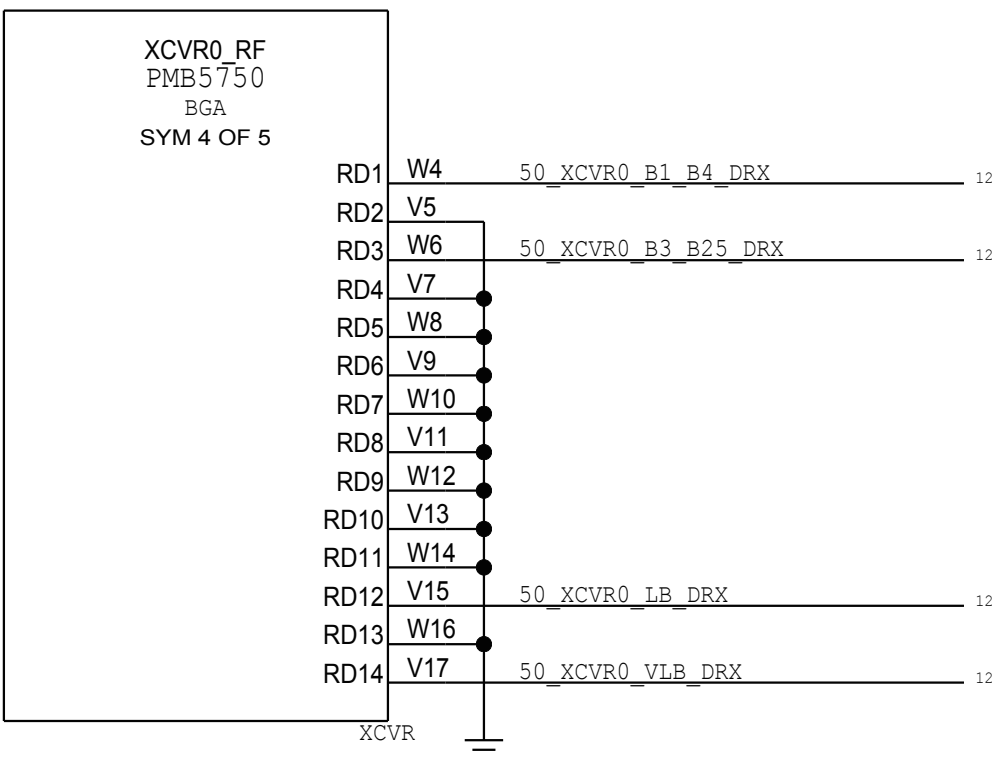


TUNER GPO

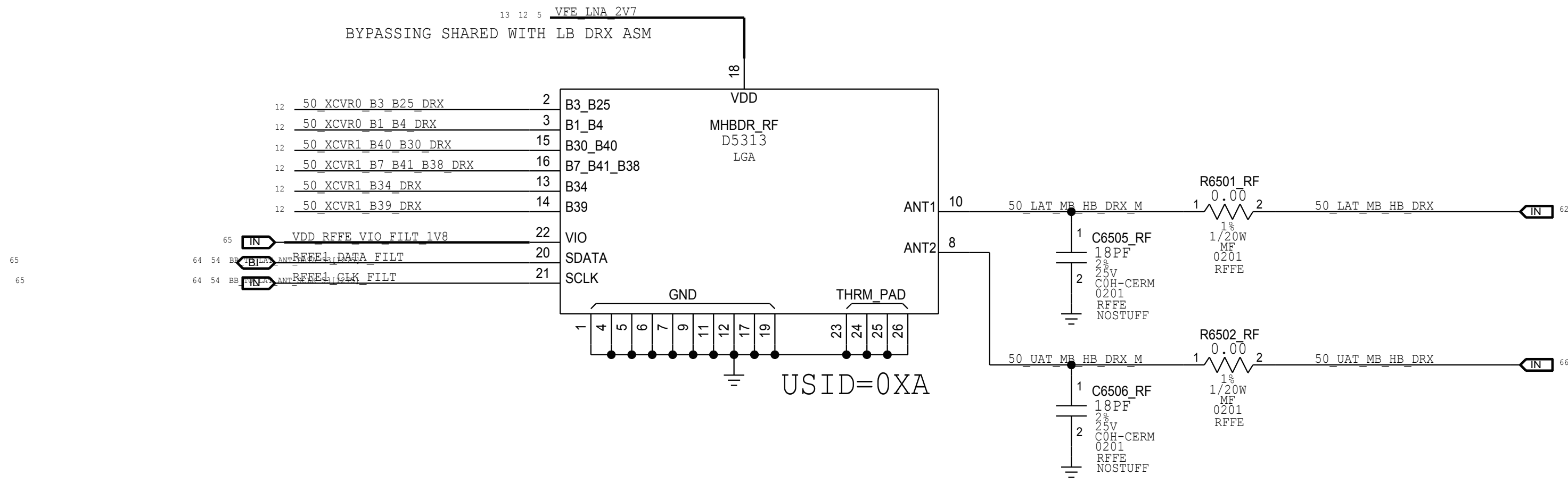
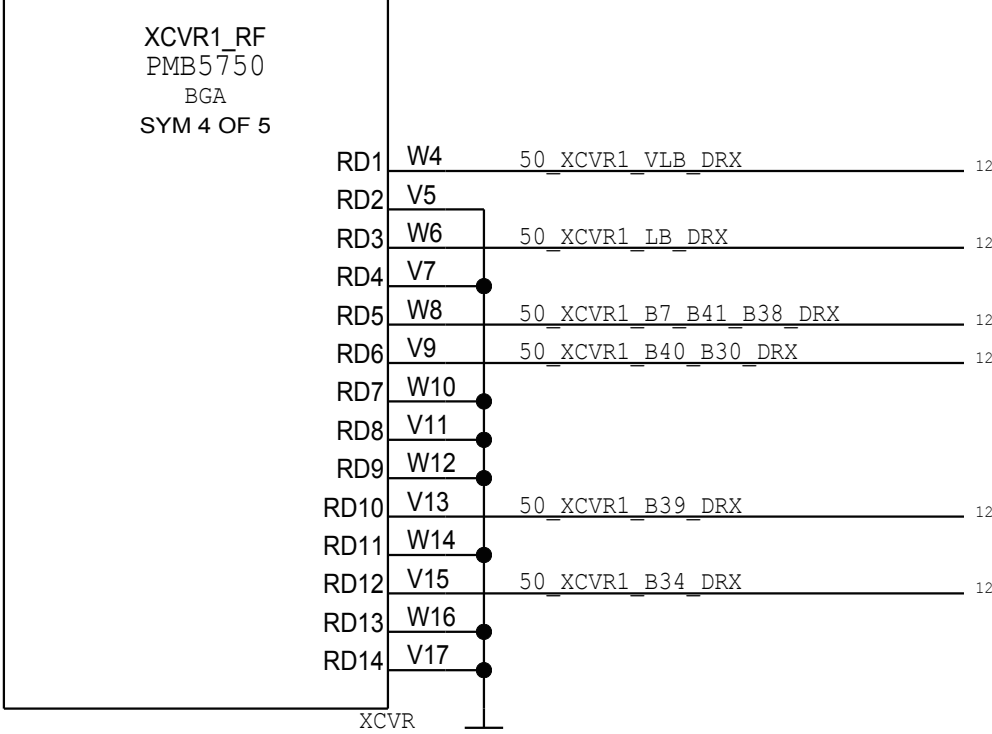


DIVERSITY RECEIVE ASM'S

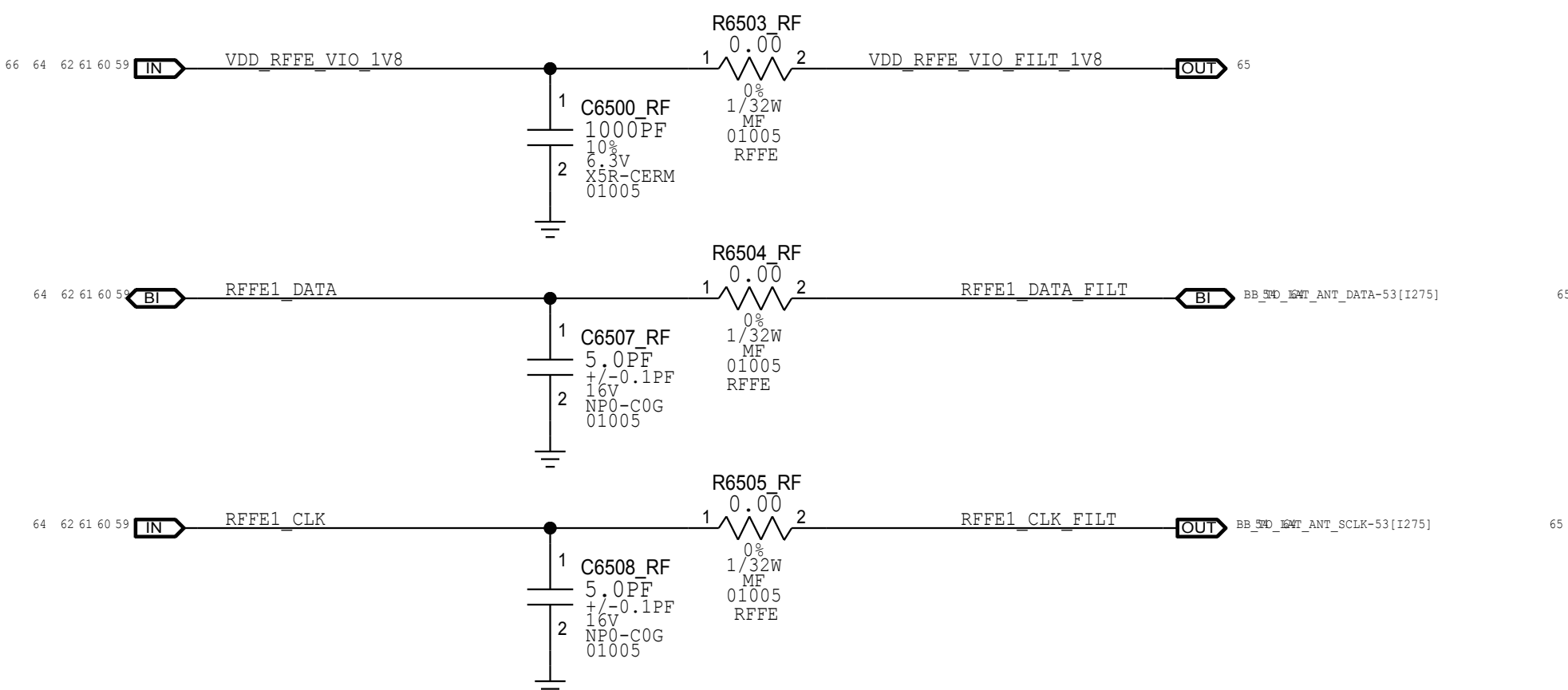
LB DRX ASM



MB HB DRX ASM



RFFE FILTERING



D



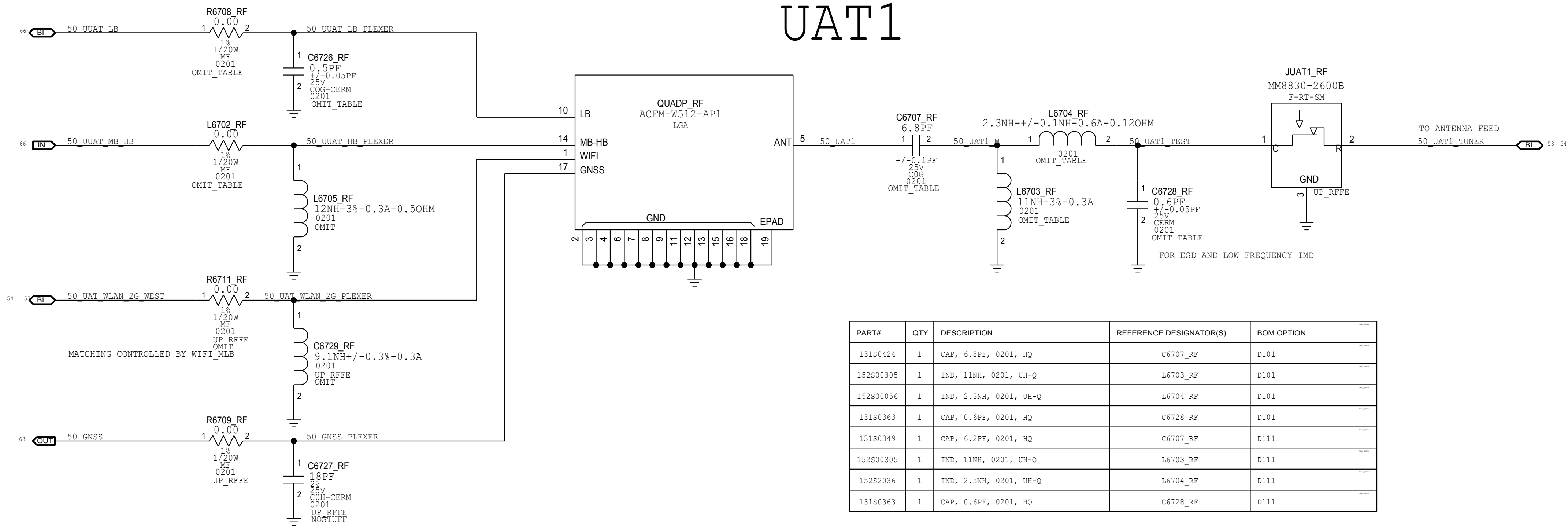
C



B



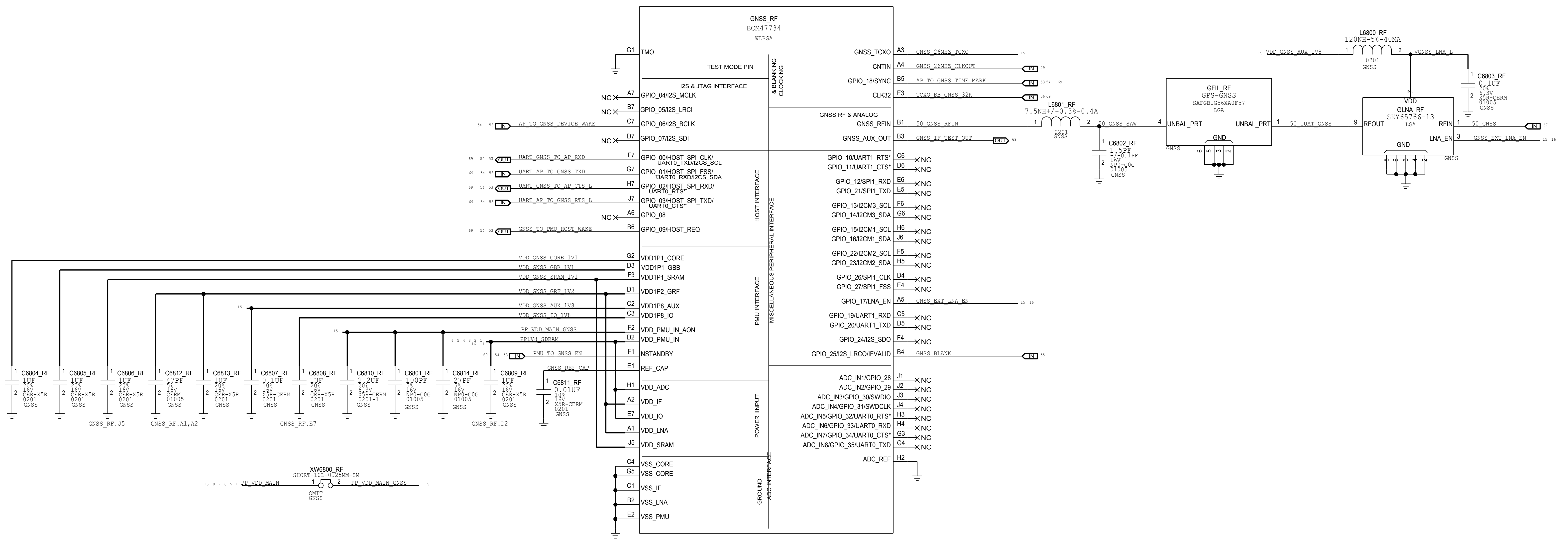
UPPER ANTENNA FEEDS



PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	BOM OPTION
118S0724	1	RES, 0OHM, 0201	R6708_RF	D101
131S0425	1	CAP, 0.5PF, 0201, HQ	C6726_RF	D101
118S0724	1	RES, 0OHM, 0201	L6702_RF	D101
152S00020	1	IND, 12NH, 0201, UH-Q	L6705_RF	D101
118S0724	1	RES, 0OHM, 0201	R6708_RF	D111
131S0425	1	CAP, 0.5PF, 0201, HQ	C6726_RF	D111
118S0724	1	RES, 0OHM, 0201	L6702_RF	D111
152S00020	1	IND, 12NH, 0201, UH-Q	L6705_RF	D111

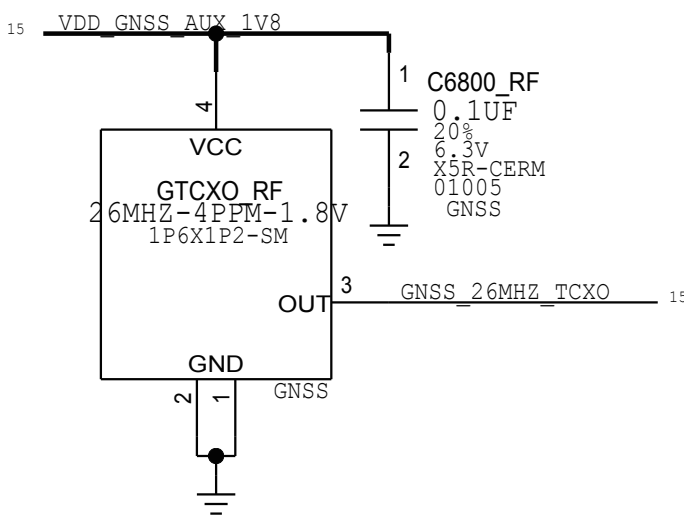
PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	BOM OPTION
131S0424	1	CAP, 6.8PF, 0201, HQ	C6707_RF	D101
152S00305	1	IND, 11NH, 0201, UH-Q	L6703_RF	D101
152S00056	1	IND, 2.3NH, 0201, UH-Q	L6704_RF	D101
131S0363	1	CAP, 0.6PF, 0201, HQ	C6728_RF	D101
131S0349	1	CAP, 6.2PF, 0201, HQ	C6707_RF	D111
152S00305	1	IND, 11NH, 0201, UH-Q	L6703_RF	D111
152S2036	1	IND, 2.5NH, 0201, UH-Q	L6704_RF	D111
131S0363	1	CAP, 0.6PF, 0201, HQ	C6728_RF	D111

GNSS LNA + RECEIVER



TCXO

KEEP AWAY FROM HEAT SOURCES



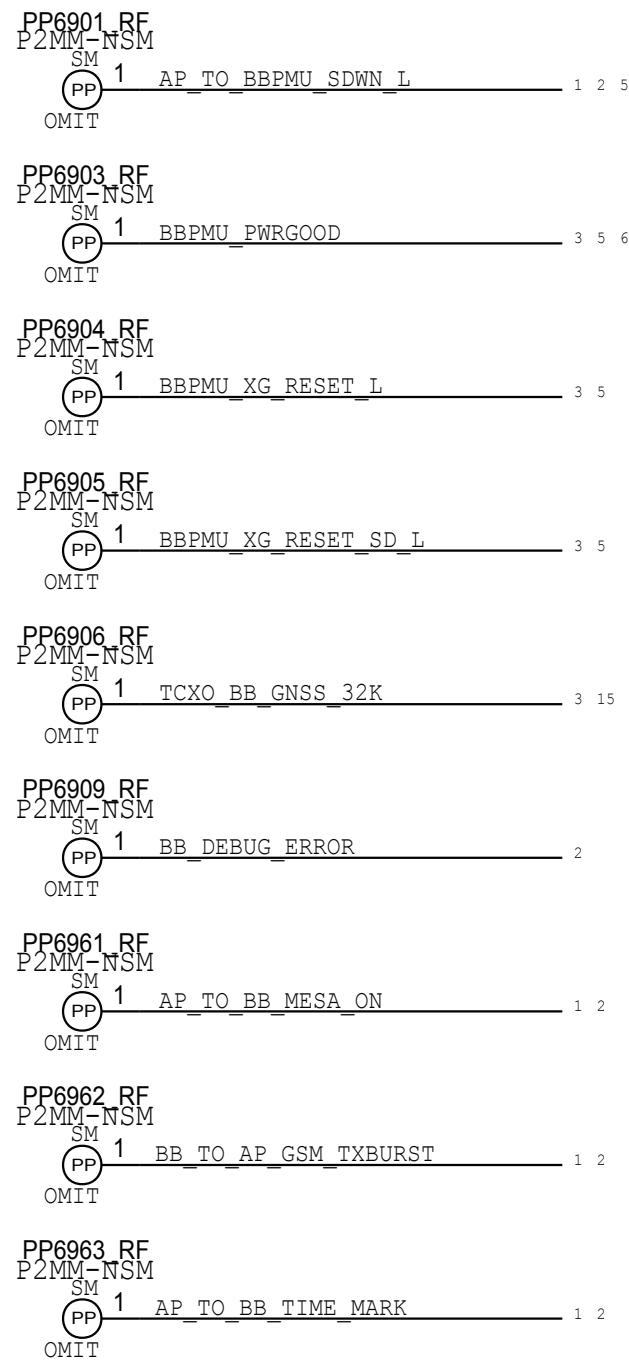
MLB TEST POINTS

SIM CARD CONNECTOR

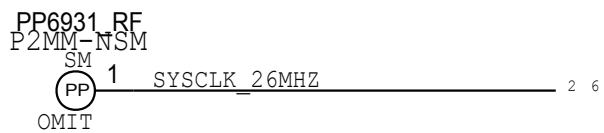
HARDWARE ID

DEV CONFIG

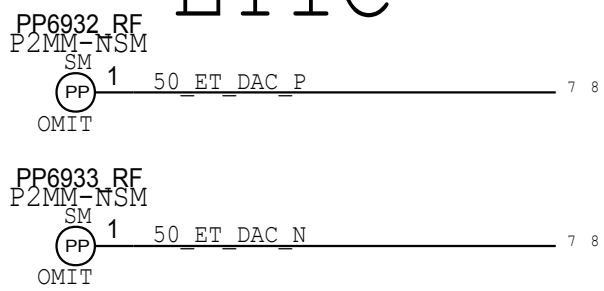
BASEBAND



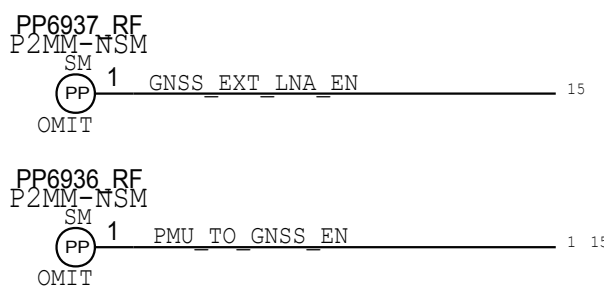
XCVR



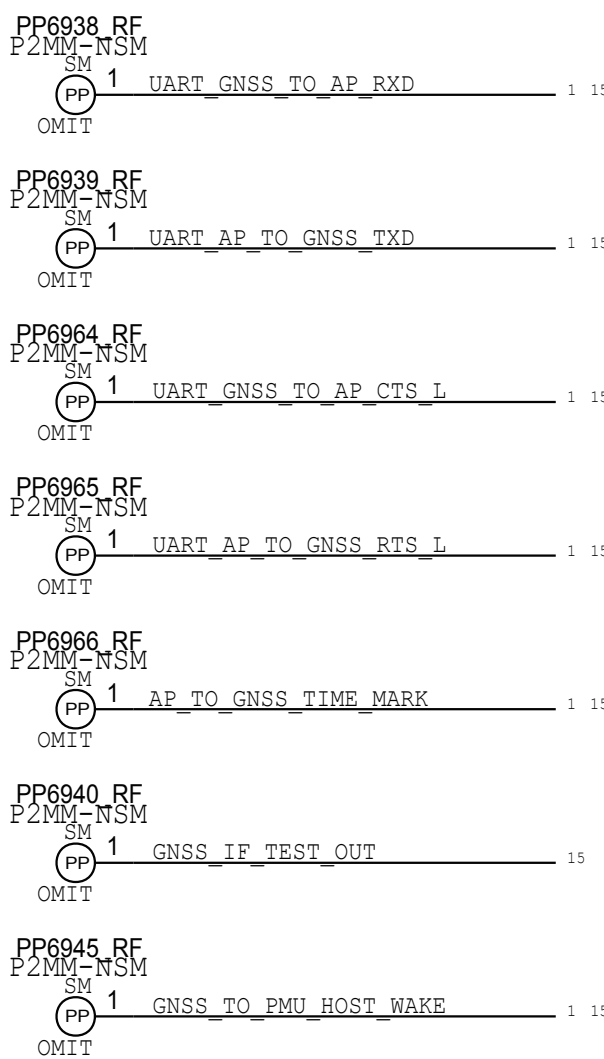
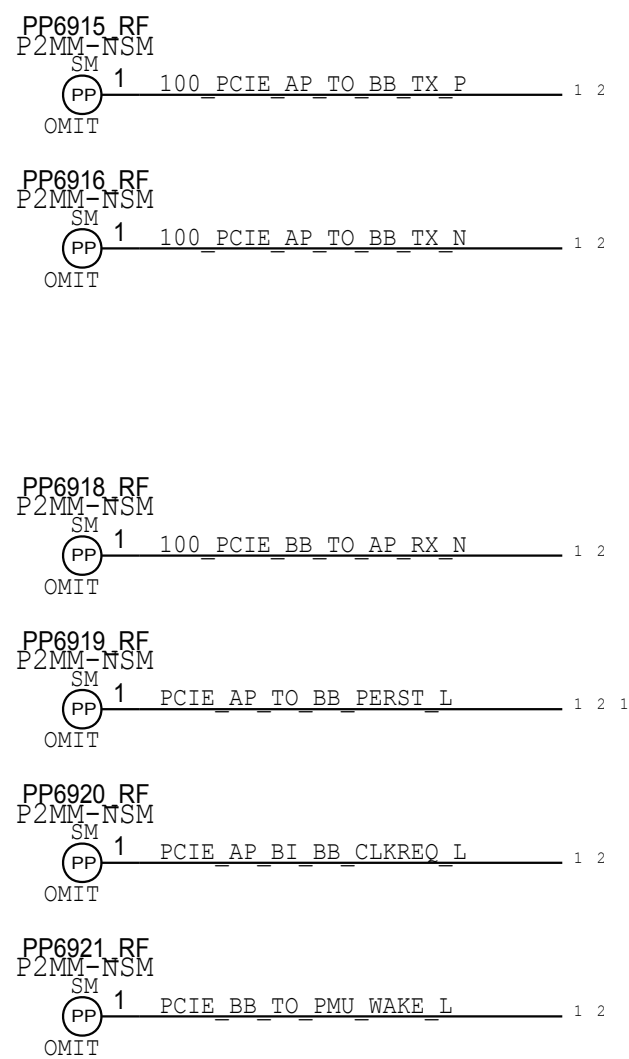
ETIC



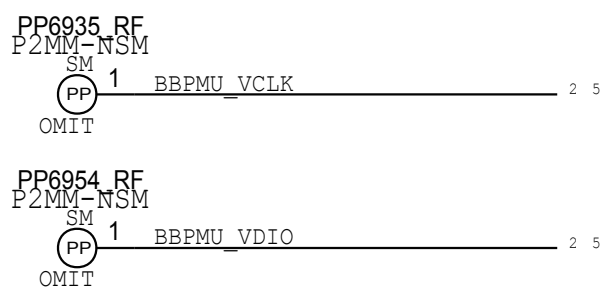
GNSS



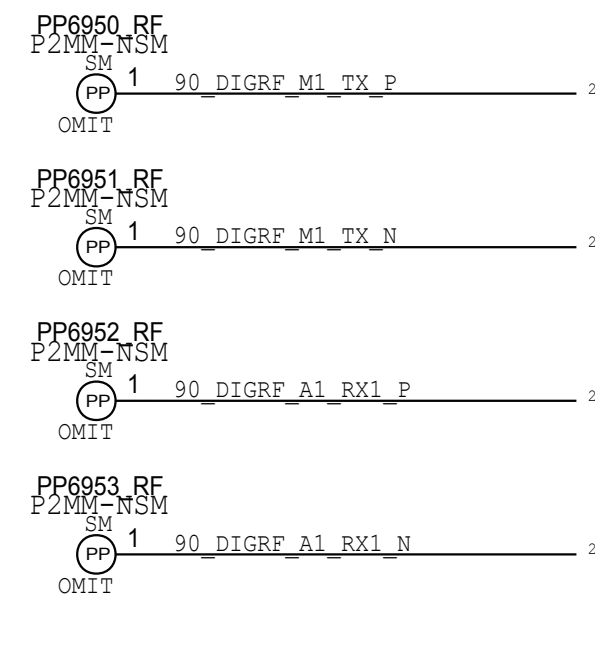
PCIE



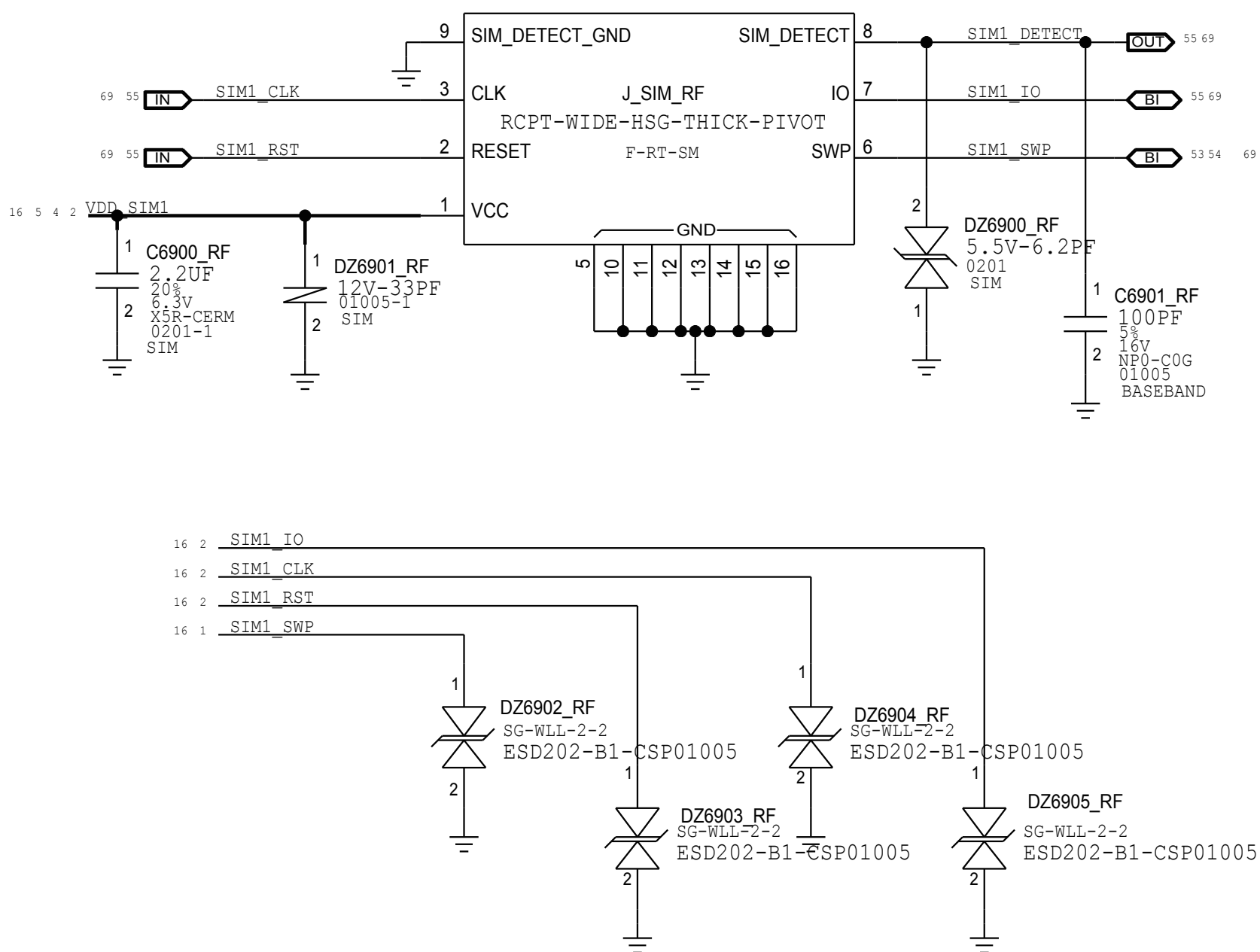
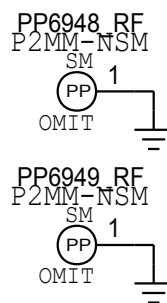
BBPMU



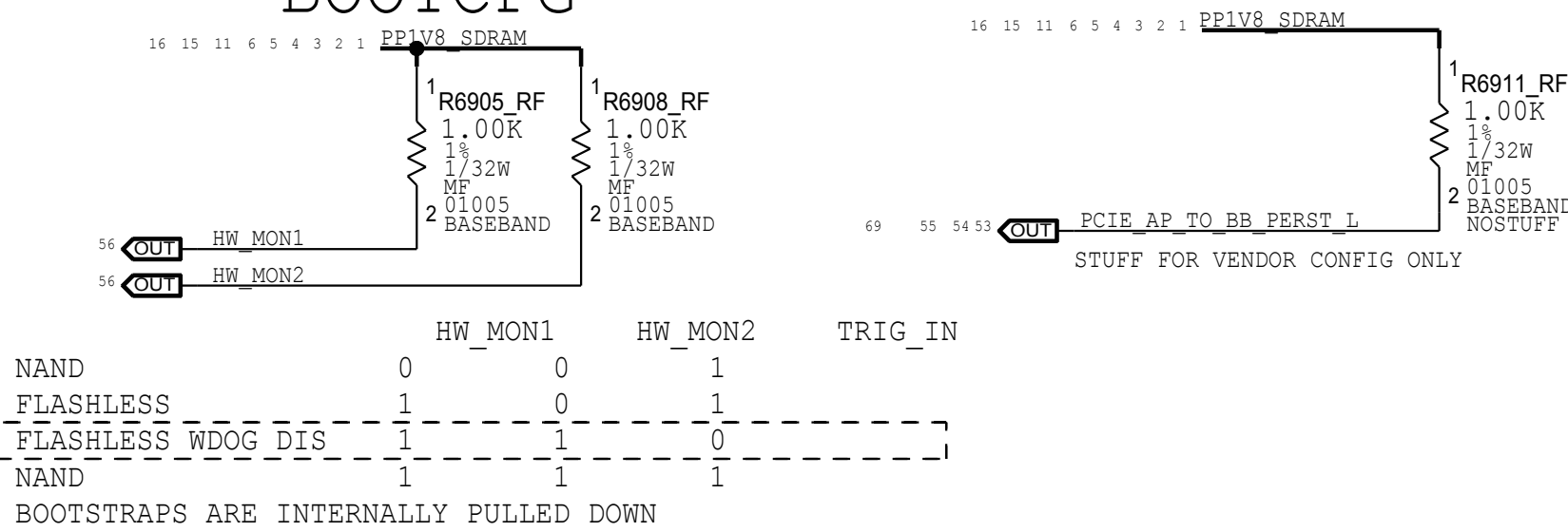
DIGRF



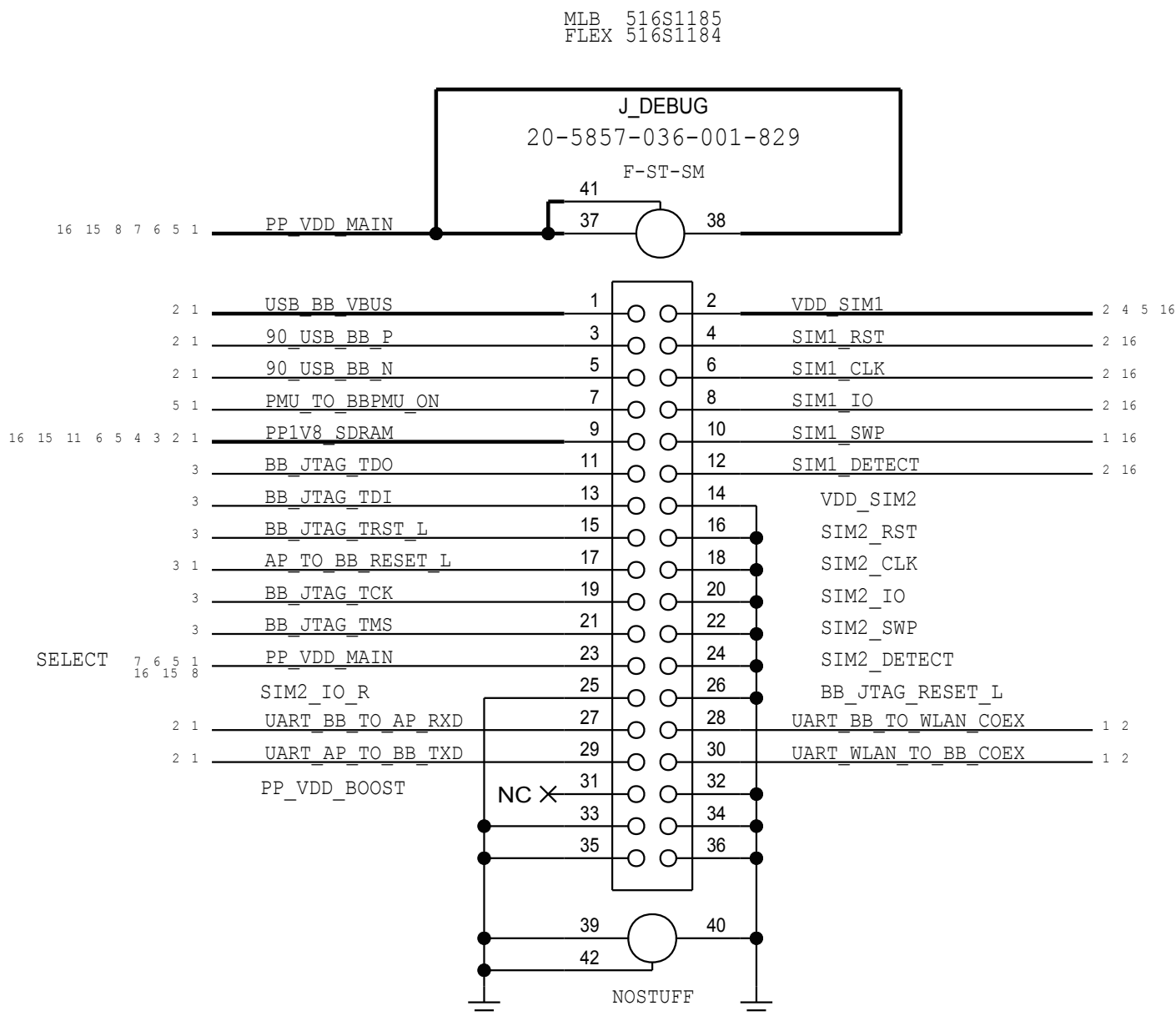
PCIE GND



BOOTCFG



DEBUG CONNECTOR



1. ALL RESISTANCE VALUES ARE IN OHMS, 0.1 WATT +/- 5%. 2. ALL CAPACITANCE VALUES ARE IN MICROFARADS. 3. ALL CRYSTALS & OSCILLATOR VALUES ARE IN HERTZ.						REV	ECN	DESCRIPTION OF REVISION	DATE
						8	0006400877	ENGINEERING RELEASED	2016-06-14

D

D10 RADIO MLB_FF

FEB 19, 2016

C

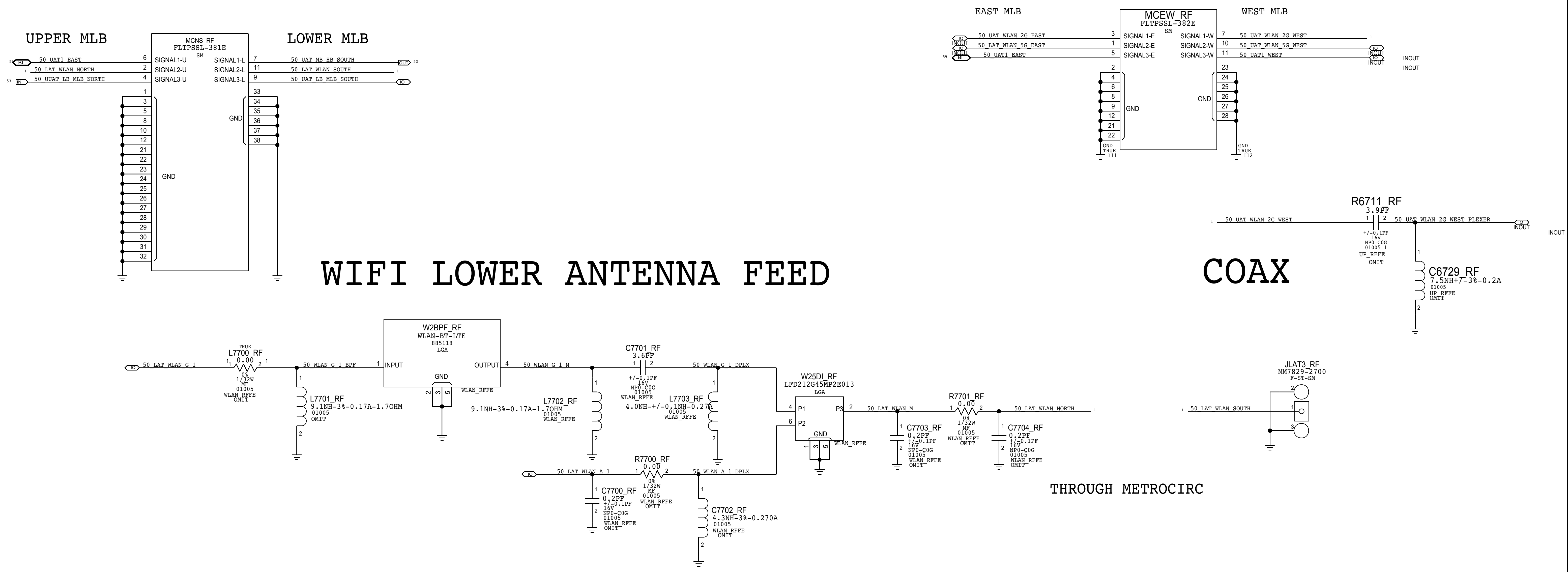
D10 NORTH-SOUTH METROCIRC

339S00086

C

D10 EAST-WEST METROCIRC

339S00110



A

ADD REV ID FOR D10/D11 HERE

